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XP200series

AC Servo Drive

User Manual

(V1.0Full version)

Product Introduction

XP200 The series of AC servo drives are general-purpose AC servo drive products with leading domestic level. This series of products adopts advanced full digital control and AC motor vector control theory, with excellent system performance and high reliability. It is widely used in servo axis drive of food processing, packaging machinery, textile machinery and other related automation industry machinery.

Features

Adopt high-performance control chip, with excellent control performance

Compact design, saving installation space

Use new industrial-grade IPM module with strong overload driving capability

Integrates speed control, position control, and torque control

Can drive various types of permanent magnet synchronous servo motors

With complete fault protection and status monitoring functions

Excellent low-speed torque characteristics and dynamic acceleration and deceleration performance

Model Specifications

XP200 -20 R L

① ② ③ ④

① Driver series:

XP100-Universal Servo Drive

② Module capacity:

20-20A

30-30A

③ Communication method:

N: No communication

R:485 communication

C: CAN bus

④ Voltage level:

L:220V

H:380V

Safety precautions

Before storing, installing, wiring, operating, inspecting or repairing the product, the user must be familiar with and comply with the following important matters to ensure the safe and correct use of the product.

⚠危险	Improper operation may cause danger and result in personal injury or death.
⚠注意	Improper operation may cause danger, personal injury, and possible damage to the equipment.
禁止	Any behavior that may result in damage to the device or inability to use it is strictly prohibited.

1.Use occasions

⚠危险
Do not expose the product to water vapor, corrosive gas, or flammable gas. Otherwise, it may cause electric shock or fire. Do not use the product in places exposed to direct sunlight, dust, salt, or metal powder. Do not use the product in places exposed to water, oil, or drug dripping.

2.Wiring

⚠危险
Please ground the ground terminal reliably. Improper grounding may cause electric shock or fire. Do not220VDriver power supply access380VOtherwise, it may cause damage to the equipment, electric shock or fire. U,V,WThe motor output terminals must not be connected to a three-phase power supply, otherwise it may cause casualties or fire.U,V,WMotor output terminals and driver terminalsU,V,WConnect them one by one, otherwise the motor may overspeed and cause equipment damage and casualties. Please tighten the power supply and motor output terminals, otherwise it may cause fire. Please refer to the wire material selection for wiring, otherwise it may cause fire.

3.operate

⚠注意
Before the mechanical equipment starts to operate, it must be adjusted to the appropriate parameter setting value. If the setting value is not adjusted to the appropriate value, the mechanical equipment may lose control or malfunction. Before starting operation, please confirm whether the emergency switch can be activated at any time to stop the machine. Please test whether the servo motor operates normally without load first, and then connect the load to avoid unnecessary losses. Do not frequently turn the power on and off, otherwise it may cause overheating inside the driver.

4.run

III

禁止
When the motor is running, do not touch any rotating parts, otherwise it will cause casualties. When the equipment is running, do not touch the driver and motor, otherwise it will cause electric shock or burns. When the equipment is running, do not move the connecting cables, otherwise it will cause personal injury or equipment damage.

5.Maintenance and Inspection

禁止
Do not touch the inside of the driver and its motor, otherwise it may cause electric shock. When the power is on, it is forbidden to disassemble the drive panel, otherwise it may cause electric shock.5Do not touch the wiring terminals within 10 minutes, otherwise the residual high voltage may cause electric shock. Do not change the wiring when the power is on, otherwise it may cause electric shock. Do not disassemble the servo motor, otherwise it may cause electric shock.

6.Scope of use

注意
The products involved in this manual are for general industrial use. Do not use them in devices that may directly endanger personal safety, such as nuclear power devices, aerospace equipment, life support and maintenance equipment, and various safety equipment. If you need to use the above, please contact the manufacturer's customer service.

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No.1Chapter Product Inspection and Installation

1.1Product Inspection

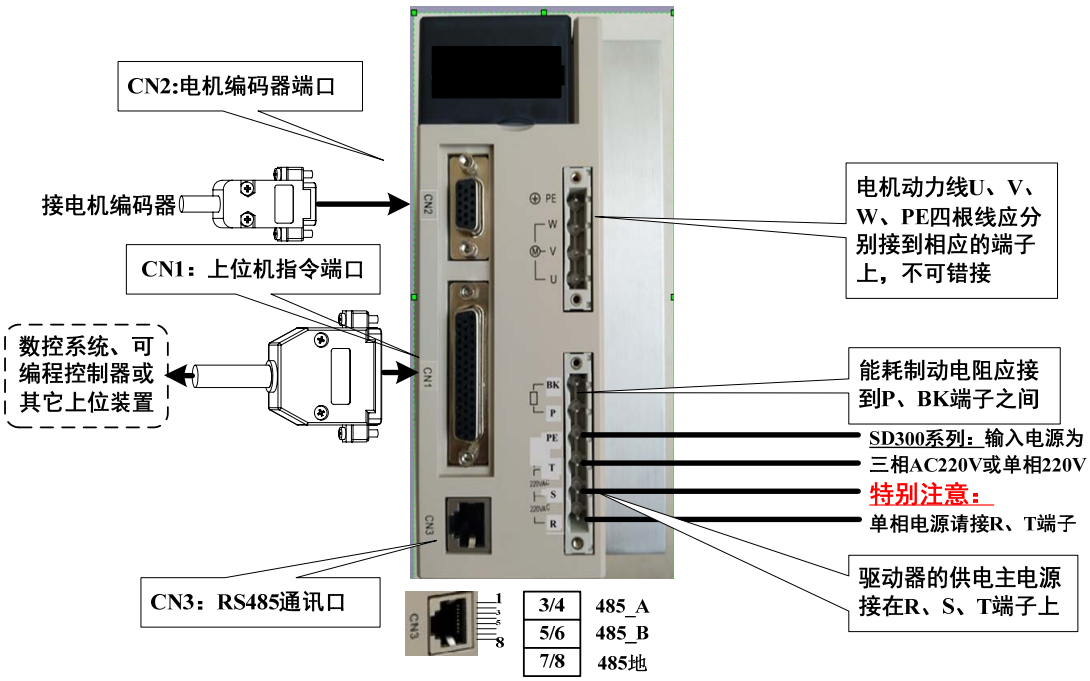
This product has been fully functionally tested before leaving the factory. To prevent product malfunctions due to negligence during delivery, please check the following items in detail after unpacking:

Check whether the servo drive and servo motor models are the same as the ordered models. Check whether the servo drive and servo motor have any damage or scratches. If they are damaged during transportation, do not connect the wires and supply power.

Check whether there are any loose components on the servo drive and servo motor, whether there are any loose screws, whether the screws are not tightened or have fallen off.

Check whether the servo motor rotor shaft can be rotated smoothly by hand. Motors with brakes cannot rotate directly. If any of the above items fails or is abnormal, please contact your dealer immediately.

1.2Product front panel



picture1.1Front panel diagram

1.3Servo drive installation

1.3.1Installation environment conditions

The installation environment of the servo drive has a direct impact on the normal function of the drive and its service life. Therefore, the installation environment of the drive must meet the following conditions:

Operating temperature:0℃～40℃; Working environment humidity:40%~80%Storage environment temperature: -40℃～50℃; Storage environment humidity:93%Below (no condensation). Vibration:0.5G the following.

Prevent from rain drops or humid environment.

Avoid direct sunlight.

Prevent oil mist and salt corrosion.

Prevent erosion by corrosive liquids and gases.

Prevent the invasion of dust, lint and metal chips.

Keep away from radioactive substances and flammable materials.

When installing several drivers in a control cabinet, please make sure that there is enough space for them to facilitate air flow and heat dissipation. Please add a cooling fan to reduce the temperature around the servo driver. The long-term safe operating temperature is40Below ℃.

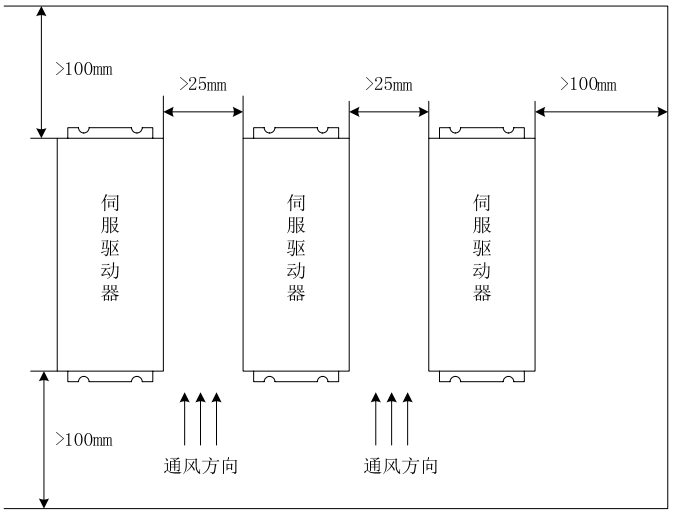
If there is a vibration source nearby (such as a punch press), please use a vibration absorber or install an anti-vibration rubber pad if it cannot be avoided.

When there are interfering devices nearby, there will be interference on the power line and control line of the servo drive, which may cause the drive to malfunction. Noise filters and other anti-interference measures can be added to ensure the normal operation of the drive. However, noise filters will increase leakage current, so an isolation transformer must be installed at the power input end of the drive.

1.3.2Installation Method

The normal installation direction of the servo drive is vertical, with the top facing upward to facilitate heat dissipation. When installing, tighten the rearM5Fix the screws. The installation distances between the servo drives and other equipment are shown in the figure on the next page. To ensure the performance and life of the drive, please leave sufficient installation distances as much as possible. A cooling fan must be installed in the electrical control cabinet to ensure that there is vertical wind to dissipate the heat of the servo drive radiator.

When installing the electrical control cabinet, prevent dust or iron filings from entering the servo drive.



picture1.2Servo drive installation diagram

1.4Servo motor installation

1.4.1Installation environment conditions

Operating temperature:0~40℃; Working environment humidity:80%Storage environment temperature: -40℃～50℃; Storage environment humidity:80%Below (no condensation). Vibration: 0.5Gthe following.

A well-ventilated place with little moisture and dust.

Environment without corrosive, flammable gas, oil gas, cutting fluid, cutting powder, iron powder, etc. Place without water vapor and direct sunlight.

1.4.2Installation Method

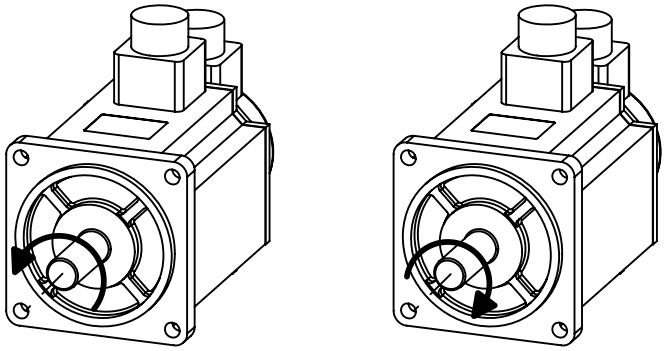
Horizontal installation: To prevent liquids such as water and oil from flowing into the motor from the motor outlet, please place the cable outlet at the bottom.

Vertical installation: If the motor shaft is installed upward and a reducer is attached, care must be taken to prevent the oil stains in the reducer from seeping into the motor through the motor shaft.

The motor shaft must be extended sufficiently. If the extension is insufficient, the motor will easily vibrate when it moves. When installing or removing the motor, do not hit the motor with a hammer, otherwise it will easily damage the motor shaft and encoder.

1.5Definition of motor rotation direction

The motor rotation direction described in this manual is defined as: facing the motor shaft extension, the rotating shaft rotates counterclockwise (CCW) For positive rotation, the shaft rotates clockwise (CW)For reverse.



picture1.3Servo motor rotation direction definition

No.2Chapter wiring

2.1System composition and wiring

2.1.1Wiring Instructions

Wiring precautions:

Use wiring materials according to the wire specifications.

Cable length, command cable3mWithin, encoder cable20mWithin.

examineR,S,TCheck whether the power supply and wiring are correct.**220VDo not connect to the servo380VPower on**. Motor outputU,V,WThe terminal phase sequence must correspond to the corresponding terminals of the driver. If the wrong connection is made, the motor may not rotate or may even run away. The motor cannot be reversed by swapping the three-phase terminals, which is completely different from an asynchronous motor.

The grounding must be reliable and single-point grounding.

The direction of the absorbing diode installed in the relay that outputs the signal must be connected correctly, otherwise it will cause a malfunction and fail to output the signal.

To prevent erroneous operation caused by noise, please add an isolation transformer and noise filter to the power supply.

Please keep the power line (power line, motor line and other strong current circuit) away from the signal line.30cmWhen wiring, do not place them in the same wiring duct.

Please install a non-fuse circuit breaker to cut off the external power supply in time when the drive fails.

2.1.2Wire specifications

Connection Terminals	symbol	Wire specifications
Main circuit power supply	R,S,T	1.5~2.5mm2
Motor connection terminals	U,V,W	1.5~4mm2
Ground terminal	PE / ⏚	1.5~4mm2
Control signal terminal	CN1	⩾0.14mm2(AWG26),With shielded wire
Encoder signal terminal	CN2	⩾0.14mm2(AWG26),With shielded wire
Braking resistor terminals	P,BK	1.5~2.5mm2

The encoder cable must use twisted pair cables. If the encoder cable is too long (>20m), which will cause insufficient power supply to the encoder. Its power and ground wires can be connected by multiple wires or using thick wires.

2.1.3Description of strong current terminal

name	Terminal Symbols	Detailed description
Main circuit power supply	R,S,T	To connect external AC power: Three-phase220VAC -15%～+10% 50/60Hz
Braking resistor Terminals	P,BK	When an external braking resistor is connected, Braking resistors in parallel Please pay attention to the selection of resistance and power to avoid burning Bad drive.
Motor connection terminals	U	Output to motorUPhase power supply
	V	Output to motorVPhase power supply
	W	Output to motorWPhase power supply
Ground terminal	PE /⏚	Motor housing grounding terminal
	PE /⏚	Driver ground terminal

2.2CN1Control signal terminal

CN1The control signal terminal provides the signal required to connect to the upper controller. Different drive models use different socket types. The signals include:

6Programmable inputs,4Programmable output; Pulse

command input (position command, speed command);

Analog command input (speed command, torque command);

Encoder signal output.

2.2.1 CN1Terminal signal description

Special Note:DI1~DI6It is a programmable input port, which is controlled by parametersPr100~Pr105Configure its functionality;DO1~DO4 It is a programmable output port, which is controlled by parametersPr108~Pr111Configure its functions; parameter configuration is detailed in5chapter 5.4.2 For detailed function description, please refer to5chapter5.5~5.6The following table shows the factory default configuration.

Terminal No.	Signal name	mark	Function
	Power supply at input terminal positive electrode	COM+	The positive power supply of the input terminal is used to drive the input terminal The optocoupler of the subDC12~24V, current ⩾ 100mA.
	DI1 Servo Enable	SON	Servo enable input terminal. SON ON: Allow the drive to work; SON OFF：Driver enable is off, the motor is in Free state. Note1：When fromOFFHitONThe motor must be stationary before starting. Note2: HitSON ONAfter that, at least wait 50ms, and then enter the command.
	DI2 Alarm clear	ACLR	Alarm clear input terminal. ARST ON: Clear system alarm; ARST OFF: Keep the system alarm. Note1: Some hardware fault alarm numbers cannot be used Method to clear, need to shut down the power for maintenance, and then turn on again electricity.
	DI3 CCWDrive Disable	CCWL	CCW(Counterclockwise) Drive inhibit input terminal. CCWL ON:CCWThe drive allows, the motor can rotate counterclockwise; CCWL OFF:CCWDrive prohibited, the motor is prohibited from rotating counterclockwise. Note1：Used for mechanical overrun, when switchOFFhour, CCWThe directional torque is maintained at0. Note2： You can set the parametersPr097Shield this function, users can use it without connecting this terminal. CCWDriver allowed.
	DI4 CWDrive Disable	CWL	CW(Clockwise) Drive inhibit input terminal. CWL ON:CWIf the drive allows, the motor can rotate clockwise; CWL OFF:CWDrive is prohibited and the motor is prohibited from rotating in clockwise direction. Note1：Used for mechanical overrun, when switchOFFhour, CW The directional torque is maintained at0. Note2： You can set the parametersPr097Shield this function, users can use it without connecting this terminal.CW Driver allowed.
	DI5 Deviation counter cleared	CLE	In position control mode (parameterPr004=0), position deviation counter clear input terminal. CLE ON: During position control, position deviation counter Cleared.

	DI6 Command pulse prohibition	INH	In position control mode (parameterPr004=0),Bit Set the command pulse prohibit input terminal. INH ON: Command pulse input is prohibited; INH OFF: Command pulse input is valid.

Terminal No.	Signal name	mark	Function
	DO1 Servo ready output	RDY	Servo ready output terminals. SRDY ON：Main power supply is normal, the driver has no alarm, and the servo is ready for outputON(output is on); SRDY OFF: The main power is not on or the driver has an alarm, the servo is ready to outputOFF(Output cutoff).
	DO2 Servo alarm output	ALM	Servo alarm output terminal. ALM ON: Servo drive no alarm, servo alarm OutputON(output is on); ALM OFF: Servo drive has an alarm, servo alarm OutputOFF(Output cutoff).
	DO3 Positioning completed output; (Position control method Down)	COIN	Positioning completed output terminals. COIN ON：When the position deviation counter value is set When the positioning range is within the range, the positioning is completed and outputON(Output conduction), otherwise the outputOFF(Output cutoff).
	DO4 Mechanical brake release	BRK	When the motor has a mechanical brake (de-energized retainer), This port can be used to control the brake. BRK ON: The brake is energized, the brake is invalid, and the motor can To run; BRK OFF: Brake cut-off, braking effective, motor Locked and cannot run. Note:BRKFunctions are controlled internally by the driver.
	Command pulsePLUS enter	PULS+	External command pulse input terminal. Note1: By parametersPr035Set the pulse input mode, Pr035=0, command pulse + symbol mode (default state state); Pr035=1,CCW/CWCommand pulse mode; Pr035=2, orthogonal pulse mode.
		PULS-	
	Command pulseSIGN enter	SIGN+	
		SIGN-	

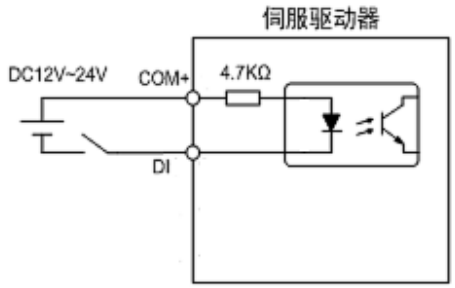
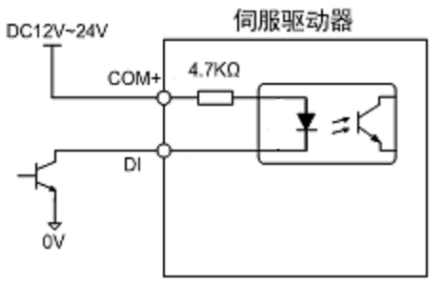
Terminal No.	Signal name	mark	Function
	Analog speed command enter	AS+	External analog command input terminal, differential mode, input impedance 10kΩ, input range -10V～＋10V.
		AS-	
	Analog Ground	GNDA	Ground for analog input.
	EncoderAbelieve Number	OA+	EncoderABZSignal differential drive output (26LS31Output, equivalent toRS422); Non-isolated output (non-insulated).
		OA-	
	EncoderBbelieve Number	OB+	
		OB-	
	EncoderZbelieve Number	OZ+	
		OZ-	
	EncoderZPhoto Gallery Electrode open circuit output	CZ	EncoderZPhase signal is output by open collector, encoderZWhen the phase signal appears, the outputON(output is on), otherwise the output OFF(Output cutoff); Non-isolated output (non-insulated); On the host computer, usuallyZThe phase signal pulse is very narrow, so please use a high-speed photocoupler to receive it.
	Encoder common ground Wire	GND	Encoder common ground.
	Shielded ground wire	FG	Shield ground terminal.

2.2.2 CN1Terminal interface type

The following will introduceCN1Each interface circuit and the wiring method with the upper control device.

1.Digital input interface (C1)

The digital input interface circuit can be controlled by switches, relays, open collector transistors, photocouplers, etc. The relays should be low current relays to avoid poor contact. External voltage range DC12V~24V.

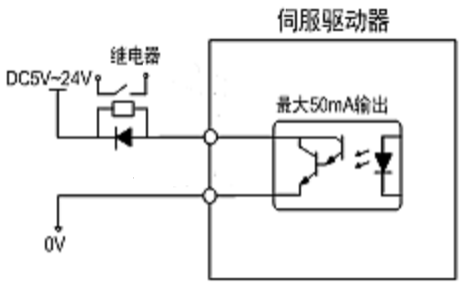
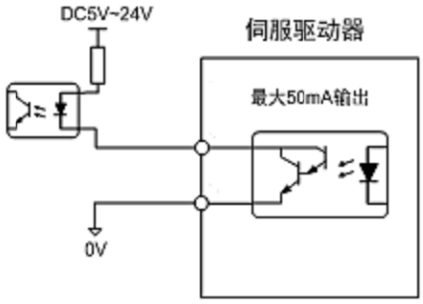
C1-1:Switch input	C1-2:Open collector transistor
	

2.Digital output interface (C2)

The output circuit uses Darlington photocoupler, which can be connected with relays and photocouplers. Note:

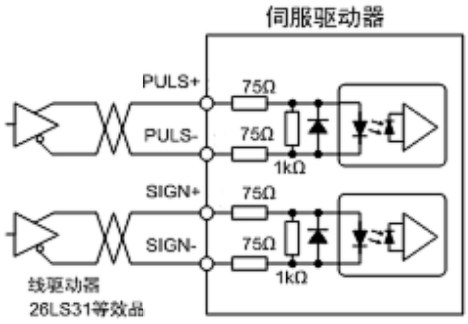
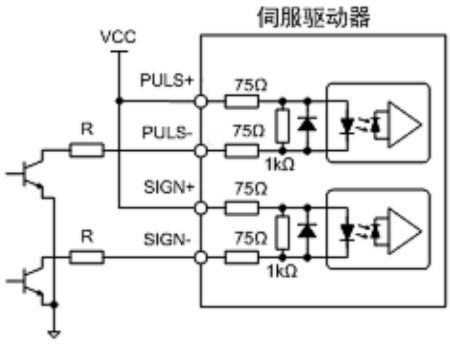
The power supply is provided by the user. If the power supply is connected in reverse, the drive will be damaged.25V, output maximum current50mA,3The total current of the circuit shall not exceed100mAWhen using an inductive load such as a relay, a diode must be added in parallel with the inductive load. If the polarity of the diode is opposite, the driver will be damaged.

When conducting, about1VThe left and right pressure drops cannot meetTTLLow level requirement, so it cannot be used withTTLThe circuits are directly connected.

C2-1:Relay Device	C2-2:Photocoupler combine Device
	
A freewheeling diode must be added.	

3.Position pulse command interface (C3)

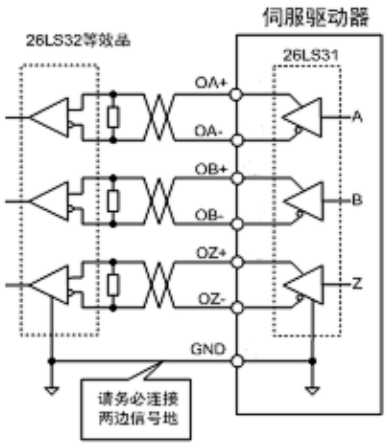
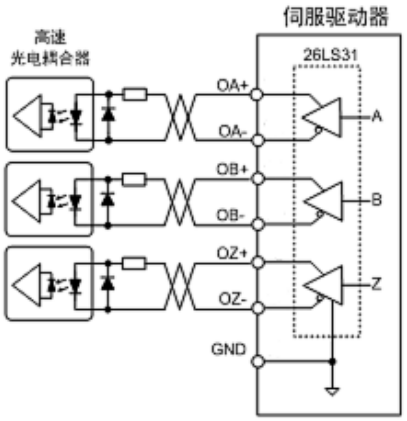
There are two connection methods: differential drive and single-ended drive. Differential drive connection is recommended. Twisted pair wires are recommended for wiring. Drive current8~15mA, by the parameterP035Set the working mode: pulse + sign, forward/reverse pulse, orthogonal pulse.

C3-1:difference drive move	C3-2:Single-ended drive move
	
Maximum pulse frequency500kHz(kpps); It is not suitable to be disturbed, so this connection method is recommended.	Maximum pulse frequency200kHz(kpps); Recommended resistorRResistance:

VCC	R
5V	82Ω~120Ω
12V	510Ω~820Ω
24V	1.5kΩ~2kΩ(Inside the drive Part has been connected Resistance

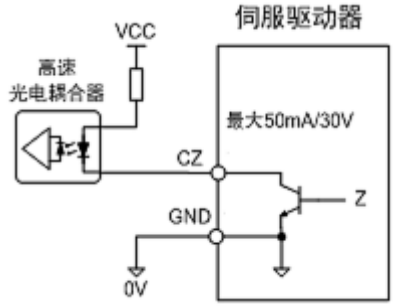
4.Encoder signal line driver output (C5)

The encoder signal is divided and then driven by the line (Line Driver)Output to the host controller.

C5-1:Long-line receiver reception	C5-2:Optocoupler receiving
	
Use of host controllerAM26LS32Equivalence If the product is used as a receiver, a terminal resistor must be connected. 220Ω~470Ω; Driver encoder signal ground (GND)must Connect to the signal ground of the host controller.	The host controller uses a high-speed photocoupler (for example like6N137), current limiting resistor value220Ωabout.

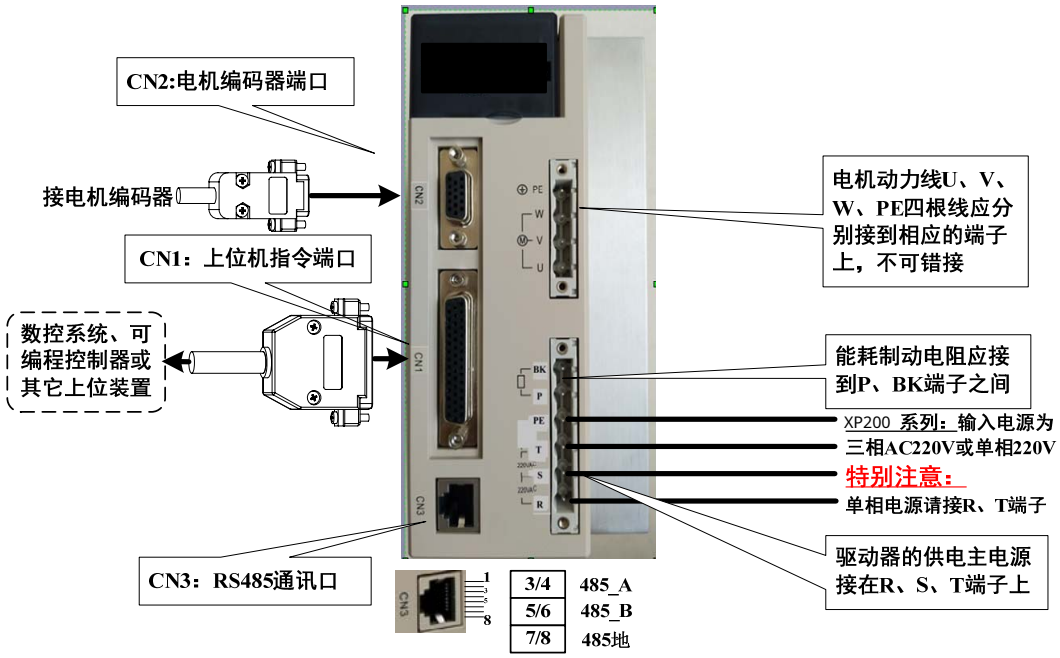
5. EncoderZSignal open collector output (C6)

The encoderZThe signal is output to the upper controller through the open collector.ZThe signal pulse width is narrow, please use a high-speed photocoupler to receive

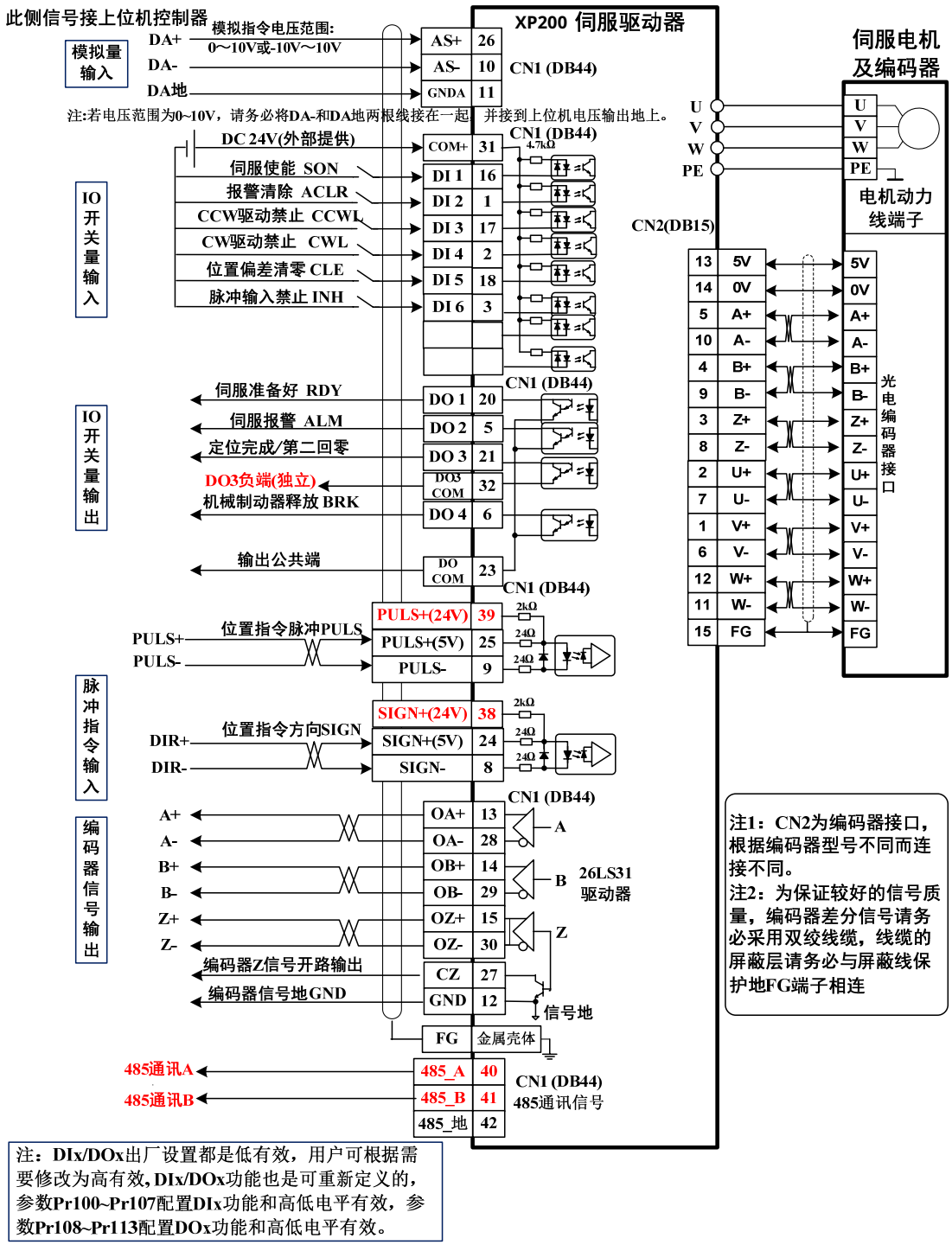
C6: EncoderZSignal open collector output

External power supply maximum30V, output maximum current50mA.

2.3Front panel terminal wiring diagram

XP200 接线示意图



2.4Control PortCN1and encoder portCN2Definition Graph



Special attention:

If the pulse command signal is24Vpower supply ,butPULS+Must receiveCN1of39foot,SIGN+Must receive CN1of38foot, otherwise the pulse command port may be damaged;

If the pulse command signal is5Vpower supply ,butPULS+Must receiveCN1of25foot,SIGN+Must receive CN1oftwenty fourOtherwise, the pulse reception may be abnormal.

2.5Braking resistor specifications

XP200series(220VServo) brake resistor configuration table

(standard220VThe servo has a built-in resistor, and an external brake resistor can be selected according to the load conditions)

Driver Model	**(Start and stop infrequently) Recommended configuration** (Only use built-in resistors)	**(Frequent start and stop) Recommended configuration** (Connect with external resistor for use or just use external resistors)
XP200-20AL	47Europe (built-in)	⩾47Ohm, power greater than500W
XP200-30AL	47Europe (built-in)	⩾47Ohm, power greater than500W
XP200-40AL	47Europe (built-in)	⩾47Ohm, power greater than500W
XP200-50AL	47Europe (built-in)	⩾47Ohm, power greater than500W
XP200-75AL	47Europe (built-in)	⩾20Ohm, power greater than1000W

No.3Chapter Panel Operation

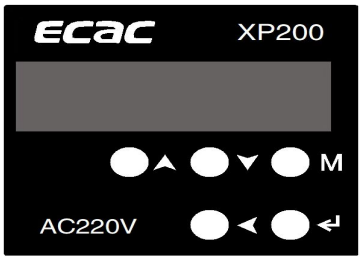
3.1Driver Panel Description

3.1.1Panel composition

XP200Series Panels by5indivualledDigital tube display,5The buttons are used to ▲, ▼, ◄, M, ↵ display various system status setting parameters, etc.

All operations are hierarchical, and are expanded layer by layer from the main menu. The operation panel is shown in the figure below:

XP200**Operation panel**

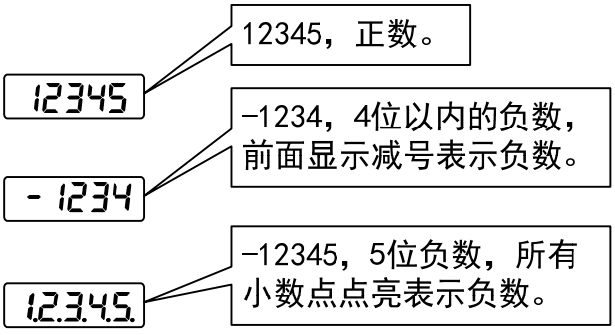


3.1.2Panel Description

symbol	name	Function
POW	Power status indicator	XP200The second digital tube from the left lights up to indicate the servo Undervoltage
RUN	Operation status indication	XP200The first digital tube from the left lights up to indicate the servo Enable
▲	Add Key	Increase the sequence number or value; long press has a repeat effect.
▼	Decrease key	Decrease the sequence number or value; long press has a repeat effect.
M	Escape key	Menu exited; operation canceled.
↵	Confirm keyEnter	Menu entry; parameter modification confirmation or operation confirmation.
◄	Shift key	Used to quickly modify parameter numbers or parameter values.

3.1.3Numerical display

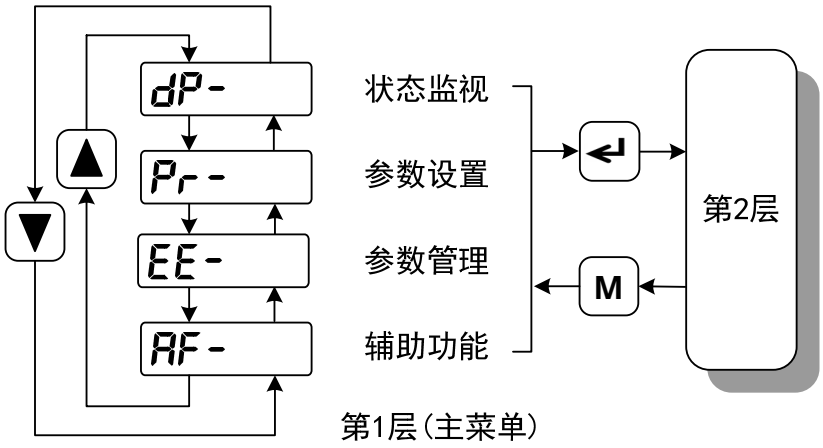
The numerical value is adopted5A digital tube display,A minus sign in front of a value indicates a negative number.5Negative numbers, If all decimal points light up, it means a negative number. Some display items have prefix characters before them. If the number of digits is too long and needs to occupy the position of the prefix characters, the prefix characters will not be displayed and only the value will be displayed. The specific display meanings are as follows:



3.2Main Menu

No.1The main menu layer has4There are several operation modes. Use the keys to change the mode. Press Key to enter

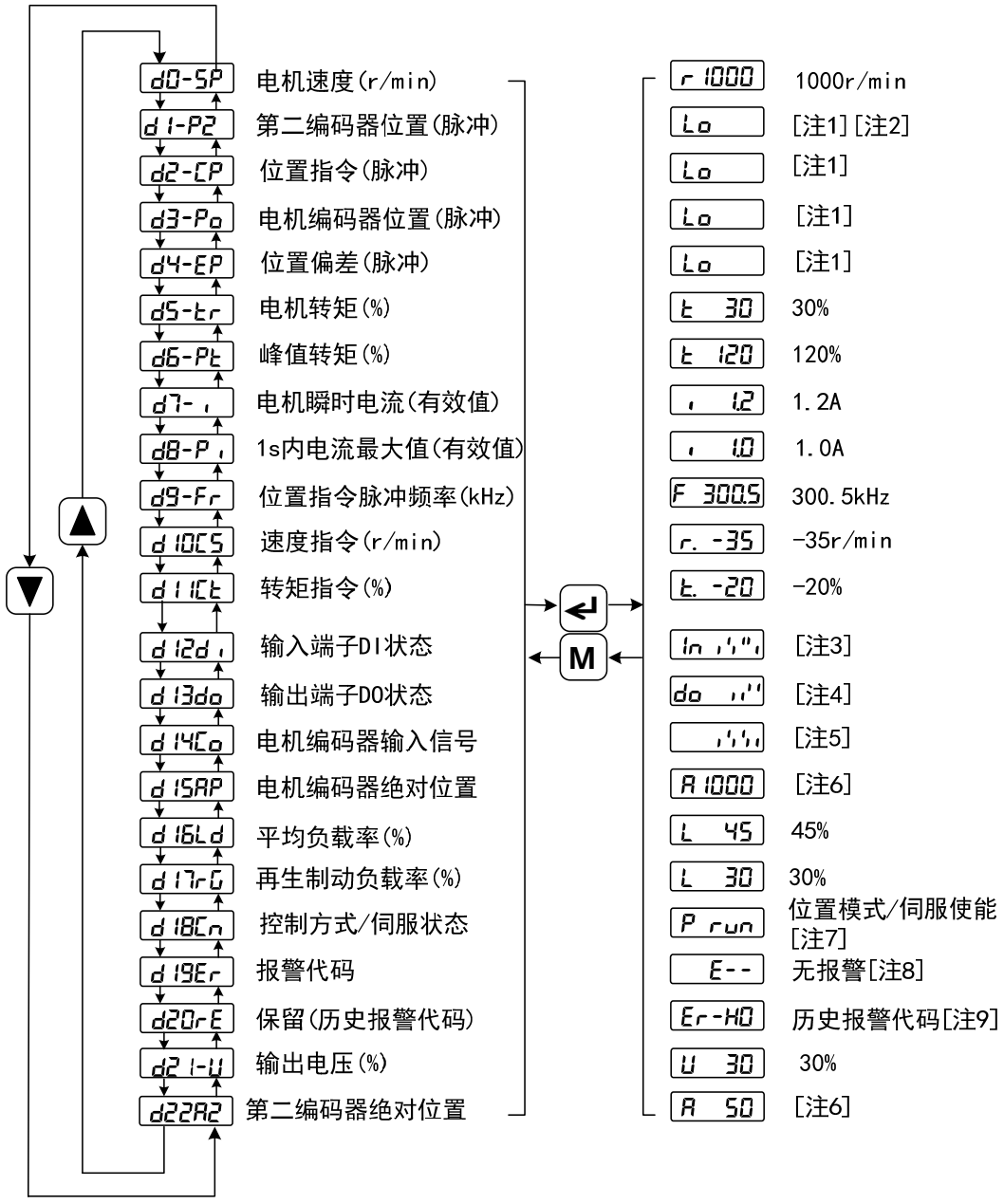
The second layer performs specific operations, and the buttons are2Return to the main menu.



3.3Condition Monitoring

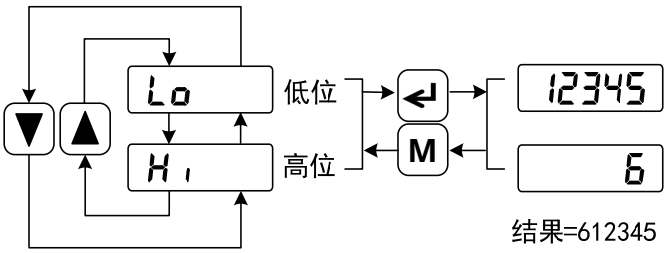
Select Status Monitor from the main menu.dP-",according to  Key to enter monitoring mode. There are multiple monitoring Item, the user uses the  keys to select the desired display item, and then press  key to enter the specific display status

The specific meanings of the status monitoring display items are as follows:



1. 32Bit binary value display [Note1]

32The binary number range is -2147483648~2147483647, expressed using a combination of low and high bits, select the low and high bits through the menu, and use the formula in the figure to synthesize the complete value.



32位数值=高位数值×100000+低位数值

2.Pulse unit2]

The pulse of the original position command refers to the number of input pulses without electronic gear conversion. The pulse unit of other items is the encoder pulse unit.2500Take the line encoder as an example:

Encoder pulse unit = encoder resolution

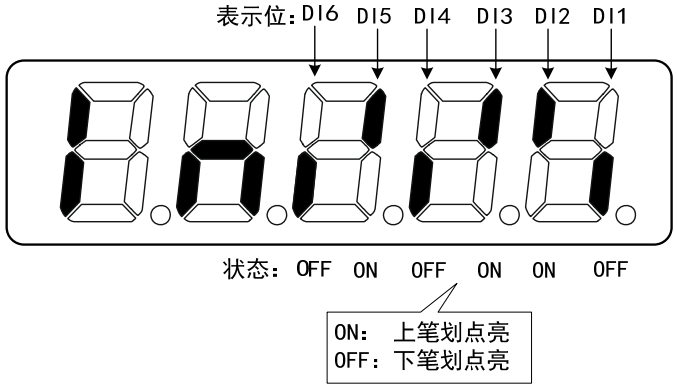
=4×Encoder line number

=4×2500(*pulse/rev*)

=10000(*pulse/rev*)

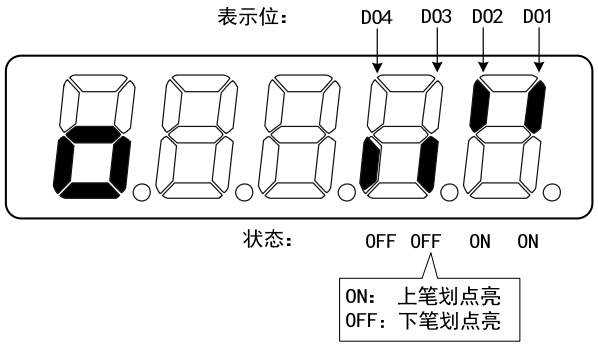
3.Input TerminalsDI[Note3]

The vertical line of the digital tube indicates the state of a bit, and the stroke on the vertical line indicatesON, the next stroke is highlightedOFF.



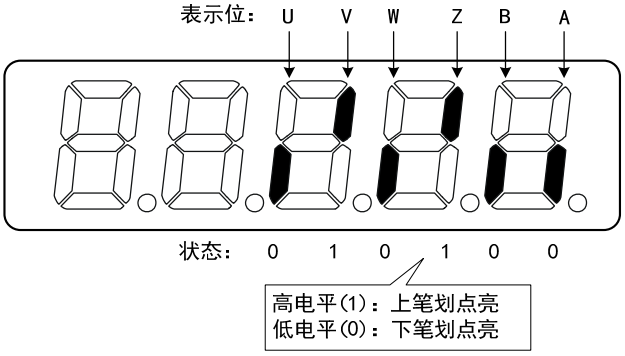
4.Output TerminalsDO[Note4]

The vertical line of the digital tube indicates the state of a bit, and the stroke on the vertical line indicatesON, the next stroke is highlightedOFF.



5.Encoder input signal [Note5]

The vertical line of the digital tube indicates the state of a bit. A lit stroke on the vertical line indicates a high level, and a lit stroke below indicates a low level. (Note: For absolute position encoders, this display is meaningless)



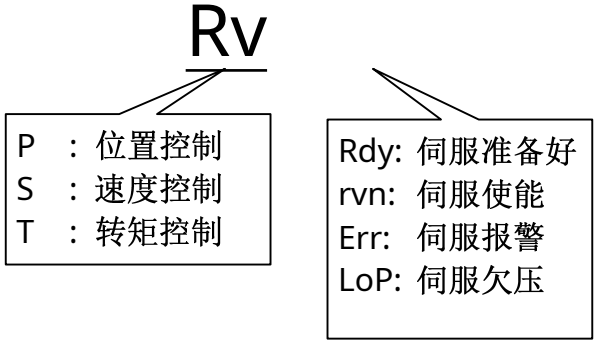
6.Rotor single turn position [Note6]

Indicates the relative position of the rotor in one revolution to the stator. One revolution is one cycle. The minimum resolution of the encoder is the unit.ZThe pulse is the origin.

2500Line encoder: Its range is0~9999（10),ZWhen the pulse occurs, the value is0.

7.Control mode/servo status [Note7]

D18CNThe displayed content represents the current control mode and servo status.XP200When the drive is powered on, it will automatically display the control mode/servo status and keep10seconds (press any key to exit this display mode), then it will automatically jump to display the motor speedR 0The specific meanings of control mode and servo status are as follows:



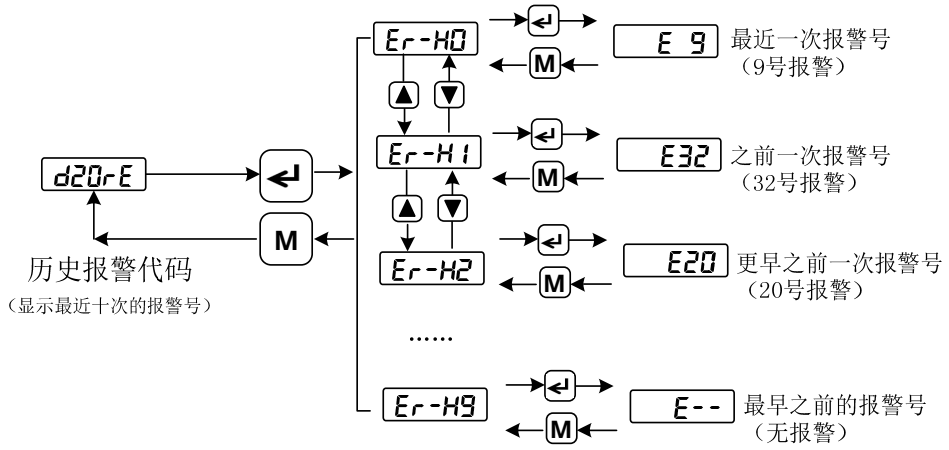
8.Alarm code [Note8]

If there is no alarm, two minus signs are displayed. If there is an alarm, the alarm number is displayed and flashes. When an alarm occurs, the display automatically enters the status monitoring and displays the alarm number, but other operations can be performed through the keyboard. When it is not in the monitoring state, the decimal point of the rightmost digital tube flashes to indicate that an alarm exists. Specific examples are as follows:

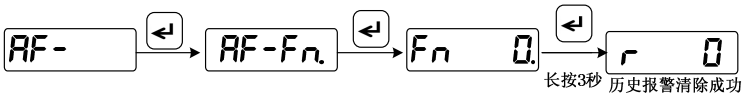


9.Historical alarm code [Note9]

The historical alarm code shows the recent10The alarm content of the time, if there is no alarm, two minus signs are displayed, if there is an alarm, the alarm number is displayed. When an alarm occurs, the driver will automatically update and store the alarm number. The operation of viewing the historical alarm code and the specific display content are as follows:



The method to clear the historical alarm code is as follows: In the absence of an alarm, first set the parametersPr119for 4, and then follow the steps below to clear all historical alarm codes.



3.4Parameter settings

Parameters are represented by parameter segment + parameter number, the hundreds digit is the segment number, and the tens and ones digits are the parameter number.Pr105, the segment number is "1", the parameter number is "05", the digital tube displays "Pr105".

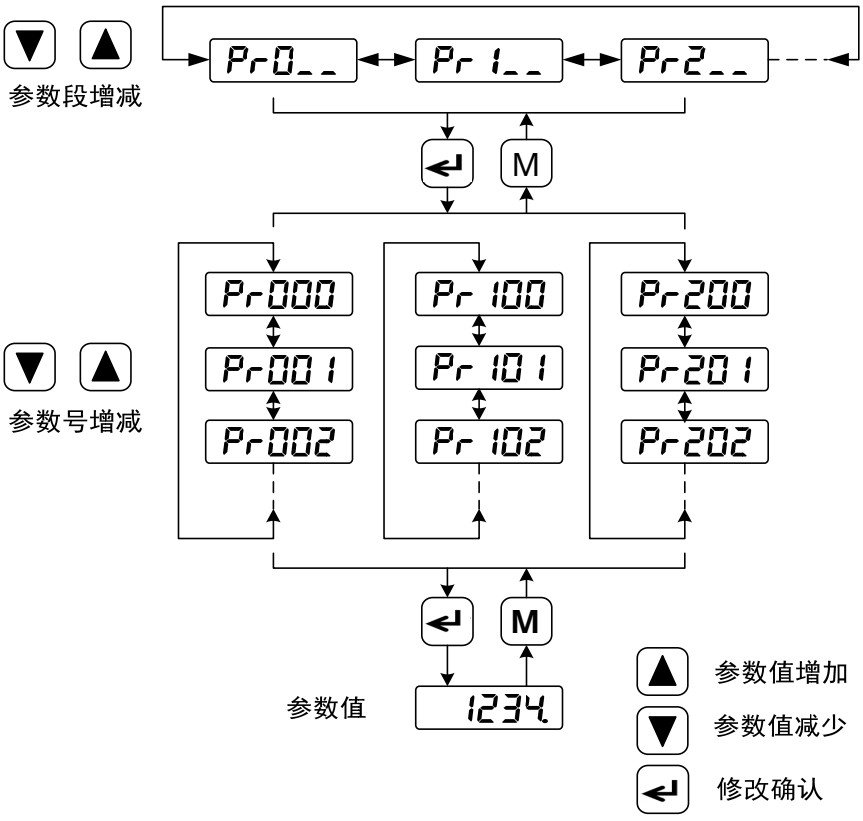
Select "Parameter Settings" from the main menu.Pr-",according to ↵ Enter the parameter setting mode. First use ▲ ▼ Use the keys to select a parameter segment. After selecting it, press ↵ key to enter the parameter number selection of this section. Then use ▲ ▼ Select the parameter number, and then press ↵ key to display the parameter value.

Use ▲ ▼ keys to modify the parameter value. Press the ▲ key once to increase or decrease the parameter.1, press and Hold the ▲ key, the parameter can be increased or decreased continuously. Use the ◄ key to change the parameter position currently being modified. (units, tens, hundreds, thousands or tens of thousands), when a parameter position is in the modified mode, the parameter The corresponding digital tube decimal point is lit. When the parameter value is modified, the digital tube decimal point flashes. ↵

The modified value will be reflected in the control immediately (some parameters need to be saved and then powered on to take effect).

You can continue to modify the parameters. After modification, press the M button to return to the parameter number selection state. If you are not satisfied with the value you are modifying, do not press ↵ Press key to confirm, press key to cancel, and the parameter will return to the original value. M

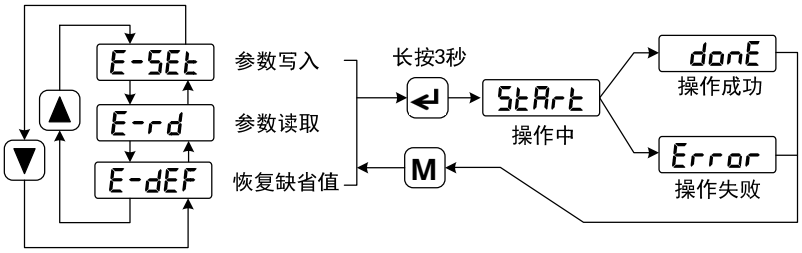
The modified parameters are not saved toEEPROMIf you need to save it permanently, please use the parameter write operation in parameter management.



3.5Parameter Management

Parameter management mainly deals with parameter tables andEEPROMIn the main menu, select Parameter Management "EE-",according to key to enter parameter management mode.

Select the operation mode, there are3Use the , keys to select a mode. After selecting an operation, press key and hold3Press for more than 1 second to activate the operation. After completion, press the key again to return to the operation mode selection state.

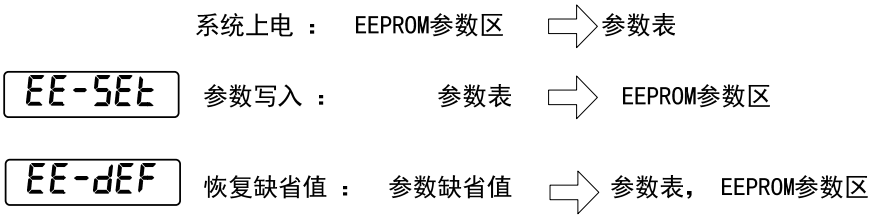


Parameter writing

Indicates writing the parameters in the parameter table intoEEPROMWhen the user modifies the parameter, only the parameter value in the parameter table is changed, and the parameter value will be restored to the original value at the next power-on. If you want to change the parameter value permanently, you need to perform a parameter write operation to write the parameter in the parameter table toEEPROMThe modified parameters will be used after power-on.

Restoring the default values

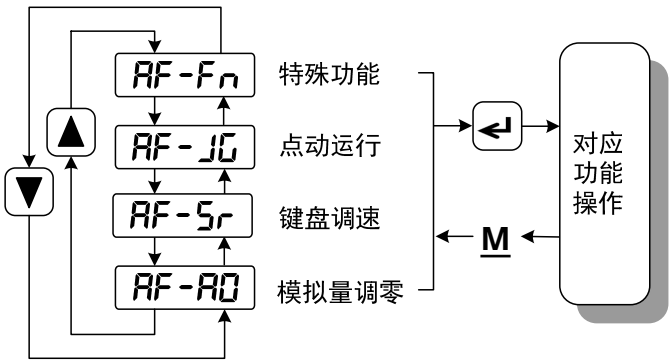
Indicates that the default values (factory values) of all parameters are read into the parameter table and written intoEEPROMThe next time you power on, the default parameters will be used. When the user has messed up the parameters and the machine cannot work properly, this operation can be used to restore all parameters to the factory settings. Because different drive models and motor models have different parameter default values, when restoring the default parameters, you must first ensure that the motor code (parameterPr002)correctness.



3.6Accessibility

Select Accessibility from the main menu.AF- ",according to Use the key to enter the auxiliary function mode.

After selecting an operation mode, press Press the key to enter the corresponding function. M Key to return to Operation mode selection status.



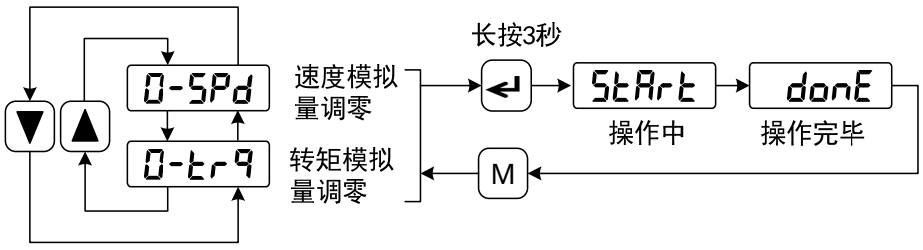
3.6.1Functional Description

3.6.2Analog zeroing

When this operation is used, the drive automatically detects the analog zero offset and writes the zero offset value into the parameter Pr047(or Pr054)This operation has saved the zero bias parameters toEEPROMTherefore, there is no need to perform parameter writing operation.

The operation method is as follows:1）The host computer outputs first0VAnalog quantity, enable the servo drive, and then select Select analog zero adjustment"AF-A0",according to Key to enter; (2) If the drive control mode is analog Speed mode, select speed analog zeroing through the menu.0-SPD”; If the drive control mode is analog torque mode, select the torque analog zero setting through the menu.0-TRQ"; (3) After selecting the operation Press and hold3Press for more than one second to activate the operation. After completion, press the key again to return to the menu selection state.

The specific display is as follows:



3.6.3Jog (JOG)run

Power on the drive first, and after confirming that there are no alarms or any abnormal conditions, enable the servo (SON) ON,RUN The indicator light is on, and the motor is energized and in zero speed state.

In the auxiliary function, select "Jog Run"AF-JG",according to Key to enter jog (JOG)Operator

The jog prompt is "J", the numerical unit isr/min, the speed command is provided by the key. Press and hold the key, the motorJOGSpeed forward (CCW)Run, release the button, the motor stops and keeps zero speed; press and hold the button, the motorJOGSpeed Reversal(CW)Run, release the button, the motor stops and maintains zero speed.

Note:JOGThe speed is determined by the parameterPr076Settings (unit isr/min), the servo enable is controlled by external switch quantity or by parameterPr098Control (set to1The internal driver is forced to enable, set0not enabled).



3.6.4Keyboard speed control

Power on the drive first, and after confirming that there are no alarms or any abnormal conditions, enable the servo (SON) ON,RUN The indicator light is on, and the motor is energized and in zero speed state.

In Accessibility, select Keyboard Speed Adjustment.AF-Sr",according to key to enter the keyboard speed adjustment mode.

The keyboard speed prompt is "r.", the numerical unit isr/min, the speed command is provided by the key. Use the and keys to change the speed command, and the motor runs at the given speed. A positive number indicates forward rotation (CCW), negative numbers indicate reversal (CW), the minimum given speed is0.1r/min.



Note: Keyboard speed control requires setting the control mode to speed control mode, that is, setting the parametersPr004for1(Power off and restart to take effect); The servo enable is controlled by external switch quantity or by parameterPr098Control (set to1The internal driver is forced to enable, set0not enabled).

3.7Restoring parameter defaults

Please use the restore default parameters (factory parameters) function in the following situations:

The parameters are messed up and the system cannot work properly;

Replace the motor, and the new motor is of a different model from the original motor;

Other reasons cause the driver and motor code (parameterPr002)Mismatch. The steps to restore the default parameters are as follows:

First confirm that the drive servo is enabled (SON)InOFFStatus, change password (parameterPr000)for 316, check the motor code (parametersPr002) is correct. If correct, proceed to step3If incorrect, perform steps 2:

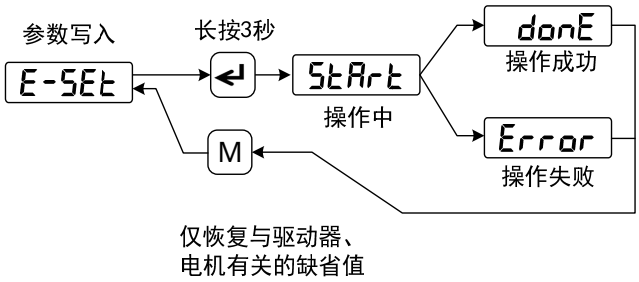
Modify the motor code (parametersPr002) is the required motor series and model. For motor code, refer to7.2 Chapter Motor Adaptation Table;

step3:

Enter parameter management and perform one of the following operations:

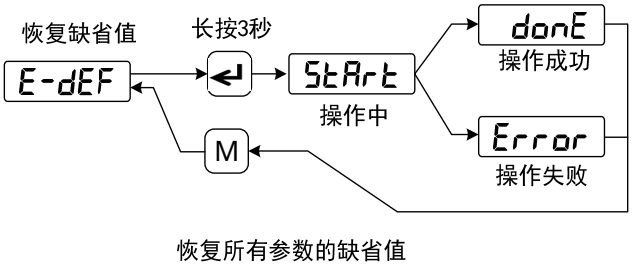
(1)Restore some parameters to default values

Only the default parameters related to the drive and motor are restored, and other user parameters are retained. Execute the parameter write operation in parameter management. This operation is only available when the password is385(or316), and the motor code has been modified. In other cases, there is only the parameter writing function.



(2)Restore all parameters to default values

Restore all parameters to default values. Parameters modified by the user are also restored to factory default values. Execute the restore default value operation in parameter management.



Turn off the power, and then turn it on again to start working.

No.4Chapter Run

4.1No-load test run

The purpose of the trial run is to confirm that the following are correct:

Driver power wiring;

Servo motor power line wiring;

Encoder wiring;

The direction and speed of the servo motor.

Special attention: This series of servo drives adopts an integrated design of control power supply and strong power supply. To use the drive smoothly, please read carefully4.7Section working sequence diagram , the host controller must strictly follow the requirements of this timing diagram.

4.1.1Wiring and inspection

Before powering on, confirm the following:

The motor is unloaded and no load is added to the motor shaft. If it has been installed on the machine, please disconnect the connector; Since the motor has impact when accelerating or decelerating, the motor must be fixed;

Is the connection correct, especially the driverU,V,WWhether with motorU,V,WWiring one-to-one, driverR,S,TIs the wiring correct?

Whether the input voltage meets the requirements indicated on the nameplate;

Is the encoder cable connected correctly?

If the servo drive is a standard application, please refer directly to8Chapter can be quickly debugged and applied.

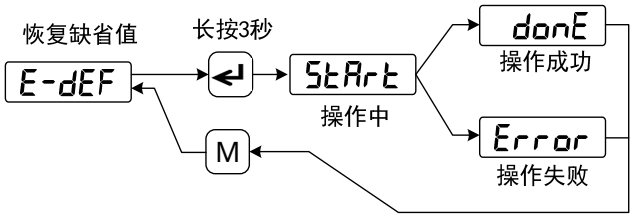
4.1.2Setting the motor code

Connect the main circuit power supply, the display panel of the driver lights up, if there is an alarm, please check the connection. Set the motor code parameters according to the following steps:

Modify the operation password (parameterPr000)for316;

2, Set the motor code according to the motor model (parameterPr002), motor model code see7.2 Energy-saving motor adaptation table;

3, enter parameter management, and perform the restore default value operation as shown below:



恢复所有参数的缺省值

Special attention Restoring the default value must be done on the drive**Disable**If theError. 4, turn off the power after the display operation is successful, wait for the driver panel display to go out, then power on again, and you can perform the trial operation. The trial operation method is as follows:

Keyboard speed adjustment test run (Fn -20); JOGJog test run;

4.1.3Keyboard speed adjustment test run (Fn -20）

1.power ups

Connect the main circuit power supply, the display panel of the driver lights up, if an alarm occurs, please check the connection.

2.operate

After confirming that there are no alarms or any abnormal conditions, perform the following operations as shown below:



The keyboard speed prompt is "r.", the numerical unit isr/min, the speed command is provided by the key. Use the and keys to change the speed command, and the motor runs at the given speed. A positive number indicates forward rotation (CCW), negative numbers indicate reversal (CW), the minimum given speed is0.1r/min.



After the keyboard speed regulation trial run is completed without any abnormality, the mechanical load can be connected and the next step of debugging can be carried out under the control instructions of the host controller.

4.1.4JOGJog test run

1.power ups

Connect the main circuit power supply, the display panel of the driver lights up, if an alarm occurs, please check the connection.

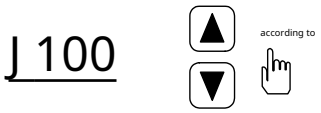
2.operate

Connect the main circuit power supply, the display panel of the driver lights up, if there is an alarm, please check the connection. After confirming that there is no alarm and any abnormal situation,Pr098Set as1, then the servo is enabled (SON)ON, at this time the motor is excited and is in zero speed state.

In the auxiliary function, select "Jog Run"AF-JG",according to  Key to enter jog (JOG)Operator

The jog prompt is "J", the numerical unit isr/min, speed command is provided by key:

Press and hold the key, the motor pressesJOGSpeed forward (CCW)Run, release the button, the motor stops and keeps zero speed; press and hold the button, the motorJOGSpeed Reversal(CW)Run, release the button, the motor stops and maintains zero speed.



JOGThe speed is determined by the parameterPr076The default speed is100r/min.**If the motor runs normally, you can proceed to the next step (joint debugging with the host controller).**

4.2Position Control

Position control is used in systems that require precise positioning, such as**CNC machine tools, textile machinery**The position command source is the pulse command, which is determined by the input terminalPULS+,PULS-andSIGN+,SIGN-Input pulse. The user needs to correctly connect the wires according to the instructions of the position control wiring diagram and the actual use requirements, and then set the position control related parameters. This manual gives a simple example of position control as follows:

4.2.1Position control wiring diagram



The parameter settings for this example are as follows:(Other parameters use factory default values)

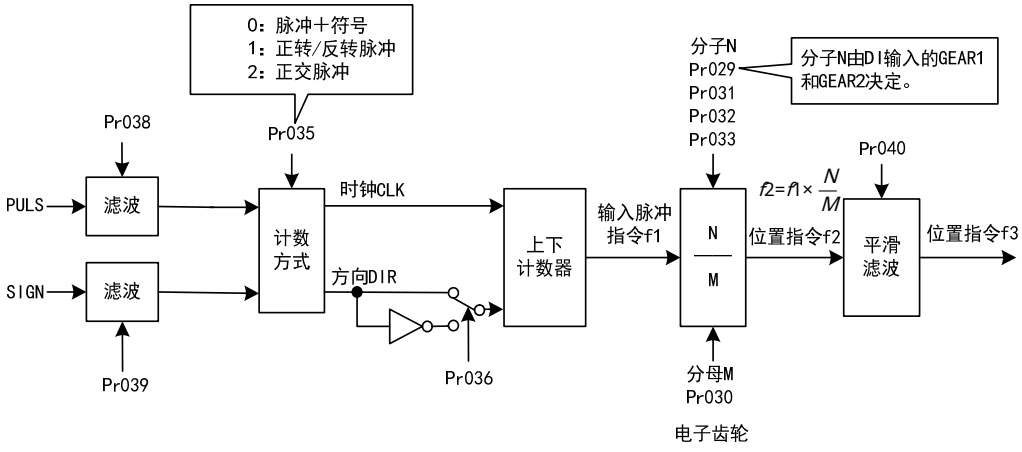
parameter	name	Setting Value	Default value	Parameter Description
Pr004	Control method	0	0	Set to position control mode
Pr097	Ignore driver disable	0 (use Driver ban end)	3 (neglect Driver ban end)	Set as0, use forward drive prohibition (CCWL)and reverse drive inhibit (CWL). If set to3Ignore the positive and negative limits. No connectionCCWL,CWL, users need to correctly set this parameter according to actual usage. Number, see4.6Festival.

4.2.2Position instructions

1.Parameters related to position instructions

parameter	name	Parameter range	Default value	unit	Applicable
Pr029	Position command pulse electronic gear 1st numerator	1~32767	1		P
Pr030	Position command pulse electronic gear denominator	1~32767	1		P
Pr035	Position command pulse input mode	0~2	0		P
Pr036	Position command pulse input direction	0~1	0		P
Pr038	Position command pulse signal filter coefficient	0~3	0		P
Pr039	Position command direction signal filter coefficient	0~3	0		P
Pr040	Exponential smoothing filter time for position command	0~10000	0	0.1ms	P

2.Command pulse transmission path



3.Command pulse input mode

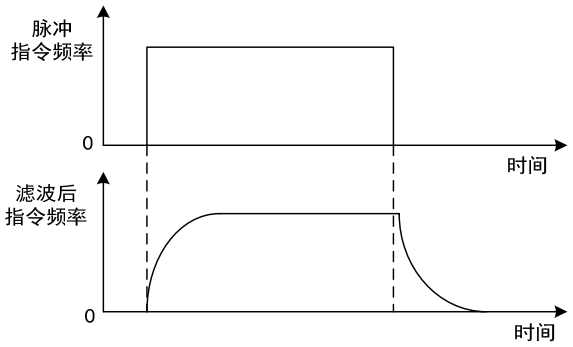
The input method is determined by the parametersPr035decision; parameterPr036Used to change the counting direction, set to1The count value is inverted.

脉冲指令形式	正转(CCW)	反转(CW)	参数Pr035
脉冲+方向	PULS SIGN		0
正转/反转脉冲	PULS SIGN		1
正交脉冲	PULS SIGN		2

Note: The arrow indicates the counting edge, andPr036=0hour

4.Smoothing Filter

As shown in the figure below, the parametersPr040It is to smooth the command pulse and has exponential acceleration and deceleration. The filter will not lose the input pulse, but there will be command delay.0The filter does not work. The parameter value is represented by0The frequency rises to63.2%The time of the position command frequency.



The filter smoothes the input pulse frequency. This filter is used in situations where the host controller has no acceleration and deceleration function, the electronic gear is relatively large, and the command frequency is low.

4.2.3Input electronic gear

(1)Basic description of electronic gear ratio

The electronic gear can be used to define the unit pulse command input to the device to move the transmission device to any distance. The pulse command generated by the host controller does not need to consider the gear ratio, reduction ratio or motor encoder line number of the transmission system. The following table is an electronic gear variable description:

variable	Variable Description	Value of this device
C	Encoder lines/bits	2500Wire
P_t	By default (electronic gear ratio is1:1） The number of pulses required for one encoder rotation (pulse/rev)	=4×*C* =4×2500 =10000(pulse/rev)
R	Reduction ratio	R=B/A,in *A*: Number of motor rotations; *B*: Load shaft rotation number
ΔP	Movement amount of one command pulse	
P_c	Number of command pulses per load shaft rotation	
Pitch	Ball screw pitch(mm)	
D	Roller diameter(mm)	

Calculation formula:

$$\text{Electronic gear ratio } (\frac{N}{M}) = \frac{\text{Encoder one-turn resolution } (P_t)}{\text{Number of command pulses per load shaft rotation } (P_c) \times \text{Reduction ratio}(R)}$$

in,

$$\text{Number of command pulses per load shaft rotation } (P_c) = \frac{\text{Amount of movement per load shaft rotation}}{\text{A command pulse movement amount } (\Delta P)}$$

Reduce the above calculation result and make the numerator and denominator less than or equal to32767The integer value is written into the parameter.

(2)Electronic gear ratio dynamic switching function

Drivers provided4Electronic gear moleculeN, can be changed online byDIInputGEAR1,GEAR2Determine the denominatorM(parameterPr030)All the same.

parameter**Pr034**It can be used to set the dynamic electronic gear application mode. The specific instructions are as follows:

Pr034Set as0: Dynamic electronic gear ratio switching is prohibited. The command pulse electronic gear ratio numerator is composed ofPr029set up; **Pr034Set as1:**Dynamic electronic gear ratio switching is allowed, and the command pulse electronic gear numerator is composed ofDIInputGEAR1 Decide;

DISignal[Note] GEAR1	Command pulse electronic gear numeratorN
0	No.1Numerator (parameterPr029)
1	No.2Numerator (parameterPr031)

Note:0expressOFF,1expressON.

Pr034Set as2: Dynamic electronic gear ratio switching is allowed, and the command pulse electronic gear numerator is composed ofDIInputGEAR1, GEAR2Decide.

DISignal[Note]		Command pulse electronic gear numeratorN
GEAR2	GEAR1	
0	0	No.1Numerator (parameterPr029)
0	1	No.2Numerator (parameterPr031)
1	0	No.3Numerator (parameterPr032)
1	1	No.4Numerator (parameterPr033)

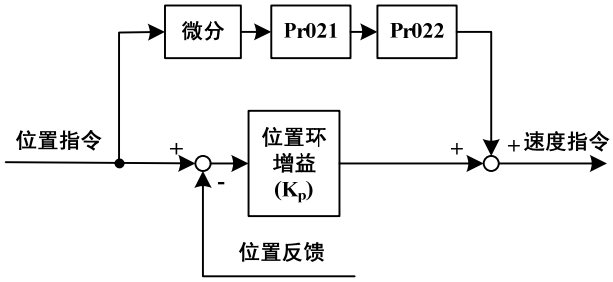
Note:0expressOFF,1expressON.

4.2.4Position control related gain

parameter	name	Parameter range	Default value	unit	Applicable
Pr009	Position loop gain	1~1000	40	1/s	P
Pr021	Position loop feedforward gain	0~100	0	%	P
Pr022	Position loop feedforward filter time constant	0.20~50.00	1.00	ms	P

Because the position loop includes the speed loop, in the order of inner loop first and outer loop later, first set the load moment of inertia ratio, then adjust the speed loop gain and speed loop integral time constant, and finally adjust the position loop gain.

The following is the position controller of the system, position loop gainK_pIncreasing can increase the position loop bandwidth, but is limited by the speed loop bandwidth. To increase the position loop gain, the speed loop bandwidth must be increased first.



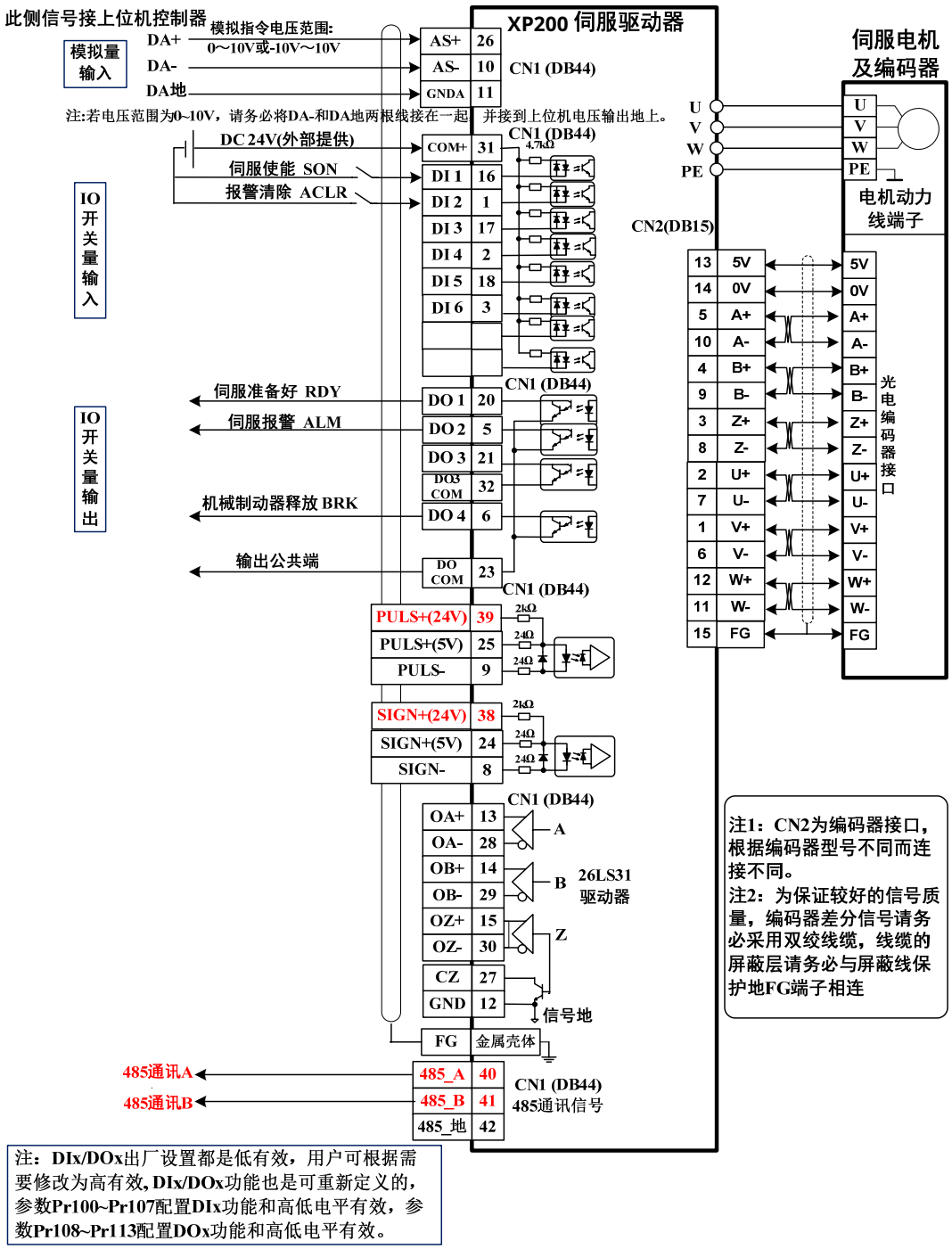
Feedforward can reduce the phase lag of position loop control, reduce the position tracking error during position control and shorten the positioning time. As the feedforward amount increases, the position control tracking error decreases, but too much will make the system unstable and overshoot. If the electronic gear ratio is greater than10It is also easy to generate noise. For general applications, you can setPr021for0%, when high response and low tracking error are required, it can be increased appropriately, but should not exceed80%At the same time, it may be necessary to adjust the position loop feedforward filter time constant (parameterPr022).

4.3Speed control

Speed control is used in situations where precise speed control is required, such as**Weaving machine, drilling machine,CNCProcessing Machine**It can also form position closed-loop control by cooperating with the host device.

4.3.1Simple example of speed control

This manual gives a simple example of speed control (analog speed command input). The following figure is the wiring diagram:



The parameter settings for this example are as follows:(Other parameters use factory default values)

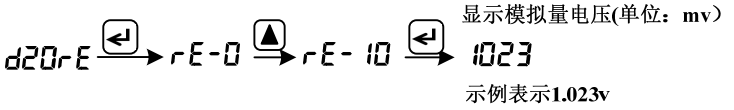
parameter	name	Setting Value	default value	Parameter Description
Pr004	Control method	1	0	Set to speed control mode
Pr025	Speed command source	0	0	Set as analog input
Pr060	Speed command acceleration between	According to demand To set	100	Zero speed to1000r/minAcceleration time between(ms)
Pr061	Speed command deceleration between	According to demand To set	100	1000r/minDeceleration time to zero speed (ms)
Pr097	Ignore driver disable	30 (use Driver ban end)	33 (suddenly Slightly drive Movement ban end)	Use forward drive prohibition (CCWL)and Reverse drive disabled (CWL)If you set To ignore, do not connectCCWL, CWL, see4.6Festival.
Pr104	Digital InputDI5Function	7	20	DI5Set to zero speed clampZCLAMP
Pr110	Digital OutputDO3achievement able	6	5	DO3Set to speed arrivalASP

4.3.2Parameters related to speed command

The parameters related to speed command are as follows:

parameter	name	Parameter range	Default value	unit	Applicable
Pr025	Speed command source	0~3	0		S
Pr046	Analog speed command gain	10~3000	300	r/min/V	S
Pr047	Analog speed command zero offset compensation	- 15000~15000	0	0.1mv	S
Pr048	Analog speed command direction	0~1	0		S
Pr049	Analog speed command filter time constant	2~500	20	0.1ms	S
Pr050	Analog speed command polarity	0~2	0		S
Pr051	Analog speed command dead zone1	0~13000	0	mv	S
Pr052	Analog speed command dead zone2	- 13000~0	0	mv	S

Analog zeroing How to operate See3.6.2You can also manually set the parameters according to the input analog quantity.Pr047 Adjust the zero offset value. The analog voltage value received by the driver can be viewed through the following menu items:



4.3.3Speed command source

The speed command has several different sources, which are determined by the parametersPr025The settings are described as follows:

Pr025	illustrate	explain
0	Analog speed command	portAS+andAS-Input analog voltage
1	Internal multi-speed command < 8OptionalPr137~Pr144 >	Depend onDIInputSP1,SP2,SP3Decision 1].
2	Analog speed command + internal multi-segment Speed command	whenSP1,SP2,SP3All forOFFWhen it is an analog instruction, the rest areSP1,SP2,SP3Decision 2].

Note1: Internal speed command:

DISignal			Speed command
SP3	SP2	SP1	
0	0	0	Internal speed1(parameterPr137)
0	0	1	Internal speed2(parameterPr138)
0	1	0	Internal speed3(parameterPr139)
0	1	1	Internal speed4(parameterPr140)
1	0	0	Internal speed5(parameterPr141)
1	0	1	Internal speed6(parameterPr142)
1	1	0	Internal speed7(parameterPr143)
1	1	1	Internal speed8(parameterPr144)

Note2: Analog speed command + internal speed command:

DISignal			Speed command
SP3	SP2	SP1	
0	0	0	Analog speed command
0	0	1	Internal speed2(parameterPr138)
0	1	0	Internal speed3(parameterPr139)
0	1	1	Internal speed4(parameterPr140)
1	0	0	Internal speed5(parameterPr141)
1	0	1	Internal speed6(parameterPr142)
1	1	0	Internal speed7(parameterPr143)
1	1	1	Internal speed8(parameterPr144)

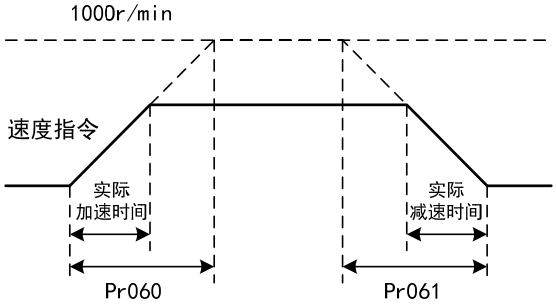
above0expressOFF,1expressONThere are twoDIInputs can provide special features:CZERO(Zero instruction), CINV(Instruction negation), whenCZEROforONWhen the speed command is forced to zero; whenCINVforON , the speed command is reversed.

4.3.4Acceleration and deceleration control

Acceleration and deceleration control are related to the following parameters:

parameter	name	Parameter range	Default value	unit	Applicable
Pr060	Speed command acceleration time	0~30000	100	ms	S
Pr061	Speed command deceleration time	0~30000	100	ms	S

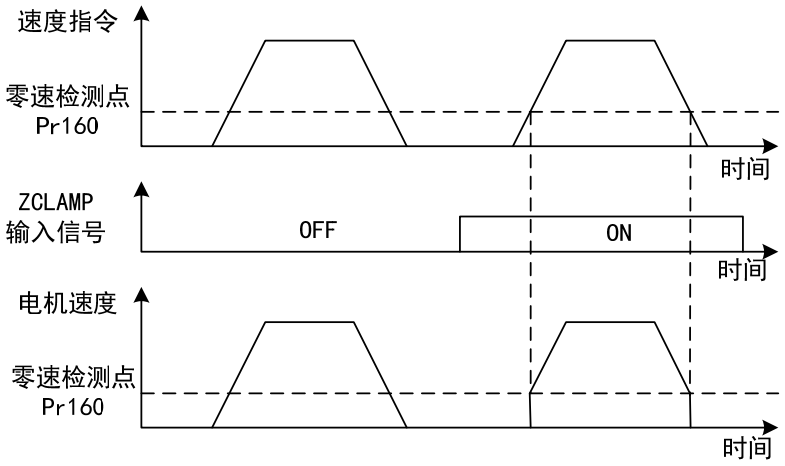
Acceleration and deceleration control can slow down the sudden change of speed and make the motor run smoothly. As shown in the figure below, the parametersPr060 Set the motor speed from zero to1000r/minThe acceleration time,Pr061Set the motor from 1000r/minIf the drive and the host device form position control, this parameter should be set to0.



4.3.5Zero speed clamp

The parameters related to zero speed clamp are as follows:

parameter	name	Parameter range	Default value	unit	Applicable
Pr160	Zero speed detection point	0~1000	10	r/min	ALL
Pr161	Zero speed detection hysteresis	0~1000	5	r/min	ALL
Pr162	Zero speed clamp mode	0~1	0		S



During speed control, even if the motor is at zero speed, external force may cause rotation and position change. If the analog speed command input is used, it is not easy to achieve an absolute zero speed command. In order to solve these two problems, you can consider using the zero speed clamp function. The zero speed clamp function is enabled when the following conditions are met:

Condition 1:Speed control mode;

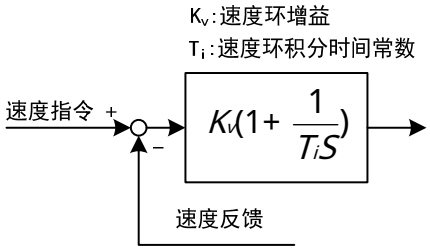
Condition 2:DIInZCLAMP(Zero speed clamp)ON; Condition 3:The speed command is lower than the parameterPr160When any of the above conditions is not met, normal speed control is performed. There are two modes for zero speed clamping:

Pr162	illustrate
0	The motor position is fixed at the moment the function is turned on. At this time, the internal position control is connected, and even if it rotates due to external force, it will return to the zero position fixed point.
1	When the function is turned on, the speed command is forced to zero speed. The internal speed is still controlled and rotation may occur due to external force.

4.3.6Speed control related gain

parameter	name	Parameter range	Default value	unit	Applicable
Pr005	1st speed loop gain	1~3000	40	Hz	P,S
Pr006	First speed loop integral time constant	1.0~1000.0	20.0	ms	P,S

The following is the speed controller of the system, increasing the speed loop gainK_vIt can improve the speed response bandwidth and reduce the speed loop integral time constant.T_i, which can increase the system rigidity and reduce the steady-state error.

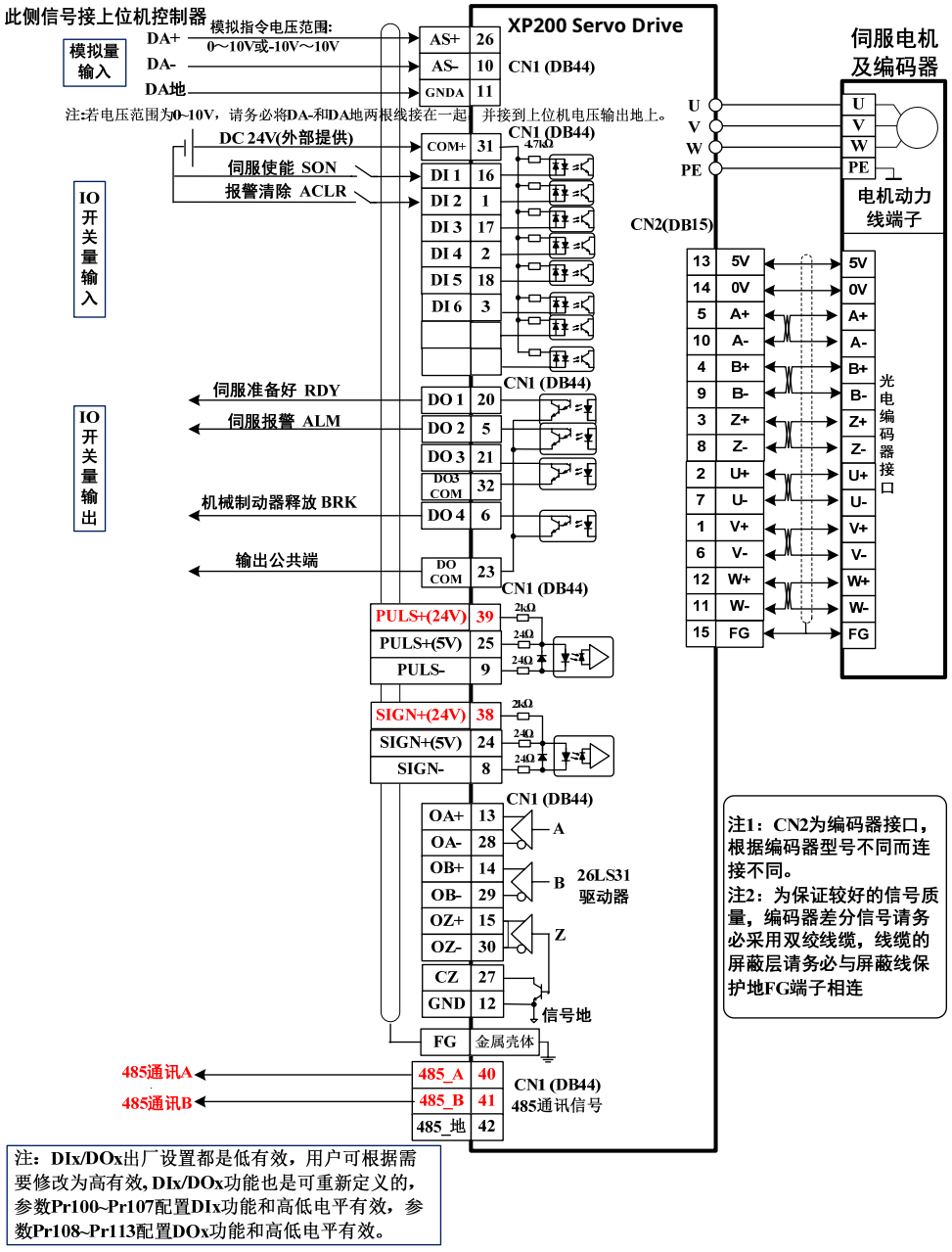


4.4Torque control

Torque control for**Printing machine, winding machine, injection molding machine**In other occasions, the motor output torque is proportional to the input command.

4.4.1Simple example of torque control

This manual gives a simple example of torque control (simulating torque command input). The following figure is the wiring diagram:



The parameter settings for this example are as follows:(Other parameters use factory default values)

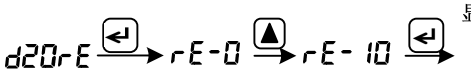
parameter	name	set up value	default value	Parameter Description
Pr004	Control method	2	0	Set to torque control
Pr026	Torque command source	0	0	Set as analog input
Pr097	Ignore driver disable	30	33	Use forward drive prohibition (CCWL)and Reverse drive disabled (CWL)If you set To ignore, do not connectCCWL,CWL , see4.6Festival.
Pr104	Digital InputDI5achievement able	5	20	DI5Set to forward torque limit TCCW
Pr105	Digital InputDI6achievement able	6	twenty one	DI6Set to reverse torque limitTCW
Pr110	Digital OutputDO3achievement able	6	5	DO3Set to speed arrivalASP

4.4.2Parameters related to torque command

The parameters related to torque command are shown in the following table:

parameter	name	Parameter range	Default value	unit	Applicable
Pr026	Torque command source	0~2	0		T
Pr053	Analog torque command gain	1~300	30	%/V	T
Pr054	Analog torque command zero offset compensation	- 1500.0~ 1500.0	0.0	mv	T
Pr055	Analog torque command direction	0~1	0		T
Pr056	Analog torque command filter time constant	20~500	20	0.1ms	T
Pr057	Analog torque command polarity	0~2	0		T
Pr075	Maximum speed limit	0~6000	3500	r/min	ALL

Analog zeroing How to operate See3.6.2You can also manually set the parameters according to the input analog quantity.Pr054 Adjust the zero offset value. The analog voltage value received by the driver can be viewed through the following menu items:



显示模拟量电压(单位：mv）

示例表示**1.023v**

4.4.3Torque command source

The torque command has several different sources, which are determined by the parametersPr026The settings are as follows:

Pr026	illustrate	explain
0	Analog torque command	portAS+andAS-Input analog voltage
1	Internal torque command	Depend onDIInputTRQ1,TRQ2Decision1].
2	Analog torque command + internal torque instruction	whenTRQ1,TRQ2All forOFFWhen it is an analog instruction, the rest areTRQ1,TRQ2Decision2].

Note1: Internal torque command:

DISignal		Torque command
TRQ2	TRQ1	
0	0	Internal torque1(parameterPr145)
0	1	Internal torque2(parameterPr146)
1	0	Internal torque3(parameterPr147)
1	1	Internal torque4(parameterPr148)

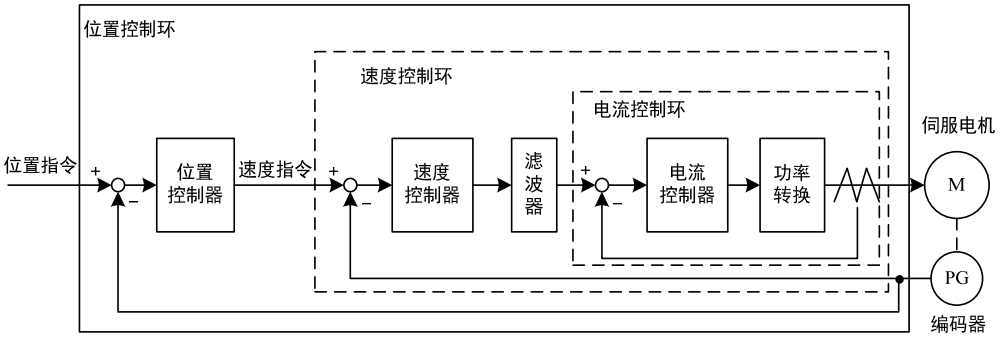
Note2: Analog torque command + internal torque command:

DISignal		Torque command
TRQ2	TRQ1	
0	0	Analog torque command
0	1	Internal torque2(parameterPr146)
1	0	Internal torque3(parameterPr147)
1	1	Internal torque4(parameterPr148)

above0expressOFF,1expressONThere are twoDIenterCZERO(Zero instruction),CINV(Instruction inversion) can provide special functions.CZEROforONWhen , the torque is forced to zero; whenCINVforON, the torque command is reversed.

4.5Gain Adjustment

The driver includes three control loops: current control loop, speed control loop and position control loop. The control block diagram is as follows:



In theory, the bandwidth of the inner control loop must be higher than that of the outer control loop, otherwise the entire control system will be unstable and cause vibration or poor response. Therefore, the relationship between the bandwidths of the three control loops is as follows:

Current loop bandwidth>Speed loop bandwidth>Position loop bandwidth Since the driver has already adjusted the current control loop to the optimal state, the user only needs to adjust the speed control loop and position control loop parameters.

4.5.1Gain parameter

The parameters related to control gain are as follows:

parameter	name	Parameter range	Default value	unit	Applicable
Pr005	Speed loop gain	1~3000	50	Hz	P,S
Pr006	Speed loop integral time constant	1~1000	20	ms	P,S
Pr009	Position loop gain	1~1000	40	1/s	P
Pr007	Torque filter time constant	1~500	25	0.1ms	ALL
Pr019	Speed detection filter time constant	5~500	25	0.1ms	P,S

The symbols are defined as follows:

K_V: Speed loop gain;

T_i: Speed loop integral time constant;

K_P: Position loop gain;

G: Load moment of inertia ratio;

J_L: load moment of inertia converted to the motor shaft;

J_M: Motor rotor moment of inertia.

1.Speed loop gainK_V

Speed loop gainK_VDirectly determines the response bandwidth of the speed loop. Under the premise that the mechanical system does not generate vibration or noise, increasing the speed loop gain value will speed up the speed response and improve the followability of the speed command. However, a setting that is too large can easily cause mechanical resonance. The speed loop bandwidth is expressed as:

$$\text{Speed loop bandwidth}(Hz) = \frac{1+G}{1+J_L/J_M} \times K_V(Hz)$$

If the load moment of inertia is greater thanGSet up correctly ($G=J_L/J_M$),Then the speed loop bandwidth is equal to the speed loop gain K_V.

2.Speed loop integral time constantT_i

The speed loop integral can effectively eliminate the speed steady-state error and quickly respond to subtle speed changes. Under the premise that the mechanical system does not generate vibration or noise, the speed loop integral time constant can be reduced.T_i, in order to increase the rigidity of the system and reduce the steady-state error. If the load inertia ratio is large or there is resonance in the mechanical system, it is necessary to confirm

The speed loop integral time constant is large enough, otherwise the mechanical system is prone to resonance.GSet up correctly ($G=J_L/J_M$), use the following formula to get the speed loop integral time constantT_i:

$$T_i(ms) \geqslant \frac{4000}{2\pi \times K_v(Hz)}$$

3. Position loop gainK_p

The position loop gain directly determines the reaction speed of the position loop. Under the premise that the mechanical system does not generate vibration or noise, increase the position loop gain value to speed up the reaction speed, reduce the position tracking error, and shorten the positioning time. However, too large a setting will cause mechanical system jitter or positioning overshoot. The position loop bandwidth cannot be higher than the speed loop bandwidth. Generally,

$$\text{Position loop bandwidth}(Hz) \leqslant \frac{\text{Speed loop bandwidth(Hz)}}{4}$$

If the load moment of inertia is greater thanGSet up correctly ($G=J_L/J_M$),Then the position loop gainK_pThe calculation is as follows:

$$K_p(1/s) \leqslant 2\pi \times \frac{K_v(Hz)}{4}$$

4.5.2Gain Adjustment Procedure

The selection of position and speed bandwidth must be determined by the rigidity of the machine and the application. The rigidity of the conveying machine connected by the belt is low, so it can be set to a lower bandwidth; the mechanical rigidity of the ball screw driven by the reducer is medium, so it can be set to a medium bandwidth; the rigidity of the direct drive ball screw or linear motor is high, so it can be set to a high bandwidth. If the mechanical characteristics are unknown, the gain can be gradually increased to increase the bandwidth until resonance occurs, and then the gain can be lowered.

In the servo gain, if one parameter is changed, the other parameters also need to be readjusted. Please do not make a large change to only one parameter. Regarding the steps for changing the servo parameters, generally follow the following principles:

Improve responsiveness	Reduce response, suppress vibration and overshoot
1.Increase speed loop gainK_v(Pr005)	1.Reduce the position loop gainK_p(Pr009)
2.Reduce the speed loop integral time constantT_i(Pr006)	2.Increase the speed loop integral time constantT_i(Pr006)
3.Increase position loop gainK_p(Pr009)	3.Reduce speed loop gainK_v(Pr005)

Gain adjustment steps for speed control

1.Set the load inertia ratio (the driver automatically adapts according to the motor model).

2.Set the speed loop integral time constant (Pr006)is a larger value.

3.Speed loop gain(Pr005)Adjust the pressure within the range that does not cause vibration or abnormal sound. If vibration occurs, adjust it slightly.

4.The speed loop integral time constant should be adjusted to a smaller value within the range where no vibration occurs. If vibration occurs, adjust it slightly larger.

5.If the gain cannot be increased due to resonance of the mechanical system, and the desired responsiveness cannot be obtained, the torque low-pass filter (Pr007)or speed detection filter (Pr019)After adjusting to suppress resonance, repeat the above steps to improve responsiveness.

Gain adjustment steps for position control

1.Set the load inertia ratio (the driver automatically adapts according to the motor model).

2.Set the speed loop integral time constant (Pr006)is a larger value.

3.Speed loop gain(Pr005)Adjust the pressure within the range that does not cause vibration or abnormal sound. If vibration occurs, adjust it slightly.

4.The speed loop integral time constant should be adjusted to a smaller value within the range where no vibration occurs. If vibration occurs, adjust it slightly larger.

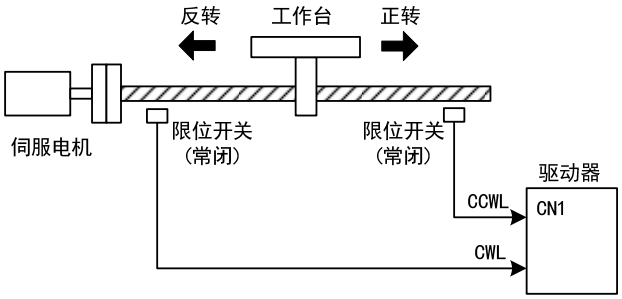
5.Increase the position loop gain (Pr009), if vibration occurs, adjust it down slightly.

6.If the gain cannot be increased due to resonance of the mechanical system, and the desired responsiveness cannot be obtained, the torque low-pass filter (Pr007)or speed detection filter (Pr019)After adjusting to suppress resonance, repeat the above steps to improve responsiveness.

7.If a shorter positioning time and smaller position tracking error are required, the position feedforward can be adjusted appropriately (Pr021, Pr022).

4.6Over travel protection

The overtravel protection function means that when the moving part of the machine exceeds the designed safe moving range, the limit switch will be activated and the motor will be forced to stop. The schematic diagram of overtravel protection is as follows:



It is recommended to use a normally closed contact for the limit switch, which is closed within the safe range and open when overtravel.CCWL)and reverse drive inhibit (CWL), through the parametersPr097It can also be set to use or ignore. If it is set to use, the limit signal must be connected; if it is set to ignore, the signal is not required. The default value of the parameter isCCWLandCWLIgnore all. If you need to use them, you must modify the parameters.Pr097Even in the overtravel state, it is still allowed to exit the overtravel state by inputting a reverse command.

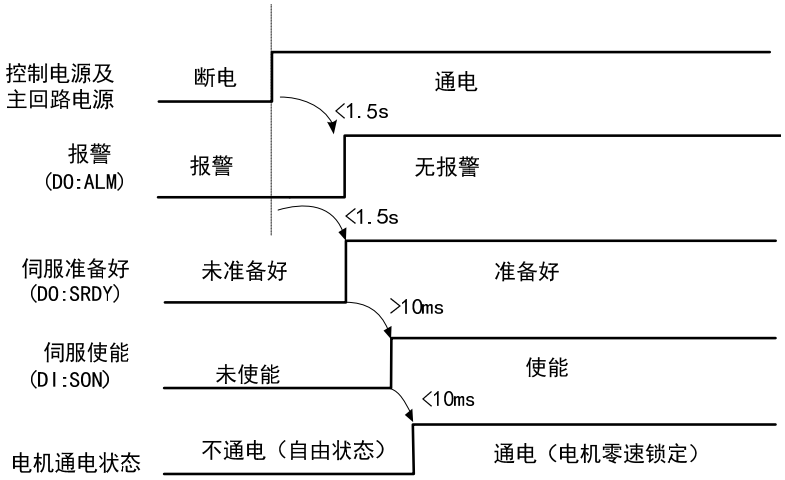
Pr097	Reverse drive prohibition (CWL)	Forward drive disabled (CCWL)
0	use	use
1	use	neglect

2	neglect	use
3(default)	neglect	neglect

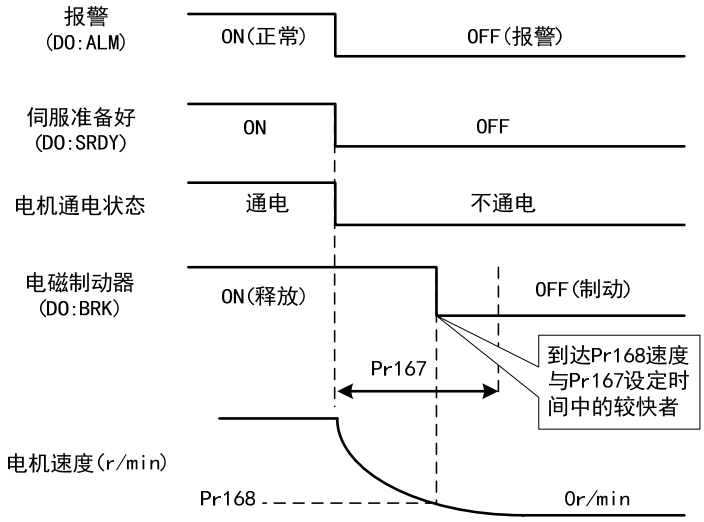
4.7Working sequence

4.7.1Power on sequence

After the main power is turned on, the delay is about1.5seconds, servo ready signal (SRDY) ON, you can now accept the servo enable (SON)Signal, detect that the servo enable is valid, the power circuit is turned on, the motor is excited, and it is in the running state. Detect that the servo enable is invalid or there is an alarm, the power circuit is turned off, and the motor is in the free state.

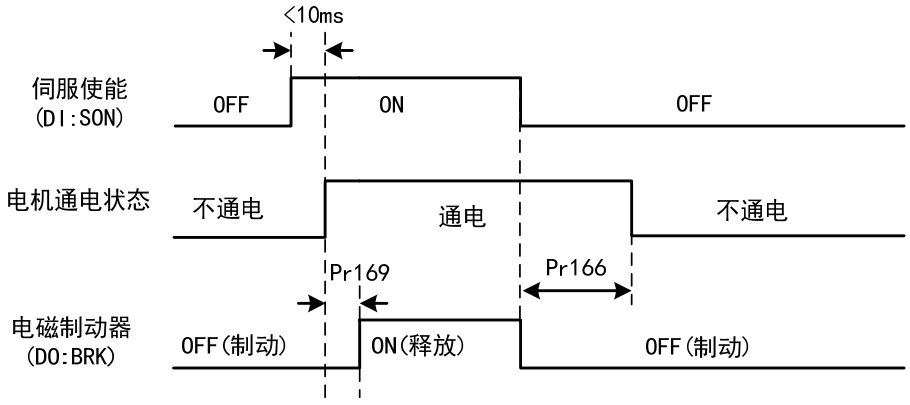


4.7.2ServoONAlarm timing



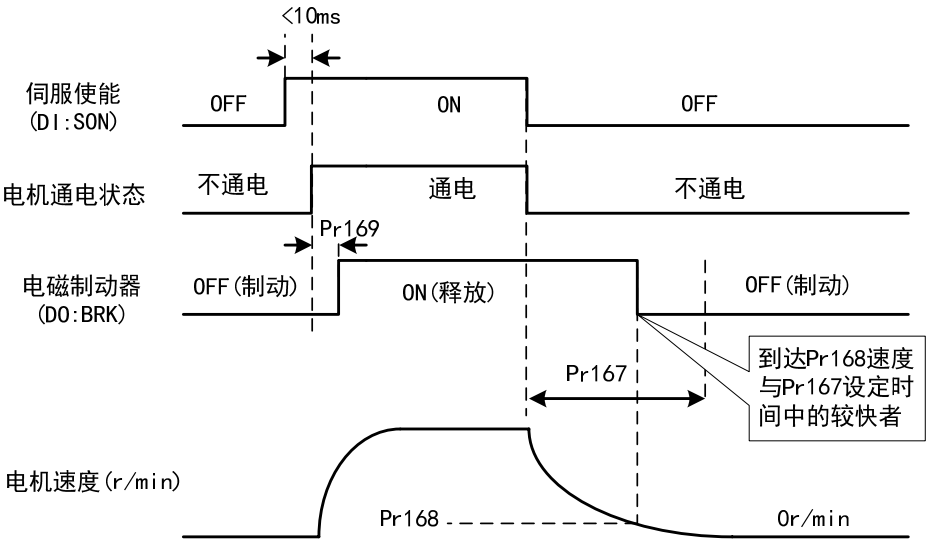
4.7.3Electromagnetic brake action sequence when the motor is stationary

When the motor speed is lower than the parameterPr165Electromagnetic brake action sequence:



4.7.4Electromagnetic brake action sequence when the motor is running

When the motor speed is higher than the parameterPr165Electromagnetic brake action sequence:


<10ms
伺服使能
(DI:SON)
OFF
ON
OFF
电机通电状态
不通电
通电
不通电
Pr169
电磁制动器
(DO:BRK)
OFF(制动)
ON(释放)
OFF(制动)
Pr167
到达Pr168速度
与Pr167设定时
间中的较快者
电机速度(r/min)
Pr168
0r/min

4.8Electromagnetic brake

Electromagnetic brakes (holding brakes, power-off brakes) are used to lock the vertical or tilted workbench connected to the motor to prevent the workbench from falling after the servo power is lost. To achieve this function, you need to purchase a motor with a brake. The brake can only be used to hold the workbench and must not be used to slow down or stop the machine movement.

4.8.1Electromagnetic brake parameters

Parameters of electromagnetic brake:

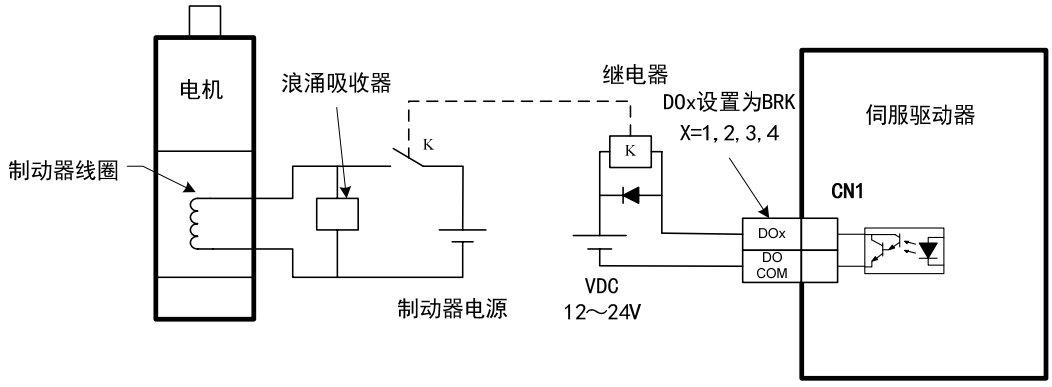
parameter	name	Parameter range	Default value	unit	Applicable
Pr165	Motor static speed detection point	0~1000	5	r/min	ALL
Pr166	Electromagnetic brake delay time when the motor is stationary	0~2000	500	ms	ALL
Pr167	Electromagnetic brake waiting time when the motor is running	0~2000	500	ms	ALL
Pr168	Electromagnetic brake action speed when the motor is running	0~3000	100	r/min	ALL
Pr169	Electromagnetic brake release delay time	0~3000	500	ms	ALL

4.8.2Use of electromagnetic brake

The figure below is the brake wiring diagram, the brake release signal of the driverBRKConnect the relay coil and the relay contacts to the brake power supply. The brake power supply is provided by the user and has sufficient capacity. It is recommended to install a surge absorber to suppress the surge voltage caused by the relay on/off action. A diode can also be used as a surge absorber, but it will cause a slight braking delay.

After the motor stops, the speed is less thanPr165)ServoOFFAt this time, the motor continues to be powered to maintain the position. The brake is released and then braked. After a period of stabilization (the time is determined by the parameterPr166Confirm), remove the power supply to the motor. The motor is running (speed greater thanPr165)ServoOFFAt this time, the motor current is cut off and the brake continues to be released. After a delay, the brake is applied. This is to slow down the motor from high speed to low speed before the mechanical brake is activated to avoid damage to the brake. The delay time is a parameterPr167Or the motor speed is decelerated to the parameterPr168The speed and time required are the minimum of the two.

Pr169：When the system changes from the disabled state to the enabled state, it defines the time from when the motor current is turned on to when the electromagnetic brake is released (DOOutput TerminalsBRK ON)Delay time.



No.5Chapter Parameters

5.1Parameter list

The applicable column indicates the applicable control mode.PFor position control,SFor speed control,TFor torque control, AllIt is applicable to position, speed and torque control. Parameter value "*" indicates that the factory default value may be different.

5.1.1 0Segment parameters

parameter	name	Parameter range	Default value	unit	Applicable
Pr000	Operation password	0~9999	315		ALL
Pr001	Drive model code	*	*		ALL
Pr002	Motor code	*	*		ALL
Pr003	Software Version	*	*		ALL
Pr004	Control method	0~8	0		ALL
Pr005	No.1Speed loop gain	1~3000	50	Hz	P,S
Pr006	No.1Speed loop integral time constant	1~1000	20	ms	P,S
Pr007	No.1Torque filter time constant	1~500	25	0.1ms	ALL
Pr009	No.1Position loop gain	1~1000	40	1/s	P
Pr019	Speed detection filter time constant	5~500	35	0.1ms	P,S
Pr021	Position loop feedforward gain	0~100	0	%	P
Pr022	Position loop feedforward filter time constant	2~500	10	0.1ms	P
Pr025	Speed command source	0~3	0		S
Pr026	Torque command source	0~2	0		T
Pr029	Position command pulse electronic gear 1st numerator	1~32767	1		P
Pr030	Position command pulse electronic gear denominator	1~32767	1		P
Pr031	Command pulse electronic gear2molecular	1~32767	1		P
Pr032	Command pulse electronic gear3molecular	1~32767	1		P
Pr033	Command pulse electronic gear4molecular	1~32767	1		P
Pr035	Position command pulse input mode	0~2	0		P
Pr036	Position command pulse input direction	0~1	0		P
Pr037	Position command pulse input signal logic	0~3	0		P
Pr038	Position command pulse signal filter coefficient	0~15	6		P
Pr039	Position command direction signal filter coefficient	0~3	0		P
Pr040	Exponential smoothing filter time for position command	0~5000	0	1ms	P
Pr046	Analog speed command gain	10~3000	300	r/min/V	S
Pr047	Analog speed command zero offset compensation	- 15000~15000	0	0.1mv	S
Pr048	Analog speed command direction	0~1	0		S

parameter	name	Parameter range	Default value	unit	Applicable
Pr049	Analog speed command filter time constant	2~500	20	0.1ms	S
Pr050	Analog speed command polarity	0~2	0		S
Pr051	Analog speed command dead zone1	0~13000	0	mv	S
Pr052	Analog speed command dead zone2	- 13000~0	0	mv	S
Pr053	Analog torque command gain	1~300	30	%/V	T
Pr054	Analog torque command zero offset compensation	1500~1500	0	mv	T
Pr055	Analog torque command direction	0~1	0		T
Pr056	Analog torque command filter time constant	2~500	20	0.1ms	T
Pr057	Analog torque command polarity	0~2	0		T
Pr060	Speed command acceleration time	0~10000	100	ms	S
Pr061	Speed command deceleration time	0~10000	100	ms	S
Pr063	EMGEmergency stop)/ALMDeceleration time	0~10000	1000	ms	ALL
Pr064	Torque limit selection	0~2	0		ALL
Pr065	Internal forward (CCW)Torque limit	0~300	300	%	ALL
Pr066	Internal reversal (CW)Torque limit	- 300~0	- 300	%	ALL
Pr067	External forward (CCW)Torque limit	0~300	100	%	ALL
Pr068	External reversal (CW)Torque limit	- 300~0	- 100	%	ALL
Pr069	Test run torque limit	0~300	100	%	ALL
Pr070	Forward(CCW)Torque overload alarm level	1~300	140	%	ALL
Pr071	ReverseCW)Torque overload alarm level	- 300～-1	- 140	%	ALL
Pr072	Torque overload alarm detection time	1~30000	5000	ms	ALL
Pr075	Maximum speed limit	0~6000	3500	r/min	ALL
Pr076	JOGRunning speed	0~5000	100	r/min	S
Pr077	Speed limit selection	0~2	0		T
Pr078	Speed limit in torque control	0~5000	1000	r/min	T
Pr079	Speed limit error during torque control	1~5000	100	r/min	T
Pr080	Position deviation detection	0~32767	400	0.01lock up	P
Pr081	First encoder type selection	0~1	0		ALL
Pr082	The first encoder line number	1~32767	2500	pulse	ALL
Pr083	The first encoder zero position offset angle	- 3600~3600	- 230	0.1Spend	ALL
Pr084	First encoder control parameters	0~11111	0		ALL
Pr097	Positive and negative limit use mode/positive and negative reverse control	0~33	33		ALL
Pr098	Forced Enable	0~1	0		ALL
Pr099	Motor series code	0~15	0		ALL

5.1.2 1Segment parameters

parameter	name	Parameter range	Default value	unit	Applicable
Pr100	Digital InputDI1Function	- twenty one~twenty one	1		ALL
Pr101	Digital InputDI2Function	- twenty one~twenty one	2		ALL
Pr102	Digital InputDI3Function	- twenty one~twenty one	3		ALL
Pr103	Digital InputDI4Function	- twenty one~twenty one	4		ALL
Pr104	Digital InputDI5Function	- twenty one~twenty one	20		ALL
Pr105	Digital InputDI6Function	- twenty one~twenty one	twenty one		ALL
Pr108	Digital OutputDO1Function	- 12~12	2		ALL
Pr109	Digital OutputDO2Function	- 12~12	3		ALL
Pr110	Digital OutputDO3Function	- 12~12	5		ALL
Pr111	Digital OutputDO4Function	- 12~12	8		ALL
Pr120	Digital InputDIMandatory1	00000~11111	00000		ALL
Pr121	Digital InputDIMandatory2	00000~11111	00000		ALL
Pr122	Digital InputDIMandatory3	00000~11111	00000		ALL
Pr123	Digital InputDIMandatory4	00000~11111	00000		ALL
Pr124	Digital InputDIMandatory5	00000~11111	00000		ALL
Pr125	Digital InputDIxFiltering	1~1000	5	ms	ALL
Pr136	EnableoffOr stop mode when alarm occurs	0~1	0		ALL
Pr137	Internal speed1	- 5000~5000	0	r/min	S
Pr138	Internal speed2	- 5000~5000	0	r/min	S
Pr139	Internal speed3	- 5000~5000	0	r/min	S
Pr140	Internal speed4	- 5000~5000	0	r/min	S
Pr141	Internal speed5	- 5000~5000	0	r/min	S
Pr142	Internal speed6	- 5000~5000	0	r/min	S
Pr143	Internal speed7	- 5000~5000	0	r/min	S
Pr144	Internal speed8	- 5000~5000	0	r/min	S
Pr145	Internal torque1	- 300~300	0	%	T
Pr146	Internal torque2	- 300~300	0	%	T
Pr147	Internal torque3	- 300~300	0	%	T
Pr148	Internal torque4	- 300~300	0	%	T
Pr150	Positioning completion range	0~32767	10	pulse	P
Pr151	Positioning completion hysteresis	0~32767	5	pulse	P
Pr152	Positioning proximity range	0~32767	500	pulse	P
Pr153	Positioning proximity hysteresis	0~32767	50	pulse	P
Pr154	Arrival speed	- 5000~5000	500	r/min	ALL
Pr155	Arrival speed difference	0~5000	30	r/min	ALL

parameter	name	Parameter range	Default value	unit	Applicable
Pr156	Arrival speed polarity	0~1	0		ALL
Pr157	Torque	- 300~300	100	%	ALL
Pr158	Reach torque differential	0~300	5	%	ALL
Pr159	Arrival torque polarity	0~1	0		ALL
Pr160	Zero speed detection point	0~1000	10	r/min	ALL
Pr161	Zero speed detection hysteresis	0~1000	5	r/min	ALL
Pr162	Zero speed clamp mode	0~1	0		S
Pr164	Emergency stop method	0~1	0		P
Pr165	Motor static speed detection point	0~1000	5	r/min	ALL
Pr166	Electromagnetic brake delay time when the motor is stationary	0~5000	500	ms	ALL
Pr167	Electromagnetic brake waiting time when the motor is running	0~5000	500	ms	ALL
Pr168	Electromagnetic brake action speed when the motor is running	0~3000	100	r/min	ALL
Pr169	Electromagnetic brake release delay time	0~5000	500	ms	ALL

5.1.3 2Segment parameters (partial)

parameter	name	Parameter range	Default value	unit	Applicable
Pr201	Motor pole pairs	1~50	4		ALL
Pr204	Motor rated current	1~4000	60	0.1A	ALL
Pr207	Motor rated speed	1~30000	2500	r/min	ALL

5.2 DIFunction List

sequence Number	symbol	DIFunction	Serial number	symbol	DIFunction
0	NULL	No function	14	TRQ2	Internal torque selection2
1	SON	Servo Enable	15	EMG	Emergency Stop
2	ACLR	Alarm clear	16	CMODE	Control mode switching
3	CCWL	Forward drive disabled	17	GAIN	Gain switching
4	CWL	Reverse drive prohibition	18	GEAR1	Electronic gear selection1
5	TCCW	Forward torque limit	19	GEAR2	Electronic gear selection2
6	TCW	Reverse torque limit	20	CLE	Position deviation clearing
7	ZCLAMP	Zero speed clamp	twenty one	INH	Pulse input disabled
8	CZERO	Zero instruction			
9	CINV	Instruction Negation	twenty three	CCW	Forward operation (start)
10	SP1	Internal speed selection1	twenty four	CW	Reverse operation
11	SP2	Internal speed selection2			

12	SP3	Internal speed selection3			
13	TRQ1	Internal torque selection1			

5.3 DOFunction List

Serial number	symbol	DOFunction	Serial number	symbol	DOFunction
0	OFF	Has been ineffective	8	BRK	Electromagnetic brake
1	ON	Always valid	9	RUN	Servo running
2	SRDY	Servo ready	10	NEAR	Positioning proximity
3	ALM	Call the police	11	TRQL	Torque limiting
4	ZSP	Zero speed	12	SPL	Speed limit
5	COIN	Positioning completed	14	PtS	Position/speed mode switching Finish
6	ASP	Speed arrival	15	Pt	Position/torque mode switching Finish
7	ATRQ	Torque reached	16	StoT	Speed/torque mode switching Finish
			20	DO3_ZOUT	Return to zeroZSignal output (only DO3Support this feature)

5.4Parameters

5.4.1 0Segment parameters

		scope	Default value	unit	Applicable
Pr000	Operation password	0~9999	315		ALL

Hierarchical management of parameters can ensure that parameters are not modified by mistake.

Some special operations require setting a suitable password.

P-002Motor code,P-099Motor series code, when modifying, you need toP-000Set as385(or316), otherwise the modification will be invalid

		scope	Default value	unit	Applicable
Pr001	Driver Code	*	*		ALL

The drive model currently in use. It is factory set and cannot be modified by the user.

		scope	Default value	unit	Applicable
Pr002	Motor code	*	*		ALL

The motor model currently used, please set it according to the actual motor model used.7.2Chapter Motor Adaptation Table.

When changing to a different type of motor, you need to modify this parameter. For specific operations, please refer to3.7chapter.

Pr003	Software Version	scope	Default value	unit	Applicable
		*	*		ALL

Software version number, cannot be modified.

Pr004	Control method	scope	Default value	unit	Applicable
		0~8	0		ALL

Parameter meaning:

0: Position control mode; 1: Speed control mode; 2: Torque control mode;

3: Position/speed mode; 4: Position/torque mode; 5: Speed/torque mode;

Set to3,4,5When the specific control method isDIInputCMODEDecide:

P004	CMODE[Note]	Control method
3	0	Position Control
	1	Speed control
4	0	Position Control
	1	Torque control
5	0	Speed control
	1	Torque control

Note:0expressOFF,1expressON.

Pr005	No.1Speed loop gain	scope	Default value	unit	Applicable
		1~3000	50	Hz	P,S

The proportional gain of the speed regulator. Increasing the parameter value can speed up the speed response. Too large a value can easily cause vibration and noise.

Pr006	No.1Speed loop integral time constant	scope	Default value	unit	Applicable
		1~1000	20	ms	P,S

The integral time constant of the speed regulator can reduce the speed control error and increase the rigidity by reducing the parameter value. If it is too small, it will easily cause vibration and noise.

Set to the maximum value (1000)Indicates canceling the integral, the speed regulator isPController.

Pr007	No.1Torque filter time constant	scope	Default value	unit	Applicable
		1~500	35	0.1ms	ALL

Torque low-pass filter to suppress mechanically induced vibrations.

The larger the value, the better the vibration suppression effect. However, a too large value will cause a slower response and may cause oscillation. The smaller the value, the faster the response, but it is limited by mechanical conditions.

When the load inertia is small, a smaller value can be set; when the load inertia is large, a larger value can be set.

Pr009	No.1Position loop gain	scope	Default value	unit	Applicable
		1~1000	40	1/s	P

Proportional gain of the position regulator; increasing the parameter value can reduce the position tracking error and improve the response. Too large a value may cause overshoot or oscillation.

Pr019	Speed detection filter time constant	scope	Default value	unit	Applicable
		5~500	35	0.1ms	P,S

The larger the parameter value, the smoother the detection, and the smaller the parameter value, the faster the detection response. A too small value may cause noise, and a too large value may cause oscillation.

Pr021	Position loop feedforward gain	scope	Default value	unit	Applicable
		0~100	0	%	P

Feedforward can reduce the position tracking error during position control and is set to100When the command pulse frequency is any, the position tracking error is always0.

Increasing the parameter value will improve the position control response, but too large a value will make the system unstable and prone to oscillation.

Pr022	Position loop feedforward filter time constant	scope	Default value	unit	Applicable
		2~500	10	0.1ms	P

The filtering of the position loop feedforward quantity is used to increase the stability of the feedforward control.

Pr025	Speed command source	scope	Default value	unit	Applicable
		0~5	0		S

In speed control, set the source of speed command.

Parameter meaning:

0: Analog speed1: Internal multi-speed<8OptionalPr137~Pr144 > 2: Analog + internal multi-speed3： The specific meaning of pulse speed is as follows:

0: Analog speed command, from the analog portAS+,AS-enter. 1: Internal speed command, determined byDIInputSP1,SP2,SP3Decide:

DISignal[Note]			Speed command
SP3	SP2	SP1	
0	0	0	Internal speed1(parameterPr137)
0	0	1	Internal speed2(parameterPr138)
0	1	0	Internal speed3(parameterPr139)
0	1	1	Internal speed4(parameterPr140)
1	0	0	Internal speed5(parameterPr141)
1	0	1	Internal speed6(parameterPr142)
1	1	0	Internal speed7(parameterPr143)
1	1	1	Internal speed8(parameterPr144)

Note:0expressOFF,1expressON.

2: Analog speed command + internal speed command:

DISignal[Note]			Speed command
SP3	SP2	SP1	
0	0	0	Analog speed command
0	0	1	Internal speed2(parameterPr138)
0	1	0	Internal speed3(parameterPr139)
0	1	1	Internal speed4(parameterPr140)
1	0	0	Internal speed5(parameterPr141)
1	0	1	Internal speed6(parameterPr142)
1	1	0	Internal speed7(parameterPr143)
1	1	1	Internal speed8(parameterPr144)

Note:0expressOFF,1expressON.

Pr026	Torque command source	scope	Default value	unit	Applicable
		0~2	0		T

In torque control, set the source of the torque command.

Parameter meaning:

0: Analog torque command, from the analog portAS+,AS-enter. 1:

Internal torque command, determined byDIInputTRQ1,TRQ2Decide:

DISignal[Note]		Torque command
TRQ2	TRQ1	
0	0	Internal torque1(parameterPr145)
0	1	Internal torque2(parameterPr146)
1	0	Internal torque3(parameterPr147)
1	1	Internal torque4(parameterPr148)

Note:0expressOFF,1expressON.

2: Analog torque command + internal torque command;

DISignal[Note]		Torque command
TRQ2	TRQ1	
0	0	Analog torque command
0	1	Internal torque2(parameterPr146)
1	0	Internal torque3(parameterPr147)
1	1	Internal torque4(parameterPr148)

Note:0expressOFF,1expressON.

Pr029	Position command pulse electronic gear1molecular	scope	Default value	unit	Applicable
		1~32767	1		P

Pr030	Position command pulse electronic gear denominator	scope	Default value	unit	Applicable
		1~32767	1		P

It is used to divide or multiply the input pulse frequency and can be easily matched with various pulse sources to achieve the pulse resolution required by the user.

Command pulse electronic gear numeratorNBy parametersPr029Setting, denominatorMBy parametersPr030Set.

Input pulse command passesN/MChange to get position instruction

Pr031	Position command pulse electronic gear2molecular	scope	Default value	unit	Applicable
		1~32767	1		P

Reference ParametersPr029,Pr030Description.

Pr032	Position command pulse electronic gear3molecular	scope	Default value	unit	Applicable
		1~32767	1		P

Reference ParametersPr029,Pr030Description.

Pr033	Position command pulse electronic gear4molecular	scope	Default value	unit	Applicable
		1~32767	1		P

Reference ParametersPr029,Pr030Description.

Pr034	Position command control word	scope	Default value	unit	Applicable
		0~11111	0		S

In position control, set the source of position command and dynamic electronic gear application mode.

Parameter meaning:

Tens digit: Set position command

0: External pulse1: Internal digital quantity

Units digit: Set the dynamic electronic gear application mode (see4.2.3Chapter Description)

0：Dynamic electronic gear ratio switching is prohibited. The electronic gear ratio numerator of the command pulse isPr029set up

1：Dynamic electronic gear ratio switching is allowed, and the command pulse electronic gear numerator isDIInputGEAR1Decide

2：Dynamic electronic gear ratio switching is allowed, and the command pulse electronic gear numerator isDIInputGEAR1, GEAR2 Decide

Pr035	Position command pulse input mode	scope	Default value	unit	Applicable
		0~2	0		P

Set the command pulse input mode, parameter meaning:

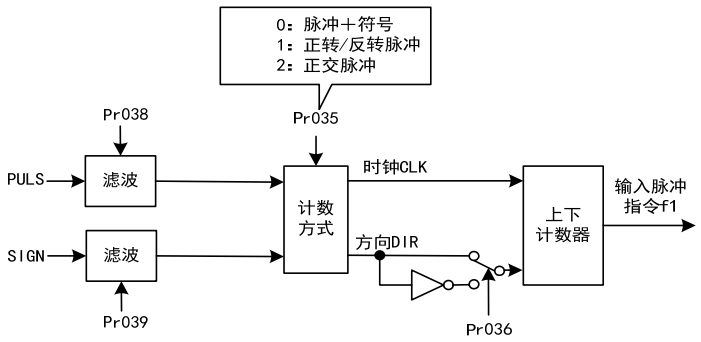
0：Pulse＋direction

1: Forward/reverse pulse

2: Orthogonal pulse

脉冲指令形式	正转(CCW)	反转(CW)	参数Pr035
脉冲+方向	PULS SIGN		0
正转/反转脉冲	PULS SIGN		1
正交脉冲	PULS SIGN		2

Note: The arrow indicates the counting edge, and the parameter setting isPr036=0,Pr037=0Command pulse input block diagram



After the parameters are modified, they must be saved and powered on again to take effect.

Pr036	Position command pulse input direction	scope	Default value	unit	Applicable
		0~1	0		P

Parameter meaning:

0: Normal direction 1: Direction reverse

Pr038	Position instructions**pulse** Signal filter coefficient	scope	Default value	unit	Applicable
		0~15	6		P

For pulse input signalPULSDigital filtering is performed. The larger the value, the larger the filtering time constant. The maximum pulse input frequency is500kHz(kpps)The larger the value, the lower the maximum pulse input frequency will be.

It is used to filter out the noise on the signal line to avoid counting errors. If the counting is inaccurate and the movement is inaccurate, the parameter value can be appropriately increased.

The timing of pulse and direction signals can be advanced or delayed.

After the parameters are modified, they must be saved and powered on again to take effect.

Pr039	Position instructions**direction** Signal filter coefficient	scope	Default value	unit	Applicable
		0~3	0		P

For pulse direction signalSIGNFor digital filtering, the larger the value, the larger the filtering time constant.

After the parameters are modified, they must be saved and powered on again to take effect.

The timing of pulse and direction signals can be advanced or delayed.

After the parameters are modified, they must be saved and powered on again to take effect.

Pr040	Exponential smoothing filter time for position command	scope	Default value	unit	Applicable
		0~5000	0	1ms	P

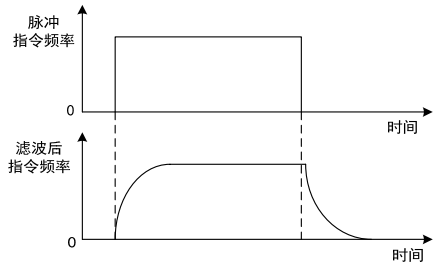
Smooth filter the command pulse, with exponential acceleration and deceleration. The filter will not lose the input pulse, but there will be command delay.0The filter is not effective when . This filter is used for:

1.The upper controller has no acceleration and deceleration function;

2.The electronic gear is relatively large (N/M>10);

3.The instruction frequency is low;

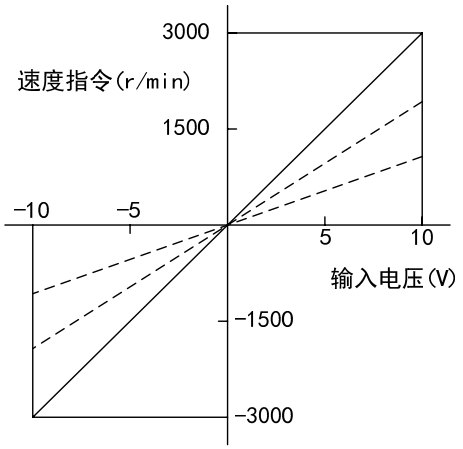
4.The motor experiences step jumps and instability during operation.



Pr046	Analog speed command gain	scope	Default value	unit	Applicable
		10~3000	300	r/min/V	S

Set the proportional relationship between the analog speed input voltage and the actual motor speed.

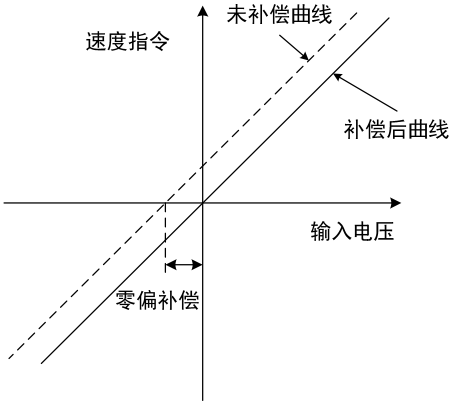
The analog input range is -10V~10V.



Pr047	Analog speed command zero offset compensation	scope	Default value	unit	Applicable
		- 15000~15000	0	0.1mv	S

The zero offset compensation of analog speed input. The actual speed command is the input analog value minus this parameter value.

The analog automatic zero adjustment function can be used, and this parameter is automatically set.3.6.2chapter.



Pr048	Analog speed command direction	scope	Default value	unit	Applicable
		0~1	0		S

Parameter meaning:

Pr048	Positive polarity (positive voltage) analog input enter	Negative polarity (negative voltage) simulation enter
0	Forward(CCW)Speed command	ReverseCW)Speed command
1	ReverseCW)Speed command	Forward(CCW)Speed command

Pr049	Analog speed command filter time constant	scope	Default value	unit	Applicable
		2~500	20	0.1ms	S

Low pass filter for analog speed input.

The larger the setting, the slower the input analog response speed, which is helpful to reduce high-frequency noise interference; the smaller the setting, the faster the response speed, but the high-frequency noise interference is greater.

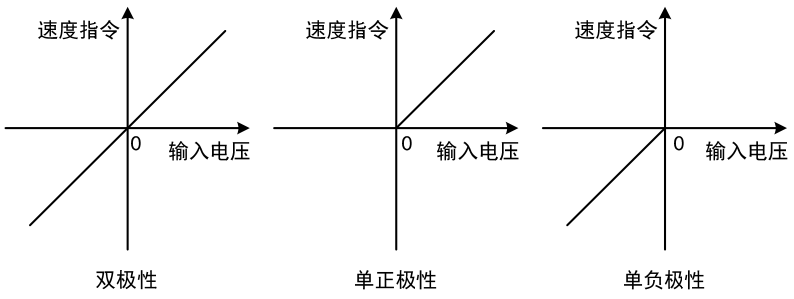
Pr050	Analog speed command polarity	scope	Default value	unit	Applicable
		0~2	0		S

Parameter meaning:

0: Bipolar.

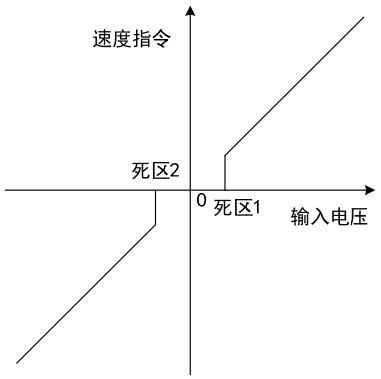
1: Single positive polarity. Input positive polarity is valid, negative polarity is forced to0. 2：

Single negative polarity. Negative input is valid, positive polarity is forced to0.



Pr051	Analog speed command dead zone1	scope	Default value	unit	Applicable
		0~13000	0	mv	S

Input voltage is in the dead zone2(parameterPr052)~Dead Zone1(parameterPr051)The command is forced to0.



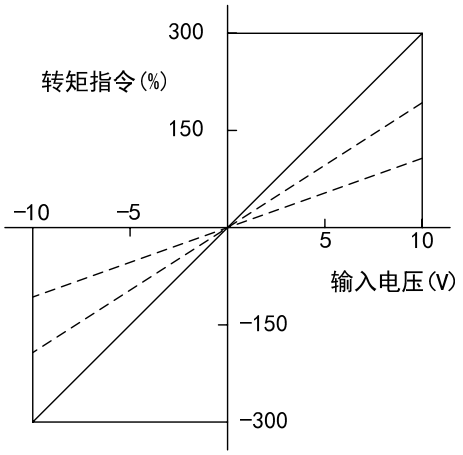
Pr052	Analog speed command dead zone2	scope	Default value	unit	Applicable
		- 13000~0	0	mv	S

Reference ParametersPr051Description.

Pr053	Analog torque command gain	scope	Default value	unit	Applicable
		1~300	30	%/V	T

Set the proportional relationship between the analog torque input voltage and the actual running torque of the motor. The unit of the set value is rated torque.1%/V;

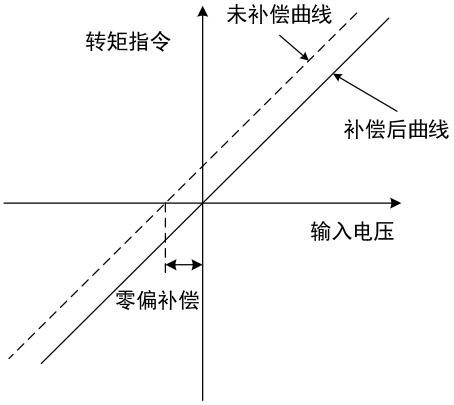
The analog input range is -10V~10V.



Pr054	Analog torque command zero offset compensation	scope	Default value	unit	Applicable
		- 1500~1500	0	mv	T

The zero offset compensation of analog torque input. The actual torque command is the input analog value minus this parameter value.

You can use the analog automatic zero adjustment function to automatically set this parameter.3.6.2chapter.



Pr055	Analog torque command direction	scope	Default value	unit	Applicable
		0~1	0		T

Parameter meaning:

Pr055	Positive polarity (positive voltage) analog input enter	Negative polarity (negative voltage) simulation enter
0	Forward(CCW)Torque command	ReverseCW)Torque command
1	ReverseCW)Torque command	Forward(CCW)Torque command

Pr056	Analog torque command filter time constant	scope	Default value	unit	Applicable
		2~500	20	0.1ms	T

Low-pass filter coefficient for analog torque input.

The larger the setting, the slower the input analog response speed, which is helpful to reduce high-frequency noise interference; the smaller the setting, the faster the response speed, but the high-frequency noise interference is greater.

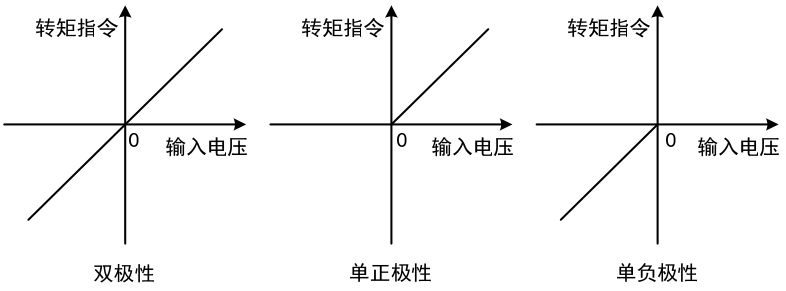
Pr057	Analog torque command polarity	scope	Default value	unit	Applicable
		0~2	0		T

Parameter meaning:

0: Bipolar.

1: Single positive polarity. Input positive polarity is valid, negative polarity is forced to0. 2:

Single negative polarity. Negative input is valid, positive polarity is forced to0.



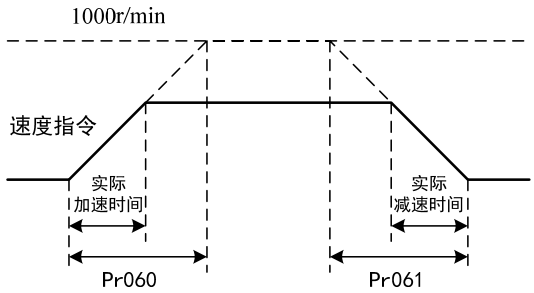
Pr060	Speed command acceleration time	scope	Default value	unit	Applicable

Pr060	Speed command acceleration time	scope	Default value	unit	Applicable
		0~10000	100	ms	S

Set the motor speed from zero to1000r/minIt is only used for

speed control mode and invalid for position control mode.

If used in speed control mode and forms a position control closed loop with the host machine, this parameter should be set to0.



Pr061	Speed command deceleration time	scope	Default value	unit	Applicable
		0~10000	100	ms	S

Set the motor from1000r/minDeceleration time to zero speed. Only

used in speed control mode, invalid in position control mode.

If used in speed control mode and forms a position control closed loop with the host machine, this parameter should be set to0.

Pr063	EMGEmergency stop)/ALMDeceleration time	scope	Default value	unit	Applicable
		0~10000	1000	ms	ALL

whenEMG(Emergency stop) mode is deceleration stop (Pr164Set as1) takes effect. When the

alarm occurs and the shutdown mode is deceleration stop (Pr136Set as1)kick in.

set upEMG(Emergency Stop) or alarm occurs when the motor1000r/minThe deceleration time to zero speed, the deceleration curve is linear.

Pr065	Internal forward (CCW)Torque limit	scope	Default value	unit	Applicable
		0~300	300	%	ALL

Setting up the motorCCWThe internal torque limit value for the direction.

This restriction applies at all times.

If the set value exceeds the maximum overload capacity allowed by the system, the actual limit is the maximum overload capacity allowed by the system.

P066	Internal reversal (CW)Torque limit	scope	Default value	unit	Applicable
		- 300~0	- 300	%	ALL

Setting up the motorCWThe internal torque limit value for the direction.

This restriction applies at all times.

If the set value exceeds the maximum overload capacity allowed by the system, the actual limit is the maximum overload capacity allowed by the system.

Pr067	External forward (CCW)Torque limit	scope	Default value	unit	Applicable
		0~300	100	%	ALL

Setting up the servo motorCCWThe external torque limit value for the direction.

Only inDIInputTCCW(Forward torque limit)ONWhen the limit is effective, the actual torque limit is the minimum value of the maximum overload capacity allowed by the system, the internal forward torque limit, and the external forward torque limit.

Pr068	External reversal (CW)Torque limit	scope	Default value	unit	Applicable
		- 300~0	- 100	%	ALL

Setting up the servo motorCWThe external torque limit value for the direction.

Only inDIInputTCW(Reverse torque limit)ONWhen the limit is effective, the actual torque limit is the smallest absolute value among the maximum overload capacity allowed by the system, the internal reverse torque limit, and the external reverse torque limit.

Pr069	Test run torque limit	scope	Default value	unit	Applicable
		0~300	100	%	ALL

Set the test run mode (speedJOGThe torque limit value in operation, keyboard speed control, and demonstration mode. It has nothing to do with the direction of rotation, and both forward and reverse rotation are limited.

Internal and external torque limits are still effective.

Pr070	Forward(CCW)Torque overload alarm level	scope	Default value	unit	Applicable
		0~300	140	%	ALL

Set forward (CCW)Torque overload value, which is the percentage of rated torque.Pr070, lasting longer thanPr072In this case, the driver alarms, and the alarm number isErr-29, the motor stops.

Pr071	ReverseCW)Torque overload alarm level	scope	Default value	unit	Applicable
		- 300~0	- 140	%	ALL

Set Invert(CW)Torque overload value, which is the percentage of rated torque.Pr071, lasting longer thanPr072In this case, the driver alarms, and the alarm number isErr-29, the motor stops.

Pr072	Torque overload alarm detection time	scope	Default value	unit	Applicable
		1~30000	10000	ms	ALL

Reference ParametersPr070andPr071Description.

Pr075	Maximum speed limit	scope	Default value	unit	Applicable
		0~6000	3500	r/min	ALL

Set the maximum speed limit allowed for the servo motor.

The direction of rotation is irrelevant.

If the set value exceeds the maximum speed allowed by the system, the actual speed will also be limited to the maximum speed.

Pr076	JOGRunning speed	scope	Default value	unit	Applicable
		0~5000	100	r/min	S

set upJOGThe speed at which the operation is performed.

Pr078	Speed limit in torque control	scope	Default value	unit	Applicable
		0~5000	1000	r/min	T

In torque control, the motor running speed is limited within this parameter.

It can prevent overspeed under light load.

When overspeed occurs, speed negative feedback is used to reduce the actual torque, but the actual speed will be slightly higher than the speed limit.

Pr079	Speed limit error during torque control	scope	Default value	unit	Applicable
		1~5000	100	r/min	T

When overspeed occurs, this parameter can adjust the speed negative feedback amount.

The smaller the parameter value, the greater the negative feedback, the steeper the speed limit curve, and the smaller the overspeed. However, too small a value may cause jitter.

Pr080	Position deviation detection	scope	Default value	unit	Applicable
		0~32767	400	0.01lock up	P

Set the position out-of-tolerance alarm detection range.

In position control mode, when the count value of the position deviation counter exceeds the pulse corresponding to this parameter value, the servo drive will give a position deviation alarm (Err 4)The unit is circle, multiply by the encoder's resolution per circle to get the number of pulses.2500Linear encoder, the resolution of each revolution of the encoder is 10000, the parameter value is400When40000encoder pulses.

Pr081	First encoder type selection	scope	Default value	unit	Applicable
		0~1	0		ALL

Set the type of motor encoder;

Parameter meaning:0: Non-wire-saving photoelectric encoder;1: Wire-saving photoelectric encoder.

Pr082	The first encoder line number	scope	Default value	unit	Applicable
		1~32767	2500	pulse	ALL

Set the number of lines of the motor encoder. The default setting is2500Linear photoelectric encoder; If it is an incremental encoder, the number of pulses per revolutionppr =Number of lines of the motor encoder *4.

Pr084	First encoder control parameters	scope	Default value	unit	Applicable
		0~11111	0		ALL

Setting the motor encoderA/BPolarity of the signal:0:A/BNo inversion 1:A/BInvert

Pr097	Positive and negative limit use mode/positive and negative reverse control	scope	Default value	unit	Applicable
		0~33	33		ALL

DIForward drive prohibition in input (CCWL)and reverse drive inhibit (CWL)For limit travel protection, using normally closed switch, input isONThe motor can only run in this direction.OFFIf the limit travel protection is not used, this parameter can be ignored, so that the drive can be operated without connecting the drive prohibition signal.

The default value is to ignore the drive prohibition. If you need to use the drive prohibition function, please modify this value first. The meaning of the single digit parameter:

Pr097 Units digit	Reverse drive prohibition (CWL)	Forward drive disabled (CCWL)
0	use	use
1	use	neglect
2	neglect	use
3	neglect	neglect

use: Input signalONWhen , the motor can run in this direction;OFFThe motor cannot run in this direction. **neglect:**The motor can run in this direction, and the drive prohibition signal has no effect, so this signal can be disconnected. The meaning of the ten-digit parameter: used in conjunction with the analog speed command in speed control mode

Pr097 Tens Character	Forward signal usage mode (Analog speed refers to make)	Reversal signal usage mode (Analog speed command)
0	use	use
1	use	Not used
2	Not used	use
3	Not used	Not used

use: The speed command size is determined by0~10VThe analog value is given, and the direction is determined by the forward and reverse input points. **Do not use:**The speed command size and direction are determined by -10V~10VAnalog reference, forward and reverse input points do not work.

Pr098	Forced Enable	scope	Default value	unit	Applicable
		0~1	0		ALL

Parameter meaning

0: Enabled byDIInputSONcontrol; 1:
Software forced enable.

Pr099	Motor series code	scope	Default value	unit	Applicable
		0~15	0		ALL

Set the motor series, different series of motor power linesUVWThe phase sequence may be different, the encoderAB The signal polarity and the offset angle of the encoder installation may also be different;

Wrong settings may cause the motor to stall or run away! Therefore, before using the driver, the user must ensure the correctness of this parameter!

Parameter meaning: For detailed setting instructions, see7Chapter Motor Adaptation Table.

5.4.2 1Segment parameters

Pr100	Digital InputDI1Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	1		ALL

Digital InputDI1Function planning, parameter absolute value represents function, symbol represents logic, function please refer to 5.5chapter.

The sign indicates the input logic, a positive number indicates positive logic, and a negative number indicates negative logic.ONTo be effective,OFF Is invalid:

Parameter Value	DIInput signal	DIresult
Positive	open circuit	OFF
	Conductivity	ON
negative number	open circuit	ON
	Conductivity	OFF

When multiple input channels have the same function selection, the function result is a logical OR relationship.Pr100and Pr101All set to1(SONfunction), thenDI1,DI2Any oneONhour,SONValid. No parameters are set.Pr100~Pr104The selected input function, that is, the unplanned function, results inOFF(Invalid). However, there are exceptions. Set the parameter Pr120~P124Can force input functionON(effective), regardless of whether the function is planned or not.

Pr101	Digital InputDI2Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	2		ALL

Digital InputDI2Function planning, reference parametersPr100Description.

Pr102	Digital InputDI3Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	3		ALL

Digital InputDI3Function planning, reference parametersPr100Description.

Pr103	Digital InputDI4Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	4		ALL

Digital InputDI4Function planning, reference parametersPr100Description.

Pr104	Digital InputDI5Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	20		ALL

Digital InputDI5Function planning, reference parametersPr100Description.

Pr105	Digital InputDI6Function	scope	Default value	unit	Applicable
		- twenty one~twenty one	twenty one		ALL

Digital InputDI6Function planning, reference parametersPr100Description.

Pr108	Digital OutputDO1Function	scope	Default value	unit	Applicable
		- 12~12	2		ALL

Digital OutputDO1Function planning, parameter absolute value represents function, symbol represents logic, function please refer to5.6chapter.

0For mandatoryOFF,1For mandatoryONThe sign represents the output logic, a positive number represents positive logic, and a negative number represents negative logic:

Parameter Value	Corresponding functions	DOOutput signal
Positive	ON	Conductivity
	OFF	Deadline
negative number	ON	Deadline
	OFF	Conductivity

Pr109	Digital OutputDO2Function	scope	Default value	unit	Applicable
		- 12~12	3		ALL

Digital OutputDO2Function planning, reference parametersPr108Description.

Pr110	Digital OutputDO3Function	scope	Default value	unit	Applicable
		- 12~12	5		ALL

Digital OutputDO3Function planning, reference parametersPr108Description.

Pr111	Digital OutputDO4Function	scope	Default value	unit	Applicable
		- 12~12	8		ALL

Digital OutputDO4Function planning, reference parametersPr108Description.

Pr120	Digital InputDIMandatory1	scope	Default value	unit	Applicable
		00000~11111	00000		ALL

The corresponding function is5Decimal representation:

Digital	Dec 5	Dec4	Dec 3	Dec 2	Dec1
Corresponding functions	CWL	CCWL	ACLR	SON	NULL

Used to forceDIThe input function is valid. If the corresponding function bit is set to1, then this function forcesON(efficient).

DISymbol meaning reference5.5Chapter.

Parameter meaning:

One of the parameters	Corresponding function[Note]	Functional Results
0	Not planned	OFF
	Planned	Determined by input signal
1	Unplanned or planned Row	ON

Note: Planned means that the parametersPr100~Pr105The selected function. Unplanned means that no parameter Pr100~Pr105Selected features.

Pr121	Digital InputDIMandatory2	scope	Default value	unit	Applicable
		00000~11111	00000		ALL

The corresponding function is5Decimal representation:

Digital	Dec 5	Dec4	Dec 3	Dec 2	Dec1
Corresponding functions	CINV	CZERO	ZCLAMP	TCW	TCCW

Other reference parametersPr120Description.

Pr122	Digital InputDIMandatory3	scope	Default value	unit	Applicable
		00000~11111	00000		ALL

The corresponding function is5Decimal representation

Digital	Dec 5	Dec4	Dec 3	Dec 2	Dec1
Corresponding functions	TRQ2	TRQ1	SP3	SP2	SP1

Other reference parametersPr120Description.

Pr123	Digital InputDIMandatory4	scope	Default value	unit	Applicable
		00000~11111	00000		ALL

The corresponding function is5Decimal representation:

Digital	Dec 5	Dec4	Dec 3	Dec 2	Dec1
Corresponding functions	GEAR2	GEAR1	GAIN	CMODE	EMG

Other reference parametersPr120Description.

Pr124	Digital InputDIMandatory5	scope	Default value	unit	Applicable
		00000~11111	00000		ALL

The corresponding function is5Decimal representation:

Digital	Dec 5	Dec4	Dec 3	Dec 2	Dec1
Corresponding functions			PC	INH	CLR

Other reference parametersPr120Description.

Pr125	Digital InputDIxFiltering (DI1~DI6Share this filter parameter)	scope	Default value	unit	Applicable
		1~1000	5	ms	ALL

DIInput digital filter time constant.

The smaller the parameter value, the faster the signal response speed; the larger the parameter value, the slower the signal response speed, but the stronger the noise filtering ability.

Pr136	EnableoffOr the shutdown mode when an alarm occurs	scope	Default value	unit	Applicable
		0~1	0		P

When the driver is enabledoffOr when an alarm occurs, this parameter is used to set the shutdown method. The meaning of the parameters is:

0: The driver directly cuts off the motor current and the motor stops freely.

1： The driver keeps the enabled state first and controls the motor toPr063The motor stops after the defined deceleration time, and then turns off the enable to cut off the motor current.

Pr137	Internal speed1	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed1, reference parametersPr025Description.

Pr138	Internal speed2	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed2, reference parametersPr025Description.

Pr139	Internal speed3	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed3, reference parametersPr025Description.

Pr140	Internal speed4	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed4, reference parametersPr025Description.

Pr141	Internal speed5	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed5, reference parametersPr025Description.

Pr142	Internal speed6	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed6, reference parametersPr025Description.

Pr143	Internal speed7	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed7, reference parametersPr025Description.

Pr144	Internal speed8	scope	Default value	unit	Applicable
		- 5000~5000	0	r/min	S

Internal speed8, reference parametersPr025Description.

Pr145	Internal torque1	scope	Default value	unit	Applicable
		- 300~300	0	%	T

Internal torque1, reference parametersPr026Description.

Pr146	Internal torque2	scope	Default value	unit	Applicable
		- 300~300	0	%	T

Internal torque2, reference parametersPr026Description.

Pr147	Internal torque3	scope	Default value	unit	Applicable
		- 300~300	0	%	T

Internal torque3, reference parametersPr026Description.

Pr148	Internal torque4	scope	Default value	unit	Applicable
		- 300~300	0	%	T

Internal torque4, reference parametersPr026Description.

Pr150	Positioning completion range	scope	Default value	unit	Applicable
		0~32767	10	pulse	P

Set the positioning completion pulse range under position control.

When the number of remaining pulses in the position deviation counter is less than or equal to the setting value of this parameter, the digital output DOofCOIN(Positioning completed)ON,otherwiseOFF.

Pr152	Positioning proximity range	scope	Default value	unit	Applicable
		0~32767	500	pulse	P

Set the positioning approach pulse range under position control.

When the number of remaining pulses in the position deviation counter is less than or equal to the setting value of this parameter, the digital output DO ofNEAR(Location nearby)ON,otherwiseOFFThe comparator has a hysteresis function, which is determined by the parametersPr153Setting. Used for the host computer to accept when positioning is about to be completed.NEARThe signal prepares for the next step. Generally, the parameter value should be greater thanPr150.

Pr154	Arrival speed	scope	Default value	unit	Applicable
		- 5000~5000	500	r/min	ALL

When the motor speed exceeds this parameter, the digital outputDOofASP(Speed Arrival)ON,otherwiseOFFThe comparator has a hysteresis function, which is determined by the parametersPr155Setting. With polarity setting function, by parameterPr156 control:

Pr156	Pr154	Comparator
0	> 0	Speed is not direction-dependent
1	> 0	Only detect forward speed
	<0	Only detect reverse speed

Pr156	Arrival speed polarity	scope	Default value	unit	Applicable
		0~1	0		ALL

Reference ParametersPr154Description.

Pr157	Torque	scope	Default value	unit	Applicable
		- 300~300	100	%	ALL

When the motor torque exceeds this parameter, the digital outputDOofATRQ(Torque reached)ON,otherwiseOFF . With polarity setting function, by parameterPr159control:

Pr159	Pr157	Comparator
0	> 0	Torque is irrelevant in direction
1	> 0	Only detects forward torque
	<0	Only detects reverse torque

Pr159	Arrival torque polarity	scope	Default value	unit	Applicable
		0~1	0		ALL

Reference ParametersPr157Description.

Pr160	Zero speed detection point	scope	Default value	unit	Applicable
		0~1000	10	r/min	ALL

When the motor speed is lower than this parameter, the digital outputDOofZSP(Zero speed)ON,otherwiseOFF.

Pr162	Zero speed clamp mode	scope	Default value	unit	Applicable
		0~1	0		S

When the following conditions are met, the zero speed clamp function is enabled:

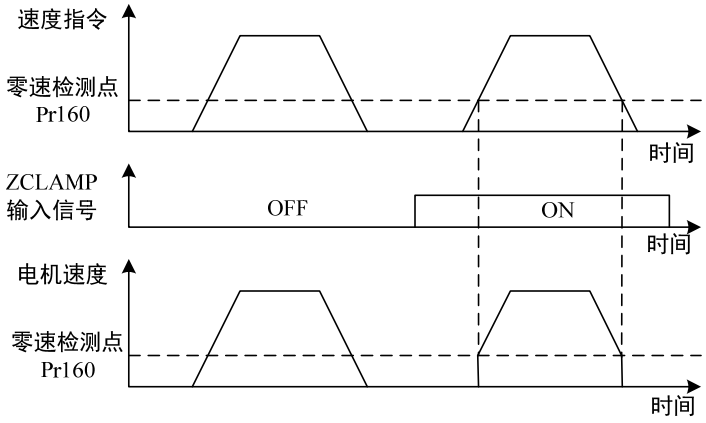
condition1: Speed control mode

condition2:DIInZCLAMP(Zero speed clamp)ON condition3: Speed command is lower than parameterPr160 When any of the above conditions is not met, normal speed control is performed.

When the zero speed clamp function is turned on, the meaning of this parameter is:

0：The motor position is fixed at the moment the function is turned on. At this time, the internal position control is connected, and even if it rotates due to external force, it will return to the zero position fixed point.

1: When the function is turned on, the speed command is forced to zero speed. The internal speed is still controlled and rotation may occur due to external force.



Pr164	Emergency stop method	scope	Default value	unit	Applicable
		0~1	0		P

When the driveDIInEMG(Emergency Stop)ONThis parameter is used to set the shutdown mode. The meaning of the parameters are:

0: The driver directly cuts off the motor current and the motor stops freely;

1：The driver remains enabled and controls the motor toPr063The device stops after the defined acceleration/deceleration time.

Pr165	Motor static speed detection point	scope	Default value	unit	Applicable
		0~1000	5	r/min	ALL

Motor stationary detection: if the motor speed is lower than the parameter value, the motor is considered stationary.

It is only used for electromagnetic brake timing judgment.

Pr166	Electromagnetic brake delay time when the motor is stationary	scope	Default value	unit	Applicable
		0~5000	500	ms	ALL

When the system changes from the enabled state to the disabled state or an alarm occurs, the motor is defined to brake from the electromagnetic brake during standstill (DOOutput TerminalsBRK OFF)The delay time from the start of the brake to the motor current cut-off. This parameter is to make the brake reliable before cutting off the current to avoid the motor's slight displacement or the workpiece falling. The parameter should not be less than the delay time of the mechanical brake.

For the corresponding timing, see4.7chapter.

Pr167	Electromagnetic brake waiting time when the motor is running	scope	Default value	unit	Applicable
		0~5000	500	ms	ALL

When the system changes from the enabled state to the disabled state or an alarm occurs, the motor current is cut off during operation to the electromagnetic brake braking (DOOutput TerminalsBRK OFF)This parameter is used to slow down the motor from high speed to low speed before applying the brake to avoid damage to the brake.

The actual action time isPr167Or the motor decelerates toPr168The time required for the value is the minimum of the two.

For the corresponding timing, see4.7chapter.

Pr168	Electromagnetic brake action speed when the motor is running	scope	Default value	unit	Applicable
		0~3000	100	r/min	ALL

Reference ParametersPr167Description.

Pr169	Electromagnetic brake release delay time	scope	Default value	unit	Applicable
		0~5000	500	r/min	ALL

When the system changes from the disabled state to the enabled state, it is defined that the motor current is turned on until the electromagnetic brake is released (DOOutput TerminalsBRK ON)For the corresponding timing, refer to4.7chapter.

5.5 DIFunctional Details

Special Note:DI1~DI6It is a programmable input port, which is controlled by parametersPr100~Pr105Configure its functionality;DO1~DO4 It is a programmable output port, which is controlled by parametersPr108~P111Configure its functions; parameter configuration is detailed in5chapter5.4.2 The specific functions are described as follows:

<table>
<tr><th>Serial number</th><th>symbol</th><th>Function</th><th>Functional Explanation</th></tr>
<tr><td>0</td><td>NULL</td><td>No function</td><td>The input status has no effect on the system.</td></tr>
<tr><td>1</td><td>SON</td><td>Servo Enable</td><td>OFF: The servo drive is not enabled and the motor does not pass current;
ON: The servo drive is enabled and current flows through the motor.</td></tr>
<tr><td>2</td><td>ACLR</td><td>Alarm clear</td><td>When an alarm occurs, if the alarm is allowed to be cleared, the input rising edge (OFFChangeON Note that only some alarms can be cleared.</td></tr>
<tr><td>3</td><td>CCWL</td><td>Forward drive prohibit</td><td>OFF：Forward rotation is prohibited (CCW)Rotation; ON：Forward rotation is allowed (CCW)Rotate. Used for mechanical limit travel protection, function is subject to parameterPr097Control. Attention Pr097The default value is to ignore this function. If you need to use this function, you need to modify it.Pr097.
<table>
<tr><td>Pr097</td><td>illustrate</td></tr>
<tr><td>0</td><td rowspan="2">When using the forward drive prohibition function, the travel
Normally closed contacts of the switch.</td></tr>
<tr><td>2</td></tr>
<tr><td>1</td><td rowspan="2">Ignore the forward drive inhibit function and the motor can run forward.
Direction running, this signal has no effect and does not need to be connected.</td></tr>
<tr><td>3(default)</td></tr>
</table>
</td></tr>
<tr><td>4</td><td>CWL</td><td>Reverse drive prohibit</td><td>OFF：Prohibit reverse (CW)Rotation; ON: Allow reversal (CW)Rotate. Used for mechanical limit travel protection, function is subject to parameterPr097Control. Attention Pr097The default value is to ignore this function. If you need to use this function, you need to modify it.Pr097.
<table>
<tr><td>Pr097</td><td>illustrate</td></tr>
<tr><td>0</td><td rowspan="2">To use the reverse drive prohibition function, you must connect the travel
Normally closed contacts of the switch.</td></tr>
<tr><td>1</td></tr>
<tr><td>2</td><td rowspan="2">Ignore the reverse drive prohibition function, the motor can reverse
Direction running, this signal has no effect and does not need to be connected.</td></tr>
<tr><td>3(default)</td></tr>
</table>
</td></tr>
<tr><td>5</td><td>TCCW</td><td>Forward torque limit</td><td>OFF:CCWDirectional torque is not affectedPr067Parameter restrictions; ON:CCW Directional torquePr067Parameter restrictions. Note that regardless ofTCCWValid or invalid,CCWThe directional torque is also affected by the parameterPr065limit.</td></tr>
</table>

<table>
<tr><th>Serial number</th><th>symbol</th><th>Function</th><th>Functional Explanation</th></tr>
<tr><td>6</td><td>TCW</td><td>Reverse torque limit</td><td>OFF:CWDirectional torque is not affectedPr068Parameter restrictions; ON:CWDirectional torquePr068Parameter restrictions. Note that regardless ofTCWValid or invalid,CWThe directional torque is also affected by the parameter Pr066limit.</td></tr>
<tr><td>7</td><td>ZCLAMP</td><td>Zero speed clamp</td><td>When the following conditions are met, the zero speed clamp function is enabled:
condition1: Speed control mode;
condition2:ZCLAMP ON; condition3: Speed command is lower than parameterPr160When any of the above conditions is not met, normal speed control is performed. For specific application, refer to the parametersPr162illustrate.</td></tr>
<tr><td>8</td><td>CZERO</td><td>Zero instruction</td><td>Under speed or torque control, the speed or torque commands are:
OFF: Normal instruction;
ON: Zero instructions.</td></tr>
<tr><td>9</td><td>CINV</td><td>Instruction Negation</td><td>Under speed or torque control, the speed or torque commands are:
OFF: Normal instruction;
ON: Invert the instruction.</td></tr>
<tr><td>10</td><td>SP1</td><td>Internal speed choose1</td><td rowspan="3">When speed control or speed limit is enabled,SP1,SP2,SP3Combination selection internal speed1~8:
<table>
<tr><th colspan="3">DISignal[Note]</th><th rowspan="2">Speed command</th></tr>
<tr><th>SP3</th><th>SP2</th><th>SP1</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>Internal speed1(parameterPr137)</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>Internal speed2(parameterPr138)</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>Internal speed3(parameterPr139)</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>Internal speed4(parameterPr140)</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>Internal speed5(parameterPr141)</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>Internal speed6(parameterPr142)</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>Internal speed7(parameterPr143)</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>Internal speed8(parameterPr144)</td></tr>
</table>
Note:0expressOFF,1expressON.</td></tr>
<tr><td>11</td><td>SP2</td><td>Internal speed choose2</td></tr>
<tr><td>12</td><td>SP3</td><td>Internal speed choose3</td></tr>
<tr><td>13</td><td>TRQ1</td><td>Internal torque choose1</td><td rowspan="2">During torque control and torque limitation,TRQ1,TRQ2Combination selection internal torque 1~4:
<table>
<tr><th colspan="2">DISignal[Note]</th><th rowspan="2">Torque command</th></tr>
<tr><th>TRQ2</th><th>TRQ1</th></tr>
<tr><td>0</td><td>0</td><td>Internal torque1(parameterPr145)</td></tr>
<tr><td>0</td><td>1</td><td>Internal torque2(parameterPr146)</td></tr>
<tr><td>1</td><td>0</td><td>Internal torque3(parameterPr147)</td></tr>
<tr><td>1</td><td>1</td><td>Internal torque4(parameterPr148)</td></tr>
</table>
Note:0expressOFF,1expressON.</td></tr>
<tr><td>14</td><td>TRQ2</td><td>Internal torque choose2</td></tr>
</table>

<table>
<tr><th>Serial number</th><th>symbol</th><th>Function</th><th>Functional Explanation</th></tr>
<tr><td>15</td><td>EMG</td><td>Emergency Stop</td><td>OFF: Allow the servo drive to work;
ON： Based on parametersPr164The motor stops running in the set way.</td></tr>
<tr><td>16</td><td>CMODE</td><td>Control Mode Switch</td><td>parameterPr004Set to3,4,5When the control mode is switched:
<table>
<tr><th>Pr004</th><th>CMODE</th><th>Control method</th></tr>
<tr><td rowspan="2">3</td><td>0</td><td>Location</td></tr>
<tr><td>1</td><td>speed</td></tr>
<tr><td rowspan="2">4</td><td>0</td><td>Location</td></tr>
<tr><td>1</td><td>Torque</td></tr>
<tr><td rowspan="2">5</td><td>0</td><td>speed</td></tr>
<tr><td>1</td><td>Torque</td></tr>
</table>
Note:0expressOFF,1expressON.</td></tr>
<tr><td>17</td><td>GAIN</td><td>Gain switching</td><td>When the parameterPr176=2When, throughGAINSwitching gain combinations: OFF:1Gain;
ON:2Gain.</td></tr>
<tr><td>18</td><td>GEAR1</td><td>Electronic gear choose1</td><td rowspan="2">parameterPr034Used to configure dynamic electronic gear application mode, see4.2.3 Detailed description of the chapter.</td></tr>
<tr><td>19</td><td>GEAR2</td><td>Electronic gear choose2</td></tr>
<tr><td>20</td><td>CLE</td><td>Position deviation Clear</td><td>Clear position deviation counter</td></tr>
<tr><td>twenty one</td><td>INH</td><td>Pulse input prohibit</td><td>OFF: Position command pulse is allowed to pass;
ON: Position command pulse is prohibited.</td></tr>
</table>

5.6 DOFunctional Details

Serial number	symbol	Function	Functional Explanation
0	OFF	Has been ineffective	Forced outputOFF.
1	ON	Always valid	Forced outputON.
2	SRDY	Servo ready	OFF: The servo main power supply is not turned on or there is an alarm; ON: The servo main power supply is normal and there is no alarm.
3	ALM	Call the police	OFF: There is an alarm; ON: No alarm.
4	ZSP	Zero speed	OFF：The motor speed is higher than the parameterPr160(Regardless of direction); ON ：The motor speed is lower than the parameterPr160(regardless of direction).
5	COIN	Positioning completed	Position control OFF：Position deviation is greater than parameterPr150; ON ：The position deviation is less than the parameterPr150.
6	ASP	Speed arrival	OFF：The motor speed is lower than the parameterPr154; ON ：The motor speed is higher than the parameterPr154. With polarity setting function, refer to parametersPr154illustrate.
7	ATRQ	Torque reached	OFF：The motor torque is lower than the parameterPr157; ON：The motor torque is higher than the parameterPr157. With polarity setting function, refer to parametersPr157illustrate.
8	BRK	Electromagnetic brake	OFF: Electromagnetic brake braking; ON: The electromagnetic brake is released.
9	RUN	Servo running	OFF: The servo motor is not powered on; ON: The servo motor is powered on and running.
10	NEAR	Positioning proximity	Position control OFF：Position deviation is greater than parameterPr152; ON ：The position deviation is less than the parameterPr152.
11	TRQL	Torque limiting	OFF: The motor torque does not reach the limit value; ON: The motor torque reaches the limit value. Torque limiting method through parametersPr064set up.
12	SPL	Speed limit	Torque control OFF: The motor speed does not reach the limit value; ON: The motor speed reaches the limit value. The speed limit method passes the parameterPr077set up.

No.6Chapter Fault and Diagnosis

6.1Alarm list

Call the police Code	Alarm Name	Alarm content	Call the police Clear
E--	No alarm	Working fine	
E 1	Speeding	The motor speed exceeds the maximum limit	Can
E 2	Power main circuit overvoltage	The main circuit power supply voltage exceeds the specified value. Check the brake circuit. Is the resistor damaged or is the resistance value appropriate?	no
E 3	Power main circuit undervoltage	The main circuit power supply voltage is lower than the specified value	no
E 4	Location out of tolerance	Position tracking deviation exceeds the set value	Can
E 5	Position Command Overclocking	The position command frequency exceeds the maximum allowable frequency	Can
E 6	Motor stall	Motor power line connection error, pole pair numberP-201mistake	Can
E 7	Driver disabled exception	CCWL,CWLThe drive travel limit switch signal is abnormal	Can
E 9	Incremental encoderABZSignal failure	EncoderABZThe signal is disturbed or disconnected	Can
E10	Incremental encoderUVWSignal failure	EncoderUVWThe signal is disturbed or disconnected	Can
E11	IPMModule failure	Power main circuitIPMThe inverter module fails.	no
E12	Overcurrent	Servo drive instantaneous current is too large	Can
E13	Overload	The average load current of the motor is too large	Can
E14	Braking peak power overload	The instantaneous load is too large during short-term braking. Check the braking resistor. Is it damaged or is the resistance value appropriate?	Can
E20	EEPROMmistake	EEPROMRead/write error or unexpected power failure when saving parameters	no
E21	Logic circuit error	Processor peripheral logic circuit failure	no
E23	ADConversion reference voltage error	ADThe sampling circuit voltage reference is not the standard value	no
E24	ADThe conversion channel is asymmetric or the zero drift is large	ADAbnormal sampling and amplification circuit	no
E29	User torque overload alarm	The motor load exceeds the value and duration set by the user	Can
E30	EncoderZSignal loss	EncoderZSignal not present	no
E31	EncoderZSignal detection abnormality	EncoderZThe signal is interfered or unstable	no
E32	EncoderUVWSignal illegal encoding	EncoderUVWSignal disconnection	no
E33	Wire-saving encoder signal error	No high impedance state during power-up sequence	no

6.2Alarm causes and solutions

E 1(Speeding

reason	examine	deal with
Motor wiringU,V,WPhase sequence error	examineU,V,Wwiring	Connect correctlyU,V,WWiring, with driver plugU,V,WOne-to-one correspondence
Motor speed overshoot	Check the running status and view the parameters. numberPr075Maximum speed limit Is the value appropriate?	Adjust the servo gain to reduce the overshoot. Adjust; when controlling the speed, you can increase the addition and subtraction Speed Time
Encoder wiring error	Check encoder wiring	Correct wiring

E 2(Main circuit overvoltage)

reason	examine	deal with
Input AC power is too high	Check the power supply voltage	Make the voltage meet the product specifications
Regenerative braking failure	Regenerative brake resistor, brake pipe Is it faulty or disconnected?	repair
Excessive regenerative braking energy	Check the brake load rate	Reduce the frequency of starting and stopping Increase acceleration and deceleration time Reduce the torque limit value Reduce load inertia Replace with a higher power drive and motor Replace with a larger brake resistor

E 3(Main circuit undervoltage)

reason	examine	deal with
Input AC power is too low	Check the power supply voltage	Make the voltage meet the product specifications

E 4(Location out of tolerance)

reason	examine	deal with
Motor wiringU,V,WPhase sequence error	examineU,V,Wwiring	Connect the motor correctlyU,V,W Wiring, with driver plugU,V,WOne-to-one correspondence
Encoder zero point change	Check encoder zero point	Reinstall the encoder and zero it

Encoder wiring error	Check encoder wiring	Correct wiring
Motor stuck	Check the motor and mechanical connections part	repair
The command pulse frequency is too high	Check input frequency, pulse distribution Frequency multiplication parameters	Reduce input frequency Adjust the pulse frequency division parameters
Position loop gain is too small	Check parametersP009,P013	Increase position loop gain
The out-of-tolerance detection range is too small	Check parametersP080	Add parametersP080Numeric
Insufficient torque	Check torque	Increase the torque limit value Increase the position command smoothing filter time Reduce the load Replace with a higher power drive and motor

E 5(Position command overclocking)

reason	examine	deal with
The command pulse frequency is too high	Check input frequency, electronic gear Is the wheel ratio setting correct?	Reduce input frequency Correct SetupP029,P030
Motor enable to command pulse Interval between issuances Too short	The host computer gives the enable to the transmitter Add a delay between pulses to few500ms(Wait for the motor to charge Points incentive)	Modify the host computer timing
The host computer instruction does not perform addition and subtraction Speed processing or acceleration and deceleration time Too short	The host computer sends position instructions The acceleration and deceleration time settings are Is it reasonable?	Increase the addition and subtraction of the instructions sent by the host computer Speed Time Properly increase the position command smoothing filter coefficientP040

E 6(Motor stalled)

reason	examine	deal with
Motor stuck	Check the motor and mechanical connections part	Maintenance
Motor wiringU,V,WPhase sequence error	examineU,V,Wwiring	Connect the motor correctlyU,V,W Wiring, with driver plugU,V,WOne-to-one correspondence
Encoder wiring error	Check encoder wiring	Correct wiring
Motor overload	Check the load currentD7-iWhether big	Reduce the load Increase torque limitP065,P065 Replace the drive with a higher power one

E 7(Driver prohibition exception)

reason	examine	deal with
When servo is enabledCCWL, CWLDrive inhibit input invalid	examineCCWL,CWLwiring	Correct inputCCWL,CWLIf the signal is not usedCCWL,CWLSignal, configurable parametersP097shield

E 9(EncoderABSignal failure)

reason	examine	deal with
Encoder wiring error	Check encoder wiring	Correct wiring
Encoder cables and connectors bad	Check cables and connectors	Replacing cables and connectors
Encoder damage	Check the encoder	Replace the encoder

E 11(Power module failure)

reason	examine	deal with
Motor wiringU,V,WShort circuit between	examineU,V,Wwiring	Connect correctlyU,V,Wwiring
Motor winding insulation damage	Check the motor	Replace the motor
Drive damage	Check the drive	The motor has no problem, but the alarm still occurs when the power is turned on again. Warning: The drive may be damaged. Replace it. Drive
Poor grounding	Check the ground wire	Proper grounding
Disturbed	Check for interference sources	Add line filters and stay away from interference sources

E12(Overcurrent)

reason	examine	deal with
Motor wiringU,V,WShort circuit between	examineU,V,Wwiring	Connect correctlyU,V,Wwiring
Motor winding insulation damage	Check the motor	Replace the motor
Drive damage	Check the drive	The motor has no problem, but the alarm still occurs when the power is turned on again. Warning: The drive may be damaged. Replace it. Drive

E13(Overload)

reason	examine	deal with
Continuous operation exceeding rated load OK	Check the load factorD16LD	Reduce the load or change to a higher power drive Device
System instability	Check whether the motor is vibrating during operation swing	Reduce system gain
Acceleration and deceleration too fast	Check whether the motor is running smoothly Shun	Increase the acceleration and deceleration time
Encoder zero point change	Check encoder zero point	Reinstall the encoder and zero it

E14(Braking peak power overload)

reason	examine	deal with
Input AC power is too high	Check the power supply voltage	Make the voltage meet the product specifications
Regenerative braking failure	Regenerative brake resistor, brake pipe Is it faulty or disconnected?	repair
Excessive regenerative braking energy	Check the brake load rate	Reduce the frequency of starting and stopping Increase acceleration and deceleration time Replace with a higher power drive and motor Replace with a larger brake resistor
Encoder Problems	The number of lines and poles is incorrect EncoderZSignal Error Encoder damage	Replace the encoder

E20 (EEPROMmistake)

reason	examine	deal with
EEPROMChip damage	Re-power on and check	If the fault persists, please replace the drive

E21(Logic circuit error)

reason	examine	deal with
Control circuit failure	Re-power on and check	If the fault persists, please replace the drive

E23(ADConversion reference voltage error)

reason	examine	deal with

Current sensor and connector problems	Check the main circuit, soft Is the connection line loose? move	Replacing the drive
ADConverter and analog amplifier circuit question	Check the control circuit	Replacing the drive

E24(ADConversion channel asymmetry or zero drift value is too large)

reason	examine	deal with
ADConverter and analog amplifier circuit question	Check the control circuit	Replacing the drive

E29(User torque overload alarm)

reason	examine	deal with
Unexpected heavy load occurs	Check load conditions	Adjusting the load
parameterPr070,Pr071,Pr072 Improper settings	Check parameters	Adjusting parameters

E30(EncoderZSignal Loss)

reason	examine	deal with
Encoder Problems	View EncodersZletter Number	Replace the encoder
Encoder cable and connector problems	Check cables and connectors Item	Replacing cables and connectors
Driver interface circuit failure	Check the control circuit	Replacing the drive

E31(EncoderZSignal detection abnormality)

reason	examine	deal with
Encoder Problems	View EncodersZletter Number	Replace the encoder
There is interference in the encoder cable	Check cables and connectors Item	Replacing cables and connectors
Driver interface circuit failure	Check the control circuit	Replacing the drive

E32(EncoderUVWSignal illegal encoding)

reason	examine	deal with
Encoder Problems	Check the encoderUVW Signal	Replace the encoder
Encoder wiring error or disconnection	Check encoder wiring	Correct wiring, including shielded wires

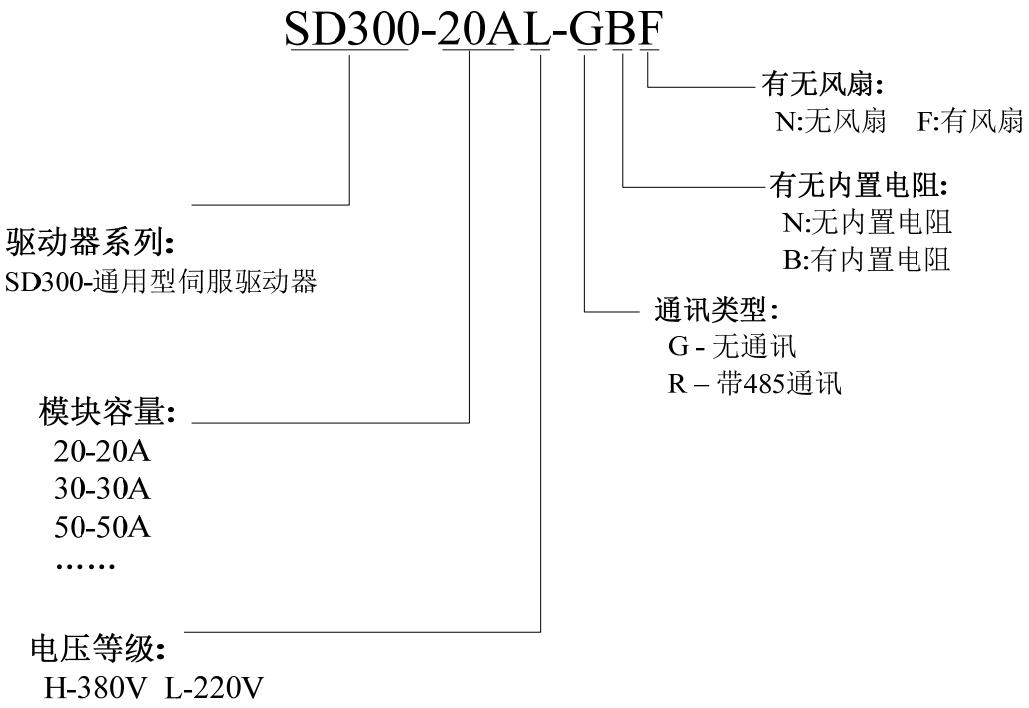
E33(Wire-saving encoder signal error)

reason	examine	deal with
Encoder Problems	Check encoder signal	Replace the encoder
The motor model/encoder type is incorrect Set	Check the motor model and confirm Is the motor configured with wire-saving type? Encoder	Reset motor model/code Device Type

No.7Chapter Specifications and Suitable Motors

7.1Driver Specifications

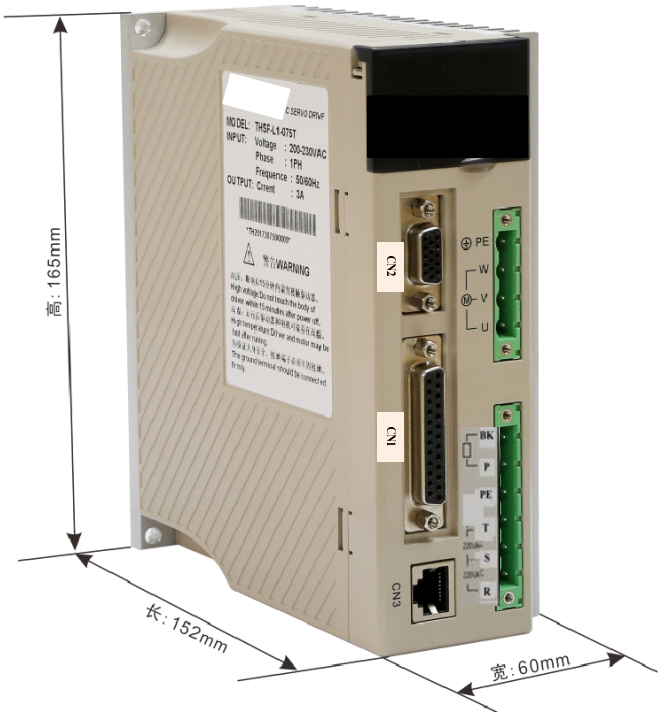
Model naming rules



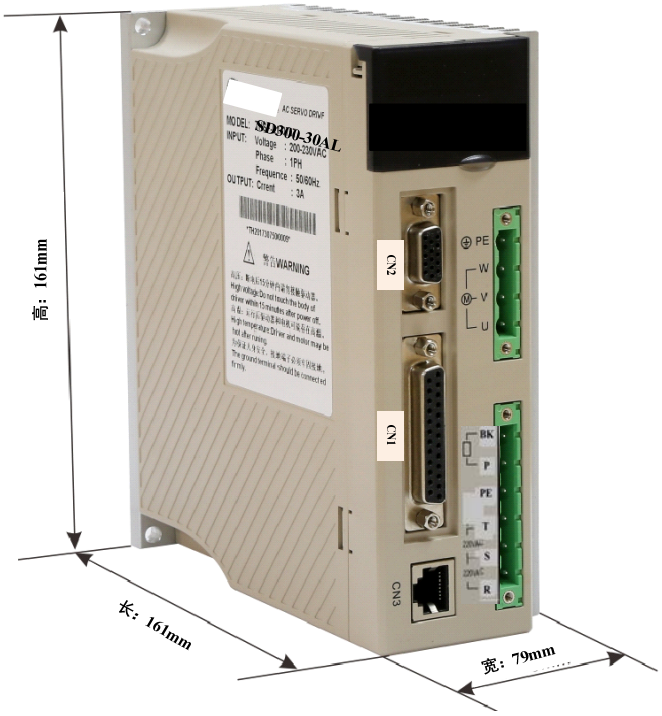
picture7.1Driver model naming convention

Driver installation dimensions

XP200 安装尺寸图



XP200 安装尺寸图



Model Specifications and Functions

surface7.1Driver model specification

model		20AL	30AL	40AL	50AL	75AL	25AH	50AH	75AH
enter power supply	Main power supply	Single phase/three phase220VAC -15%~+10% 50/60Hz					Three-phase380VAC -15%~+10% 50/60Hz		
ring territory	temperature	Work:0℃ ～40℃					Storage:-40℃ ～50℃		
	humidity	Work:40%~80%(No condensation)					Storage:93%Below (no condensation)		
	Atmospheric pressure	86kPa~106 kPa							
Protection level		IP20							
Control Mode		Position, speed, torque, position/speed, speed/torque, position/torque							
Digital Input		6Programmable input terminals (photoelectric isolation)							
Digital Output		4Programmable output terminals (photoelectric isolation)							
coding Device Signal Output	Signal Type	A,B,ZDifferential output,ZSignal open collector output							
Bit Place	Input frequency	Differential Input:≤500kHz（kpps); Single-ended input:≤200kHz（kpps）							
	Instruction Mode	Pulse+direction; forward/reverse pulse; quadrature pulse							
	Electronic gear ratio	1~32767/1~32767							
speed Spend	Simulation instructions	±10VDC, input impedance10kΩ							
	Instruction plus slow down	Parameter settingsPr060/Pr061							
	Instruction source	Analog quantity, internal speed command							
change Moment	Simulation instructions	- 10V～ +10V, input impedance10kΩ							
	Torque limit	Parameter settingsPr065/Pr066							
	Instruction source	Analog quantity, internal torque command							
Monitoring function		Speed, current position, position deviation, motor torque, motor current, command pulse frequency wait							
Protection function		Overspeed, overvoltage, overcurrent, overload, abnormal braking, encoder abnormality, position error, command overfrequency, etc.							
special sex	Speed frequency response	⩾800Hz							
	Speed Fluctuation	＜±0.03%(load0~100%); ＜±0.02%(power supply-15%～ +10%)							

7.2Motor Adaptation Table

surface7.2XP200**-20AL/30ALDriver compatible motor model**

Motor model code (Pr002)	Adapter driver (AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated Current (A)	Rated Torque (Nm)	Overload multiple
20	XP200-20AL	40ST-M00130	0.05	0.61	0.16	3
20		40ST-M00330	0.1	0.9	0.32	3
twenty one		60ST-M00630	0.2	1.2	0.6	3
twenty two		60ST-M01330	0.4	2.8	1.3	3
twenty three		60ST-M01930	0.6	3.5	1.9	3
27		80ST-M01330	0.4	2.0	1.3	3
28		80ST-M02430	0.75	3.0	2.4	3
29		80ST-M03520	0.73	3.0	3.5	3
30		80ST-M04025	1.0	4.4	4	2.5
31		90ST-M02430	0.75	3	2.4	3
32		90ST-M03520	0.73	3	3.5	3
33		90ST-M04025	1.0	4	4	2.5
34	XP200-20AL	110ST-M02030	0.6	2.5	2	3
35	XP200-30AL	110ST-M04030	1.2	5	4	3
39	XP200-20AL	110ST-M04020	0.8	3.5	4	3
36	XP200-30AL	110ST-M05030	1.5	6	5	2.5
37	XP200-30AL	110ST-M06020	1.2	4.5	6	3
38	XP200-30AL	110ST-M06030	1.8	6	6	2.5
44	XP200-30AL	130ST-M04025	1	4	4	2.5/3
45	XP200-30AL	130ST-M05025	1.3	5	5	2/3
46	XP200-30AL	130ST-M06025	1.5	6	6	2.5
41	XP200-30AL	130ST-M06030	1.8	7	6	2.2
47	XP200-30AL	130ST-M07720	1.6	6	7.7	2.5
47	XP200-30AL	130ST-M07725	2.0	7.5	7.7	2
48	XP200-30AL	130ST-M07730	2.4	9	7.7	1.6
40	XP200-20AL	130ST-M10010	1.0	4.5	10	2.2
49	XP200-30AL	130ST-M10015	1.5	6	10	2.5
43	XP200-30AL	130ST-M10020	2	8.5	10	1.7
50	XP200-30AL	130ST-M10025	2.6	10	10	1.5
51	XP200-30AL	130ST-M15025	3.8	13.5	15	1.2
52	XP200-30AL	130ST-M15015	2.3	9.5	15	1.6

surface7.3XP200-**40AL/50AL/75ALDriver compatible motor model**

Motor Model Code Pr002	Adapter driver (AC 220V)	Servo motor model (220V)	Rated power (kW)	Rated Current (A)	Rated Torque (Nm)	Overload multiple
0	XP200-40AL	130ST-M07720	1.6	6	7.7	3
0	XP200-40AL	130ST-M07725	2.0	7.5	7.7	3
1	XP200-40AL	130ST-M07730	2.4	9	7.7	3
2	XP200-40AL	130ST-M10015	1.5	6	10	3
3	XP200-40AL	130ST-M10025	2.6	10	10	3
4	XP200-40AL	130ST-M12020	2.4	10	12	3
5	XP200-40AL	130ST-M15015	2.3	9.5	15	3
6	XP200-40AL/50AL	130ST-M15025	3.9	13.5	15	2.2/2.6
7	XP200-40AL/50AL	150ST-M15020	3.0	14	15	2.2/2.6
8	XP200-50AL	150ST-M15025	3.8	17	15	2.1
9	XP200-50AL	150ST-M18020	3.6	17	18	2.1
10	XP200-50AL/75AL	150ST-M23020	4.7	twenty one	twenty three	1.6/2.1
11	XP200-50AL/75AL	150ST-M27020	5.4	twenty four	27	1.5/1.9
12	XP200-40AL	180ST-M17215	2.7	10.5	17.2	2.8
13	XP200-40AL	180ST-M19015	3	12	19	2.5
14	XP200-50AL	180ST-M21520	4.5	16	21.5	2.2
15	XP200-40AL	180ST-M27010	2.9	12	27	2.5
16	XP200-50AL	180ST-M27015	4.3	16	27	2.2
17	XP200-50AL	180ST-M35010	3.7	16	35	2.2
18	XP200-75AL	180ST-M35015	5.5	twenty four	35	1.8
19	XP200-75AL	180ST-M48015	7.5	32	48	1.5

No.8Chapter Quick Commissioning Guide

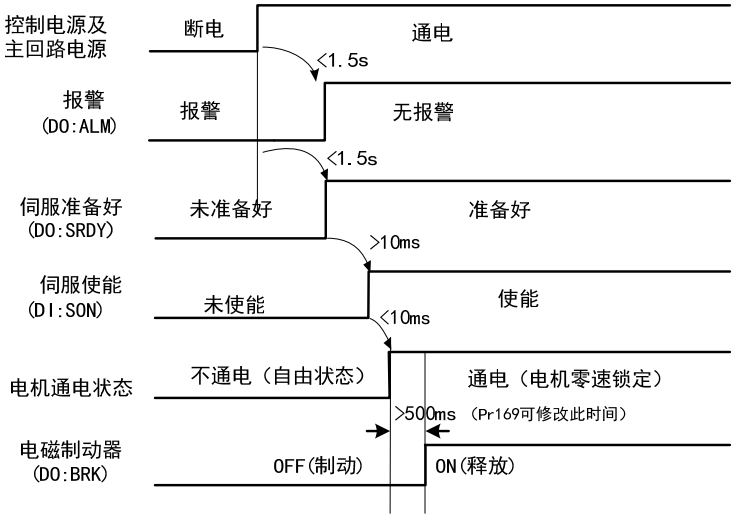
8.1Quick Debugging Notes

1. Make sure the wiring is correct

- Power TerminalsR/S/Tand motor power terminalsU/V/W, cannot be connected reversely, and cannot be loose;
- LThe series input voltage must be three-phase or single-phase220V,HThe series input voltage must be three-phase380V;
- Check whether the motor connection cable is short-circuited and must be reliably grounded;
- Check the command cable and motor encoder cable, and make sure they are reliably connected;
- The wiring of the same motor must correspond to the same driver.

2. Determine the power-on sequence

- XP200The strong current and control current of the series servo drives adopt an integrated design. When the power is turned on, the strong current and control current are turned on at the same time. When the power is turned off, the internal strong current is cut off first, and the display and control circuits are automatically powered off after a few seconds of delay discharge. Please wait patiently;
- If the servo motor is equipped with an electromagnetic brake (holding brake), the brake is not controlled by the servo driver. Please make sure that the servo is enabled.0.5The safety and position accuracy of the equipment can be guaranteed only if the brake is powered on (brake released) for more than 1 second.
- In order to use the servo drive smoothly, please read the following timing diagram carefully (please refer to this manual for more detailed timing diagrams).4.7Festival):



8.2Position control mode (quick debugging)

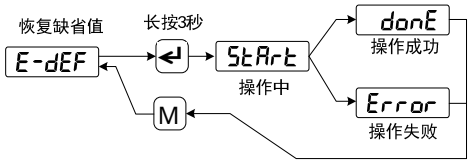
example:XP200-30ALServo drive, adaptor130ST-M15015Servo motor (position control) **1:**After power on, the panel display lights up and the power terminals are measured.R,S,TSingle phase or three phase220VVoltage, ensure the power supply is normal;

2:Do not connect the servo enable signal for the time being and check whether there is an alarm. If there is no alarm, the system is working normally and you can proceed to the next step.

3:Start adapting the motor control parameters:

a), enter parameter modification mode "Pr-", modify the operation password (parameterPr000)for316, and then modify Motor code (parametersPr002) is the required motor model. For the motor model code, refer to7.2Chapter Motor Adaptation Table,This examplePr002Set as52;

b), enter parameter management mode "EE-", perform the restore default operation as shown in the following figure;



恢复所有参数的缺省值

c)After the default value restoration operation is successful, set or check several key parameters of position control (as shown in the following table) 8.1 As shown), after confirmation, the host computer can give an external enable signal**SON**(or internal enable:Pr098 Set as1), give a pulse signal, observe the motor operation effect, and modify the gain appropriately to adjust the operation characteristics.

surface8.1Key parameter settings for position control mode

parameter	name	Setting Value	Parameter Description
Pr004	Control method	0(Default value)	Set to position control mode
Pr029	Electronic gear molecule	1(Default value)	Electronic gear molecule
Pr030	Electronic gear denominator	1(Default value)	Electronic gear denominator
Pr035	Pulse input mode	0(Default value)	0：Pulse＋direction 1: Forward/reverse pulse 2: Orthogonal pulse
Pr036	Pulse input direction	0(Default value)	0: Normal direction 1: Direction reversed
Pr005	**Speed loop proportional gain**	50(Default value)	To increase rigidity or reduce tracking error
Pr006	Speed loop integral time constant	20(Default value)	Poor, can be appropriately increasedPr005/
Pr009	**Position loop proportional gain**	40(Default value)	Pr009, each adjustment amount is5
Pr007	**Torque filter time constant**	25(Default value)	The motor generates vibration or noise when running
Pr019	**Speed detection filter time constant**	35(Default value)	When The total amount is5

8.3Speed control mode (quick commissioning)

example:XP200-30ALServo drive, adaptor130ST-M10015Servo motor (speed control) **1:**After power is turned on, the display panel lights up and the power terminals are measured.R,S,TSingle phase or three phase220VVoltage, ensure the power supply is normal;

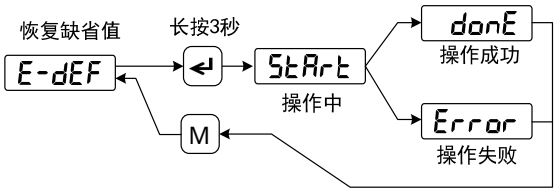
2:Make sure the speed analog differential input or single-ended input wiring is correct;

3:Do not connect the servo enable signal for the time being and check whether there is an alarm. If there is no alarm, the system is working normally and you can proceed to the next step.

4:Start adapting the motor control parameters:

a), enter parameter modification mode "PR-", modify the operation password (parameterPr000)for316, and then modify Motor code (parametersPr002) is the required motor model. For the motor model code, refer to7.2Chapter Motor Adaptation Table,This examplePr002Set as49;

b), enter parameter management mode "EE-", perform the restore default operation as shown in the following figure;



恢复所有参数的缺省值

c)After the default value restoration operation is successful, set several key parameters of speed control (such as8.2),

Then the menu switches to "E-SETLong press  key3Execute parameter saving operation in seconds, and disconnect after successful saving

After power-on restart and confirmation, the host computer can give an external enable signal**SON**(or internal enable:Pr098 Set as1), after the enable indicator light is on, the analog quantity automatic zero adjustment operation can be performed (see3.6.2Chapter), give analog commands, observe the motor operation effect, and modify the gain appropriately to adjust the operation characteristics.

surface8.2Key parameter settings for speed control mode

parameter	name	Setting Value	Parameter Description
Pr004	Control method	1	Set to speed control mode
Pr025	Speed command source	0(Default value)	0: Analog speed 1: Internal multi-speed<8Can selectPr137~Pr144Settings>
Pr046	Analog speed command gain	300(Default value) Set as needed	1VAnalog voltage corresponding to the conversion Speed (Unit:r/min/V)
Pr047	Analog speed command zero offset compensation	0(Default value) Set as needed	unit:0.1mv
Pr048	Analog speed command direction	0(Default value) Set as needed	0: Normal direction 1: Direction reversed
Pr060	Speed command acceleration time	100(Default value) Set as needed	0~1000r/minAcceleration time (unit:ms)
Pr061	Speed command deceleration time	100(Default value) Set as needed	1000~0r/minDeceleration time (unit:ms)
Pr005	**Speed loop proportional gain**	50(Default value)	To increase rigidity, adjust Pr005, each adjustment amount is5
Pr006	Speed loop integral time constant	20(Default value)	If the load inertia is large, IncreasePr006, each adjustment amount is 10
Pr007	**Torque filter time constant**	25(Default value)	The motor generates vibration or noise when running
Pr019	**Speed detection filter time constant**	35(Default value)	When The total amount is5

8.4Torque control mode (quick commissioning)

example:XP200-20ALServo drive, adaptor80ST-M02430Servo motor (torque control) **1:**After power is turned on, the display panel lights up and the power terminals are measured.R,S,TSingle phase or three phase220VVoltage, ensure the power supply is normal;

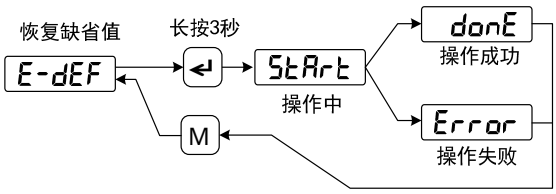
2:Make sure the torque analog differential input or single-ended input wiring is correct;

3:Do not connect the servo enable signal for the time being and check whether there is an alarm. If there is no alarm, the system is working normally and you can proceed to the next step.

4:Start adapting the motor control parameters:

a), enter parameter modification mode "Pr-", modify the operation password (parameterPr000)for316, and then modify Motor code (parametersPr002) is the required motor model. For the motor model code, refer to7.2Chapter Motor Adaptation Table,This examplePr002Set as28 ;

b), enter parameter management mode "EE-", perform the restore default operation as shown in the following figure;



恢复所有参数的缺省值

c)After the default value restoration operation is successful, set several key parameters of torque control (such as8.3),

Then the menu switches to "E-SETLong press  key3Execute parameter saving operation in seconds, and disconnect after successful saving

After power-on restart and confirmation, the host computer can give an external enable signal**SON**(or internal enable:Pr098 Set as1), after the enable indicator light is on, the analog quantity automatic zero adjustment operation can be performed (see3.6.2Chapter), give analog commands, observe the motor operation effect, and modify the gain appropriately to adjust the operation characteristics.

surface8.3Key parameter settings for torque control mode

parameter	**name**	Setting Value	Parameter Description
Pr004	Control method	2	Set to torque control mode
Pr026	Torque command source	0(Default value)	0: Analog torque 1：Internal multi-stage torque<4indivual OptionalPr145~Pr148set up >
Pr053	Analog torque command gain	30(Default value) Set as needed	1VAnalog voltage corresponding to Percentage of motor rated torque Ratio (Unit:1%/V)
Pr054	Analog torque command zero bias compensate	0(Default value) Set as needed	unit:0.1mv
Pr055	Analog torque command direction	0(Default value) Set as needed	0: Normal direction 1: Direction reversed
Pr078	Speed limit in torque control	1000(default value) Set as needed	(unit:r/min)

8.5Typical debugging problems and solutions

①, **Error message when restoring default parameters**

a, confirm the operation password (Pr000) is set correctly, you need to set it to316; b, The default restoration operation must be performed when the drive is disabled;

②, **The host computer enables, but the drive does not enable**

a, Check the power supplyR/S/TWhether the voltage is too low and the wiring is correct; b,examineCN1Interface, +24VWhether the input is correct; c,examineCN1 InterfaceSONIs the signal0VConnect;

d, If the driver is still not enabled through the above measures, you can use the internal enable (Pr098Set as1)Again try.

③,**Appear"E9/E10/E30/E31/E32/E33”Fault alarm**

Note: The photoelectric encoder at the end of the servo motor is a typical fragile component and needs special protection!

a, The above alarms indicate that there is a problem with the encoder or the encoder connection cable; b1. Check whether the shielding layer is well grounded at both ends and whether there is water or impurities in the plug; c, Is the connecting line too long?5VCause attenuation;

d2. Confirm whether it is an interference problem and whether there are strong magnetic or electric lines nearby. If so, isolate them as much as possible.

④, **Noise or vibration occurs during the operation of the servo motor**

a, appropriately increase the filter coefficientPr007andPr019, each adjustment amount is5;

bIf increasing the filter coefficient does not produce a significant effect, you need to reduce the speed loop proportional gain and position loop proportional gain. Adjust to smallPr005andPr009,Each adjustment is5;

cIf the above measures do not improve the situation, please check whether there is interference in the encoder cable and the screen. Check whether both ends of the shielding layer are well grounded.

⑤, **The servo motor vibrates during operation**

a, Determine whether the load and inertia of the servo motor are within the allowable range of the motor. The load exceeds the motor's rated value by too much (the load torque is greater than3times, the inertia is greater than5times), please reselect a motor with a larger specification;

b, appropriately increase the speed loop proportional gainPr005, each adjustment amount is5;

cIf you increasePr005The effect is not obvious, you can adjust the filter coefficient appropriatelyPr007/P-019, each adjustment amount is5;

dIf the effect is not obvious when the filter coefficient is reduced, the position loop gain can be appropriately reduced.Pr009, each adjustment amount is 5;

eIf the above measures do not improve the situation, please check whether the encoder connection line and command connection line are

Interference, check whether the shielding layer is well grounded at both ends.

⑥, the drive appearsE 2orE11Alarm number

Determine whether the drive alarms when powered on or when there is frequent acceleration and deceleration with large inertia:

a1. If the alarm sounds when the power is turned on, it can be determined that the hardware circuit of the driver is faulty;

b, If the alarm occurs when there is frequent acceleration and deceleration due to large inertia, first check the monitoring menud7-Iandd8-pI,observe

Check whether the instantaneous value and maximum value of the current exceed the allowable range of the driver and motor. If so, reduce the inertia acceleration accordingly (speed mode: increase the acceleration time).Pr060and deceleration timePr061; Position mode: increase the acceleration and deceleration time of the host controller) to control the current within the allowable range of the drive and motor, and check whether the fault disappears.

⑦, when the servo motor starts, the driver displays "E 5/E12”

a,Appear"E 5”This indicates that the pulse command frequency sent by the host computer is too fast, exceeding the response capacity of the servo motor.

It is recommended to increase the acceleration and deceleration time of the host computer, or appropriately increase the position command smoothing time.

Pr040, each adjustment amount is10;

b, the above measures are invalid or the host computer cannot modify, pleasePr116Set as32shield"E 5”Call the police; c, If the drive displays "E12”Overcurrent alarm indicates that the motor load is too large when starting. Please check

Is the drive too small or damaged?

⑧, the driver displays "E 4”

a, If the low speed operation is normal, the high speed operation will appear "E 4”, please check firstPr075Maximum speed limit

If it is too low, adjust the position loop gain appropriately.Pr009(Each adjustment amount is5), or appropriately increase the position error detection rangePr080;

b, If you only give the position command, no matter what the speed is, the motor will appear as soon as it runsE 4”, please confirm

Whether the motor is blocked or damaged, and whether the driver's high-voltage circuit is damaged;

c, If the "E 4”, please confirm whether the driver and motor selection are too small; d, If the "E 4” Please check whether the encoder connection line and command connection line exist.

Interference, check whether the shielding layer is well grounded at both ends.

⑨, the driver runs normally, but the host computer displays "position tracking error is too large"

a, determine the detection threshold of the upper computer position tracking error exceeding the limit, and set the drive position error detection range

WaiPr080Set it to a value lower than the threshold and observe whether the drive has "E 4” ;

bIf the drive does not appearE 4”, it means that the encoder signal sent back to the host computer on the command connection line

If there is interference, please check whether the connecting wires are welded reliably and whether the shielding layer is well grounded at both ends;

cIf the drive displays "E 4”, it means that the driver and motor have response problems. Please first confirm

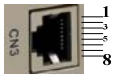
Check the integrity of the drive and motor, and then increase the position loop and speed loop gains.Pr009/Pr005; dIf the above measures have no obvious effect, consider selecting a larger specification drive and motor.

No.9Chapter Communication Function Description

9.1Functional Overview and Wiring Diagram

XP200The servo drive hasRS-485Serial communication interface, use this interface toMODBUSThe protocol can control the servo system, change parameters, monitor the servo system status and other functions. Communication portCN3 The wiring instructions are as follows:

XP200-20AL/30ALThe communication port of the driver uses a standard network cable port, and its appearance and pin distribution are as follows:



3/4	485_A
5/6	485_B
7/8	485地

Special Notes:RS485The communication port can also beCN1of40/41/42Connection, see2.4Section diagram description.

9.2Communication parameters

Pr181	Drive communicationIDNumber	scope	Default value	unit
		- 1~247	- 1	

useRS-485During communication, the upper controller is the host and the servo drive is the slave. The communication address of the servo drive needs to be set to a different communication station number through this parameter. The setting range of the station number address is -1~247, The default value is -1 Indicates that the communication function is turned off. The setting value is greater than0Indicates that the communication function is open.Before using the communication function, this parameter must be set to the required station number. This station number represents the absolute address of this drive in the communication network. A group of servo drives can only be set with one station number. If the station number is set repeatedly, normal communication will not be possible.

Pr182	MODBUSCommunication baud rate	scope	Default value	unit
		0~3	2	

This parameter selectsRS-485The baud rate of communication. Different values correspond to different baud rates. The selected communication baud rate must be consistent with the communication baud rate of the upper controller. The specific setting values are as follows:

Parameter meaning:

0: The baud rate is4800bps 1: The baud rate is9600bps

2: The baud rate is19200bps 3: The baud rate is38400bps

Pr183	MODBUSCommunication data mode selection (Currently only supportsRTUData Grid Mode)	scope	Default value	unit
		0~5	1	

This parameter selectsRS-485The data mode of communication. The selected data mode must be consistent with the communication protocol of the upper controller. The specific parameter values have the following meanings:

0: Data bits -8Bit Check digit - None Stop Bits -1Bit
1: Data bits -8Bit Parity bit - even parity (Even） Stop Bit -1Bit
2: Data bits -8Bit parity bit - odd parity (Odd） Stop Bits -1Bit
3: Data bits -8Bit Check digit - None Stop Bits -2Bit
4: Data bits -8Bit Parity bit - even parity (Even） Stop Bit -2Bit
5: Data bits -8Bit parity bit - odd parity (Odd） Stop Bit -2Bit

9.3 MODBUSCommunication Protocol

useRS-485During serial communication, each servo drive must be configured in advance in parameters.Pr181The communication station number is set on the computer, and the computer or host controller controls the corresponding servo drive according to the station number; the baud rate needs to refer to the communication method of the host controller to set the parametersPr182;MODBUSThe communication protocol currently only supportsRTU（Remote Terminal Unit) mode, the user can set the parameters according to the requirements of the upper controller. Pr183After the above parameters are set, you need to save the parameters and restart the drive to make them effective.MODBUSThe specific content of the communication.

Communication data structure:

RTUmodel:

MODBUSThe application layer protocol defines a simple protocol data unit (PDU), as shown in the figure below, this protocol data unit does not depend on the underlying communication layer.

Address field	Function code	Data Field	Verification Code

MODBUSThe address field is used as the beginning of the frame, and the content of the address field is the valid address value (0~247), the host places the slave address value in the address field of the request information to determine the recipient of the request information. After the qualified slave receives the information and completes the corresponding processing, it places its own address value in the address field of the response information, so that the host knows which slave sent the response.

The address field is followed by the function code, which indicates what operation the slave will perform. The function code is followed by the data field, which contains the parameters of the request and response. The format, length and meaning of the data field vary depending on the function code.

The checksum is used to verify the validity of information and ensure the reliability of information transmission. RTUMode is used16BitCRC（Cyclical Redundancy Check)check.

RTUIn this mode, each8One byte of data is pressed2indivual4Position16Send in binary code, for example:1bytedata64H.

RTUmode is a bit-oriented transmission mode with no less than3.5The idle time of characters is the beginning of the frame, followed by the address field. The network device monitors the communication bus. When it continuously monitors3.5After the idle time of characters, the address field information can be received. When the received address field information is judged to be valid, the subsequent information will continue to be received, and then the corresponding operation will be performed according to the function code and additional information. If a response message is required

The response information is then encapsulated and sent to the host. The last byte sent is approximately3.5After an idle time of characters, the information frame ends and a new information frame can be sent.RTUThe mode relies on analog synchronization information to maintain frame synchronization. The entire information frame must be transmitted as a continuous data stream at one time. If a continuous data stream is being transmitted, the receiving device detects that there is1.5If the interval time is more than one character, it is considered that a frame of data has been received, and the next received character is regarded as the beginning of the next frame.

Under normal circumstances, the interval between information frames is at least3.5characters, that is, a frame of data must be sent at least3.5The next frame of data can be sent only after the idle time of characters. **RTUThe information frame format in this mode is:**

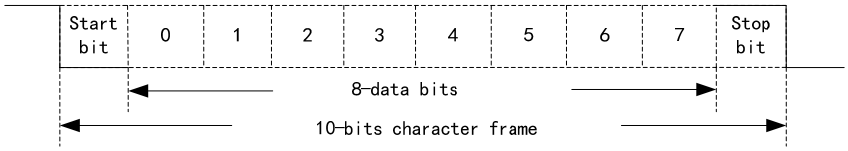
START	Address field (ADR)	Function code (CMD)	Data Field (DATA)	Check code (CRC)	END
T1-T2-T3-T4	8 bits	8 bits	n*8 bits	16 bits	T1-T2-T3-T4

Note:T1-T2-T3-T4Indicates that the minimum time interval with the previous frame is3.5Character time. RTUIn this mode, the format of sending each byte is:

Start bit	Data bit (low bit first)	Parity bit	Stop bits
1 bit	8 bits	1 bit or no bit	1 bit or 2 bits

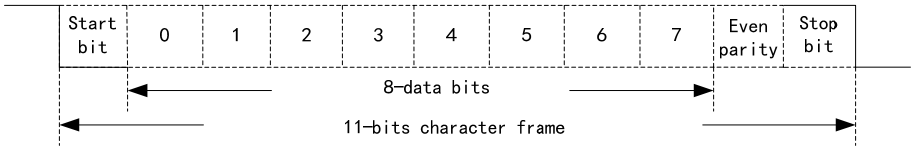
Character structure:

10bitsCharacter box (for8bitsCharacters are not checked)

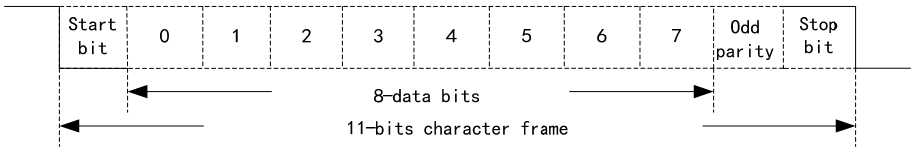


(8Bit data + no check +1Stop bits)

11bitsCharacter box (for8bitCharacter plus checksum)



(8bit data + even parity +1Stop bits)



(8bit data + odd parity +1Stop bits)

Communication information frame format:

Communication information frame format The items in the box are described as follows:

START(Communication start)

RTUMode: The minimum time interval with the previous frame is3.5Character time. ADR(

Correspondence Address)

The legal communication address range is1arrive32As shown below:16(hexadecimal10H)Communicate with the servo drive:

RTUmodel:ADR = 10H

CMD(Function code) andDATA(Data field) The format of the data characters depends on the function code. Commonly used function codes are described as follows: **Command code03H**: Read the parameter value of the drive

Command code04H: Read the operating status of the drive (such as motor speed, position, current, torque and other variable values) **Command code06H:**Write parameters to the drive (single write) **Command code41H:**Write the drive parameters to EEPROM(Parameters saved) **CRC(RTUMode) frame check calculation:**

RTUmodel:

RTUMode adoptionCRC (Cyclical Redundancy Check)Frame check,CRCThe frame check calculation is described in the following steps:

Step 1: Initialize a contentFFFFHof16 bitsRegister, calledCRCRegister. Step 2: Compare the first byte of the command information with16-bitsCRCThe low byte of the register is XORed.

And save the result backCRCregister.

Step 3: CheckCRCThe lowest bit of the register (LSB), if this bit is0, then shift right one position; if this position is1,

butCRCThe register value is shifted right by one bit and thenA001HPerform XOR operation. Step 4: Return to step 3 until step 3 has been executed.8Step 5: Repeat steps 2 to 4 for the next byte of the command information until all bytes are completed.

The above process, at this timeCRCThe contents of the register areCRCFrame check. **illustrate:**CalculateCRC After the frame check, in the command information, you must first fill inCRC, and then fill in CRCPlease refer to the following example.

For example: Read the station number as01HServo drive0part05parameter.ADRCalculated from the last byte of data CRCThe final contents of the register are3794H, then the command information is as follows, it should be noted that: Byte94HShould be in bytes37HPreviously transmitted.

ADR	01H
CMD	03H
Starting data Location	00H (High Byte)
	05H (Low Byte)
Number of data	00H (High Byte)
	02H (Low Byte)
CRC Low	94H (High Byte)

CRC High	37H (Low Byte)

END(End of communication):

RTUMode: The minimum time interval with the next frame is3.5Character time.

Function code (CMD):

Function code is1indivualbyte（8bits), whose range is1~255.

Function code (HEX)	Specific meaning	ModbusThe protocol specification contains righteous
03H	Read the drive parameter value (can be Read multiple parameters when	Read Holding Register
04H	Read the operating status of the drive (such as power Machine speed, position, current, torque, etc. Variable values, multiple states can be read at the same time state)	Read Input Register
06H	Write the drive parameters (single write enter)	Write Single Register
10H	Write parameters to the drive (multiple write enter)	Write Multiple Register
08h	Diagnostic function	Diagnostic
41H	Write the drive parameters to EEPROM(Parameters saved)	

Various exceptions and errors may occur during communication. In order to make these exceptions and errors identifiable by the host, the corresponding exception codes are defined as shown in the following table. When the slave receives information from the host and finds an error or cannot complete a normal response, the slave will send an error response frame to the host, that is, the highest position of the function code byte1, the data field places the corresponding exception code (Exception Code).

Exception Code（HEX）	Mean
01h	ILLEGAL FUNCTION
02h	ILLEGAL DATA ADDRESS
03h	ILLEGAL DATA VALUE
04h	SLAVE DEVICE FAILURE
05h	ACKNOWLEDGE
06h	SLAVE DEVICE BUSY

9.4Writing and reading parameters

For details of all servo drive parameters, please refer to the parameter section of the commissioning manual. Parameters are divided into parameter sections. Each parameter is used16 bitsThe communication address of each parameter is determined by the parameter number.

(scope0~249), for example: parameterPr005The correspondence address is005(decimal), parameterPr108 The correspondence address is108(decimal), parameterPr204The correspondence address is204(decimal), and the other parameters are similar.

All parameters described in the parameter section can be read through communication, but only some parameters are open for users to modify and write. The parameters listed in the commissioning manual are allowed to be modified and written. Other parameters not described are servo drive reserved parameters (users cannot write to reserved parameters, otherwise it may cause abnormal operation of the servo drive). If users need to modify other parameters, please contact our technical staff.

9.5Monitor status address description

The internal state of the servo drive can beRS-485Communication port read and write operations are invalid. 16bitData storage, providing0~39common40The reading of the state variable can be realized through the following two function codes:

（1) Function code (04H), address range0x0000~0x0027 （2) Function code (03H), address range0x1000~0x1027 Function code (04H) The address (hexadecimal) of the read status quantity is as follows:

0x0000: Motor speed, unit "r/min”; 0x0001：Original position command (input pulse) low16bit; 0x0002: Original position command (input pulse) high16bit; 0x0003：Position command (pulse) low16bit; 0x0004：Position command (pulse) high16bit; 0x0005：Current motor position (pulse) low16bit; 0x0006: Current motor position (pulse) high16bit; 0x0007：Position deviation (pulse) low16bit; 0x0008：Position deviation (pulse) high16bit;

0x0009: Motor torque, unit "%" (percentage of rated torque); 0x000A: Peak torque, unit "%" (1sPeak torque within);10>

0x000B: Motor instantaneous current, unit "0.1A”（12: Indicates the real-time current of the motor 1.2A); 0x000C: Motor peak current, unit "0.1A” （152:express1sThe maximum internal current is15.2A);

0x000D: Position command pulse frequency, unit "0.1kHz”（3000:express300kHz); 0x000E: Speed command, unit "r/min”; 0x000F: Torque command, unit "%"; <15> 0x0010: Speed simulation command voltage, unit "mV”; 0x0011: Torque simulation command voltage, unit "mV”; 0x0012: Input terminalDIStatus, [Note1]; 0x0013: Output terminalDOStatus, [Note2];

0x0014：Motor encoder single turn absolute position (pulse) low16bit; <20>

0x0015: Motor encoder single-turn absolute position (pulse) high16bit;

0x0016: The multi-turn position of the motor encoder (when there is no multi-turn information, read 0value); 0x0017: Regenerative braking load rate, unit "%";

0x0018: Average load rate, unit "%";
0x0019: Output voltage, unit "%"; <25>
0x001A: Alarm code;
0x001B: Motor speed, unit "0.1r/min”; 0x001C：
Second encoder position (pulse) low16bit;
0x001D: Second encoder position (pulse) high
16bit; 0x001E:Retain<30>DC bus voltage
0x001F: Motor absolute position low16(multi-turn + single-turn data splicing32absolute position)
0x0020: Motor absolute position high16(multi-turn + single-turn data splicing32absolute position)
0x0021:reserve
0x0022:reserve
0x0023:Retain<35>
0x0024:reserve
0x0025:reserve
0x0026:reserve
0x0027:Retain<39>

Function code (03H) The address (hexadecimal) of the read status quantity is as follows:

0x1000: Motor speed, unit "r/min”; 0x1001：Original
position command (input pulse) low16bit; 0x1002: Original
position command (input pulse) high16bit; 0x1003：Position
command (pulse) low16bit; 0x1004：Position command
(pulse) high16bit; 0x1005：Current motor position (pulse)
low16bit; 0x1006: Current motor position (pulse) high16bit;
0x1007：Position deviation (pulse) low16bit; 0x1008：
Position deviation (pulse) high16bit;

0x1009: Motor torque, unit "%" (percentage of rated torque);
0x100A: Peak torque, unit "%" (1sPeak torque within);10>
0x100B: Motor instantaneous current, unit "0.1A”（12: Indicates the real-time current of the motor
1.2A); 0x100C: Motor peak current, unit "0.1A” （152:express1sThe maximum internal current is15.2A);
0x100D: Position command pulse frequency, unit "0.1kHz”（3000:express300kHz);
0x100E: Speed command, unit "r/min”; 0x100F: Torque command, unit "%"; <15>
0x1010: Speed simulation command voltage, unit "mV”; 0x1011: Torque simulation
command voltage, unit "mV”; 0x1012: Input terminalDIStatus, [Note1]; 0x1013: Output
terminalDOStatus, [Note2];

0x1014：Motor encoder single turn absolute position (pulse) low16bit; <20>

0x1015: Motor encoder single-turn absolute position (pulse) high16bit;

0x1016: The multi-turn position of the motor encoder (when there is no multi-turn information, read

0value); 0x1017: Regenerative braking load rate, unit "%";

0x1018: Average load rate, unit "%";

0x1019: Output voltage, unit "%"; <25>

0x101A: Alarm code;

0x101B: Motor speed, unit "0.1r/min”; 0x101C：

Second encoder position (pulse) low16bit;

0x101D: Second encoder position (pulse) high

16bit; 0x101E:Retain<30>DC bus voltage

0x101F: Motor absolute position low16(multi-turn + single-turn data splicing32absolute position)

0x1020: Motor absolute position high16(multi-turn + single-turn data splicing32absolute position)

0x1021:reserve

0x1022:reserve

0x1023:Retain<35>

0x1024:reserve

0x1025:reserve

0x1026:reserve

0x1027:Retain<39>

[Note1]：The data read out from this address is16bit,inbit5~bit0Corresponding toDI6~DI1The input state of1”Indicates input high level,0”Indicates input low level;bit15~bit6This is a reserved bit. **[Note2]**: The data bits read out at this address16bit,inbit5~bit0expressDO6~DO1The output state, "1"Indicates output high level,0”Indicates output low level;bit15~bit6This is a reserved bit.

9.6Communication Examples

9.6.1Read status (CMD=04H）

Use the read status function code (04H) can read all monitoring status quantities in the servo drive. The communication frame format is as follows:

Note: The maximum number of states read each time is8indivual.

The request frame format is as follows:

Address field 1 byte	Function code 1 byte	Starting address value 2 bytes, the high byte is in forward	Read status count 2 bytes, high byte first	CRCCheck code 2 bytes, low byte first
0x01	0x04	0x00,0x00	0x00,0x01（**N**）	CRC_Lo, CRC_Hi

Normal response:

Address field 1 byte	Function code 1 byte	Number of bytes occupied by the status 1byte	Status value,2N bytes, High byte first	CRCCheck code 2 bytes, low byte first
0x01	0x04	0x02（**2N**）	0xXX, 0xXX,...	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0x84	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

Here is an example:

The host sends a frame:

ADDR	CODE	STADDR_H	STADDR_L	RNUM_H	RNUM_L	CRC_L	CRC_H
0x01	0x04	0x00	0x01	0x00	0x02	0xXX	0xXX

Slave response frame:

ADDR	CODE	BYTE_NUM	DATA1_H	DATA1_L	DATA2_H	DATA2_L	CRC_L	CRC_H
0x01	0x04	0x04	0xXX	0xXX	0xXX	0xXX	0xXX	0xXX

STADDR: Read status starting address

RNUM: The number of states read

BYTE_NUM: The number of bytes occupied by the read status (for exampleRNUMfor2,Right now2indivual16bit data, then BYTE_NUM for2*2 = 4byte)

DATAX: The status value read back,XRepresents the serial number,DATA1is the first data,DATA2The second data

In this example, the frame sentSTADDR = 0x0001,RNUM = 0x02, indicating that from the starting address0x0001Start, read2Status data (that is, the read address is0x0001and0x0002The two status values of (for their specific meanings, see the description in the previous section).

In response frameBYTE_NUM = 0x04, indicating that the two status values read occupy4bytes, followed byDATA1, DATA2That is, the address is0x0001and0x0002The status value of .

9.6.2Read parameters (CMD=03H)

Use the read parameter function code (03H) can read all parameters in the driver. The communication frame format is as follows:

Note: The maximum number of parameters read each time is8indivual.

The request frame format is as follows:

Address field 1 byte	Function code 1 byte	Starting address value 2 bytes, the high byte is in forward	Read the number of parameters 2 bytes, high byte first	CRCCheck code 2 bytes, low byte first
0x01	0x03	0x00,0x00	0x00,0x01 **(N)**	CRC_Lo, CRC_Hi

Normal response:

Address field 1 byte	Function code 1 byte	Number of bytes occupied by parameters 1byte	Parameter value,2N bytes, high byte first	CRCCheck code 2 bytes, low byte first
0x01	0x03	0x02 **(2N)**	0xXX,0xXX,...	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0x83	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

Here is an example:

The host sends a frame:

ADDR	CODE	STADDR_H	STADDR_L	RNUM_H	RNUM_L	CRC_L	CRC_H
0x01	0x03	0x00	0x01	0x00	0x02	0xXX	0xXX

Slave response frame:

ADDR	CODE	BYTE_NUM	DATA1_H	DATA1_L	DATA2_H	DATA2_L	CRC_L	CRC_H
0x01	0x03	0x04	0xXX	0xXX	0xXX	0xXX	0xXX	0xXX

STADDR: Read parameter starting address

RNUM: Number of parameters read

BYTE_NUM: The number of bytes occupied by the parameters read (for exampleRNUMfor2,Right now2indivual16bit data, then BYTE_NUM for2*2 = 4)

DATAX: The parameter value read back,XRepresents the serial number,DATA1is the first parameter,DATA2is the second parameter

In this example, the frame sentSTADDR = 0x01,RNUM = 0x02, indicating that from the starting address1Start, read2 parameters (that is, the read address is0x01and0x02Two parameters ofP-001andP-002).

In response frameBYTE_NUM = 0x04, indicating that the two parameter values read occupy4bytes, followed byDATA1, DATA2That is, the address is0x01and0x02Parameter value.

9.6.3Write a single parameter (CMD=06H)

Use the write parameter function code (06H) can write the parameters listed in the debugging manual. The communication frame format is described as follows: **Notice:**

Function code06HOnly one parameter can be written at a time. Before writing a parameter, please refer to the parameter manual to confirm the specific meaning of the parameter. Wrongly writing a parameter value may cause the servo drive to run abnormally!

The request frame format is as follows:

Address field 1 byte	Function code 1 byte	Parameter number 2 bytes, high byte first	Parameter value written 2 bytes, high byte first	CRCCheck code 2 bytes, low byte first
0x01	0x06	0x00,0x00	0x00,0x64	CRC_Lo, CRC_Hi

Normal response:

Address field 1 byte	Function code 1 byte	Parameter number 2 bytes, high byte first	Parameter value written 2 bytes, high byte first	CRCCheck code 2 bytes, low byte first
0x01	0x06	0x00,0x00	0x00,0x64	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0x86	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

Here is an example:

The host sends a frame:

ADDR	CODE	WADDR_H	WADDR_L	VALUE_H	VALUE_L	CRC_L	CRC_H
0x01	0x06	0x00	0x01	0x01	0x02	0xXX	0xXX

Slave response frame:

ADDR	CODE	WADDR_H	WADDR_L	VALUE_H	VALUE_L	CRC_L	CRC_H
0x01	0x06	0x00	0x01	0x01	0x02	0xXX	0xXX

WADDR：Write parameter address

VALUE: The value to be written

In this example, the frame sentADDR = 0x01,VALUE = 0x0102, which means that the value0x0102Write address is0x01 parameters.

The response frame is the same as the send frame.

9.6.4Write multiple parameters (CMD=10H)

Use the write parameter function code (10H) can write the parameters listed in the debugging manual. The communication frame format is described as follows: **Notice** :

Function code10HMultiple parameters can be written at a time (the maximum number of parameters written at a time is10Before writing a parameter, please refer to the parameter manual to confirm the specific meaning of the parameter. Wrongly writing a parameter value may cause the servo drive to run abnormally!

The request frame format is as follows:

Address field 1 byte	Function code 1 byte	Starting address value, 2 bytes, High byte first	Write parameter number,2 bytes, High byte first （1~122）	Parameter word Number of sections,1byte （0~255）	Parameter value, 2N bytes, High byte first	CRCCheck code 2 bytes, Low byte first
0x01	0x10	0x00,0x00	0x00,0x02 （N）	0x04 （2N）	0xXX, 0xXX, ...	CRC_Lo, CRC_Hi

Normal response:

Address field 1 byte	Function code 1 byte	The starting address value, 2 bytes, High byte first	Write parameter number,2 bytes, High byte first	CRCCheck code 2 bytes, low byte first
0x01	0x10	0x00,0x00	0x00,0x02 （N）	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0x90	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

Here is an example:

The host sends a frame:

ADR	CODE	SADRH	SADRL	WNUMH	WNUML	WNUMB	D1_H	D1_L	D2_H	D2_L	CRCL	CRCH
0x01	0x10	0x00	0x01	0x00	0x02	0x04	0xXX	0xXX	0xXX	0xXX	0xXX	0xXX

Slave response frame:

ADR	CODE	SADRH	SADRL	WNUMH	WNUML	CRCL	CRCH
0x01	0x10	0x00	0x01	0x00	0x02	0xXX	0xXX

SADR: Write parameter starting addressSADRH(High Byte) WNUM: SADRL(low byte)

The number of parameters to be writtenWNUMH(High Byte) WNUML(low byte)

WNUMB: The number of bytes occupied by the written parameters (for exampleWNUMfor2,Right now2indivual16bit data, thenWNUMBfor 2*2 = 4

)

D1_H: The first1The high byte value of the parameter D1_L: The first1The low byte value of the parameter

D2_H: The first2The high byte value of the parameter D2_L: The first2The low byte value of the parameter

In this example, the frame sentSADR = 0x01,WNUM = 0x02, indicating that from the starting address1Start, write2parameters (that is, the address is0x01and0x02Two parameters ofP-001andP-002).

9.6.5Diagnostic function (CMD=08H) After the host sends a diagnostic frame, the slave (servo drive) will respond with the same data. Tip: The diagnostic function can be used to determine MODBUSIs the communication normal? The host sends a frame:

ADDR	CODE	DATA_H	DATA_L	CRC_L	CRC_H
0x01	0x08	0x12	0x34	0xXX	0xXX

Slave response frame:

ADDR	CODE	DATA_H	DATA_L	CRC_L	CRC_H
0x01	0x08	0x12	0x34	0xXX	0xXX

The slave's response frame data of the diagnostic command should be exactly the same as the frame data sent by the host, so the diagnostic command can be used to detect whether the communication is normal.DATACan be any custom data, such as the host sends0xABCD,The slave responds 0xABCD

9.6.6Save parameters (CMD=41H)

Use the save parameter function code (41H) can store the drive parameters inEEPROM, the communication frame format is described as follows:

Note: After sending this command, the servo drive needs some time to save the parameters toEEPROM It is recommended to wait after sending this command.5SThen perform other operations. The request frame format is as follows:

Address field 1 byte	Function code 1 byte	CRCCheck code 2 bytes, low byte first
0x01	0x41	CRC_Lo, CRC_Hi

Normal response:

Address field 1 byte	Function code 1 byte	CRCCheck code 2 bytes, low byte first
0x01	0x41	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0xC1	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

9.6.7Servo Enable/Disable (CMD=42H）

Request/Normal Response:

Address field 1 byte	Function code 1 byte	Enable shutdown flag 1byte	CRCCheck code 2 bytes, low byte first
0x01	0x42	0x55(Enable)or 0xAA(Off)	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0xC2	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi

9.6.8Alarm clear (CMD=43H）

Request/Normal Response:

Address field 1 byte	Function code 1 byte	CRCCheck code 2 bytes, low byte first
0x01	0x43	CRC_Lo, CRC_Hi

Abnormal response:

Address field 1 byte	Function code 1 byte	Exception code 1byte	CRCCheck code 2 bytes, low byte first
0x01	0xC3	0x01 or 02 or 03 or 04	CRC_Lo, CRC_Hi